48TF008-014
48TM008-014
Single-Package Rooftop Heating/Cooling Units

Installation, Start-Up and Service Instructions

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SAFETY CONSIDERATIONS

Installation and servicing of air-conditioning equipment can be hazardous due to system pressure and electrical components. Only trained and qualified service personnel should install, repair, or service air-conditioning equipment.

Untrained personnel can perform basic maintenance functions of cleaning coils and filters and replacing filters. All other operations should be performed by trained service personnel. When working on air-conditioning equipment, observe precautions in the literature, tags and labels attached to the unit, and other safety precautions that may apply.

Follow all safety codes. Wear safety glasses and work gloves. Use quenching cloth for unbrazing operations. Have fire extinguishers available for all brazing operations.

⚠ CAUTION

Ensure voltage listed on unit data plate agrees with electrical supply provided for the unit.

⚠ WARNING



Disconnect gas piping from unit when leak testing at pressure greater than $^1/_2$ psig. Pressures greater than $^1/_2$ psig will cause gas valve damage resulting in hazardous condition. If gas valve is ever subjected to pressure greater than $^1/_2$ psig, it *must* be replaced before use. When pressure testing field-supplied gas piping at pressures of $^1/_2$ psig or less, a unit connected to such piping must be isolated by manually closing the gas valve(s).

⚠ WARNING

Before performing service or maintenance operations on unit, turn off main power switch to unit, install lockout tag. Electrical shock could cause personal injury.

INSTALLATION

Unit is shipped in the vertical discharge configuration. To convert to horizontal configuration, remove screws from side duct opening covers and remove covers. Using the same screws, install covers on vertical duct openings with the insulation side down. Seals around duct openings must be tight. See Fig. 1.

Confirm before installation of unit that voltage, amperage and circuit protection requirements listed on unit data plate agree with power supply provided.

Step 1 — Provide Unit Support

ROOF CURB — Assemble and install accessory roof curb in accordance with instructions shipped with curb. See Fig. 2. Install insulation, cant strips, roofing felt, and counter flashing as shown. *Ductwork must be attached to curb.* If gas or electrical connections are to be routed through the curb, attach the accessory thru-the-curb service connection plate to the roof curb in accordance with the accessory installation instructions. Connections must be installed before unit is set in roof curb.

IMPORTANT: The gasketing of the unit to the roof curb is critical for a watertight seal. Install gasket supplied with the roof curb as shown in Fig. 2. Improperly applied gasket can also result in air or water leaks and poor unit performance.

Curb should be level. Unit leveling tolerances are shown in Fig. 3. This is necessary for unit drain to function properly. Refer to Accessory Roof Curb Installation Instructions for additional information as required.

If gas or electrical connections are to be routed through the bottom of the unit, attach accessory thru-the-bottom service connections to the basepan in accordance with the accessory installation instructions.

ALTERNATE UNIT SUPPORT — When the curb or adapter cannot be used, support unit with sleepers using unit curb or adapter support area. If sleepers cannot be used, support long sides of unit with a minimum of three 4-in. x 4-in. pads, 2 at the corners and one at the unit's center of gravity. If more than 3 pads are used, equally space the pads along the sides of the unit.

SLAB MOUNT (Horizontal Units Only) — Provide a level concrete slab that extends a minimum of 6 in. beyond unit cabinet. Install a 6-in. gravel apron in front of condenser coil air inlet to prevent grass and foliage from obstructing airflow.

NOTE: Horizontal units may be installed on a roof curb if required.

Step 2 — Field Fabricate Ductwork — Secure all ducts to roof curb and building structure on vertical units. *Do not connect ductwork to unit.* For horizontal applications, field-supplied flanges should be attached to horizontal discharge openings and all ductwork secured to the flanges. Insulate and weatherproof all external ductwork, joints, and roof openings with counter flashing and mastic in accordance with applicable codes.

Ducts passing through an unconditioned space must be insulated and covered with a vapor barrier.

If a plenum return is used on a vertical unit, the return should be ducted through the roof deck to comply with applicable fire codes.

A minimum clearance is not required around ductwork. Cabinet return-air static shall not exceed –.35 in. wg with EconoMi$er2 or .45 in. wg without economizer.

These units are designed for a minimum heating operation continuous return-air temperature of 50 F (dry bulb), or an intermittent operation down to 45 F (dry bulb), such as when used with a night set-back thermostat.

To operate at lower return-air temperatures, a field-supplied outdoor-air temperature control must be used to initiate both stages of heat when the temperature is below 45 F. Indoor comfort may be compromised when these lower air temperatures are used with insufficient heating temperature rise.

Step 3 — Install External Trap for Condensate Drain — The unit's $^3/_4$-in. condensate drain connections are located on the bottom and side of the unit. Unit discharge connections do not determine the use of drain connections; either drain connection can be used with vertical or horizontal applications.

When using the standard side drain connection, make sure the plug (Red) in the alternate bottom connection is tight before installing the unit.

To use the bottom drain connection for a roof curb installation, relocate the factory-installed plug (Red) from the bottom connection to the side connection. See Fig. 4. The piping for the condensate drain and external trap can be completed after the unit is in place.

All units must have an external trap for condensate drainage. Install a trap a minimum of 4-in. deep and protect against freeze-up. If drain line is installed downstream from the external trap, pitch the line away from the unit at 1 in. per 10 ft of run. Do not use a pipe size smaller than the unit connection ($^3/_4$ in.).

The center drain plug looks like a star connection, however it can be removed with a $^1/_2$ in. socket drive extension.

Step 4 — Rig and Place Unit — Inspect unit for transportation damage. File any claim with transportation agency. Keep unit upright and do not drop. Spreader bars are not required if top crating is left on unit. Rollers may be used to move unit across a roof. Level by using unit frame as a reference. See Tables 1A and 1B and Fig. 5 for additional information. Operating weight is shown in Tables 1A and 1B and Fig. 5.

Lifting holes are provided in base rails as shown in Fig. 5, 6A and 6B. Refer to rigging instructions on unit.

⚠ CAUTION

All panels must be in place when rigging and lifting.

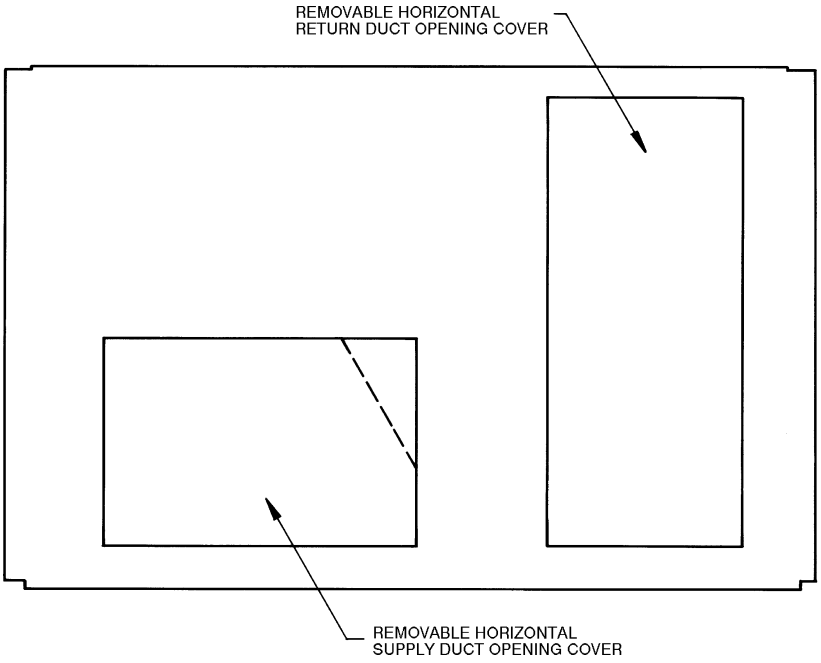


Fig. 1 — Horizontal Conversion Panels

CONNECTOR PKG. ACCY.	B	C	D ALT DRAIN HOLE	GAS	POWER	CONTROL
CRBTMPWR001A00	2′-8 7/16″ [827]	1′-10 15/16″ [583]	1 3/4″ [44.5]	3/4″ [19] NPT	3/4″ [19] NPT	1/2″ [12.7] NPT
CRBTMPWR002A00					1 1/4″ [31.7]	
CRBTMPWR003A00				1/2″ [12.7] NPT	3/4″ [19] NPT	1/2″ [12.7] NPT
CRBTMPWR004A00				3/4″ [19] NPT	1 1/4″ [31.7]	

ROOF CURB ACCESSORY	"A"	UNIT SIZE 48TF,TM
CRRFCURB003A00	1′-2″ [356]	008-014
CRRFCURB004A00	2′-0″ [610]	

NOTES:
1. Roof curb accessory is shipped disassembled.
2. Insulated panels: 1-in. thick polyurethane foam, 1 3/4 lb density.
3. Dimensions in [ ] are in millimeters.
4. Roof curb: 16-gage steel.
5. Attach ductwork to curb (flanges of duct rest on curb).
6. Service clearance 4 ft on each side.
7. Direction of airflow.
8. Connector packages CRBTMPWR001A00 and 2A00 are for thru-the-curb gas type. Packages CRBTMPWR003A00 and 4A00 are for thru-the-bottom type gas connections.

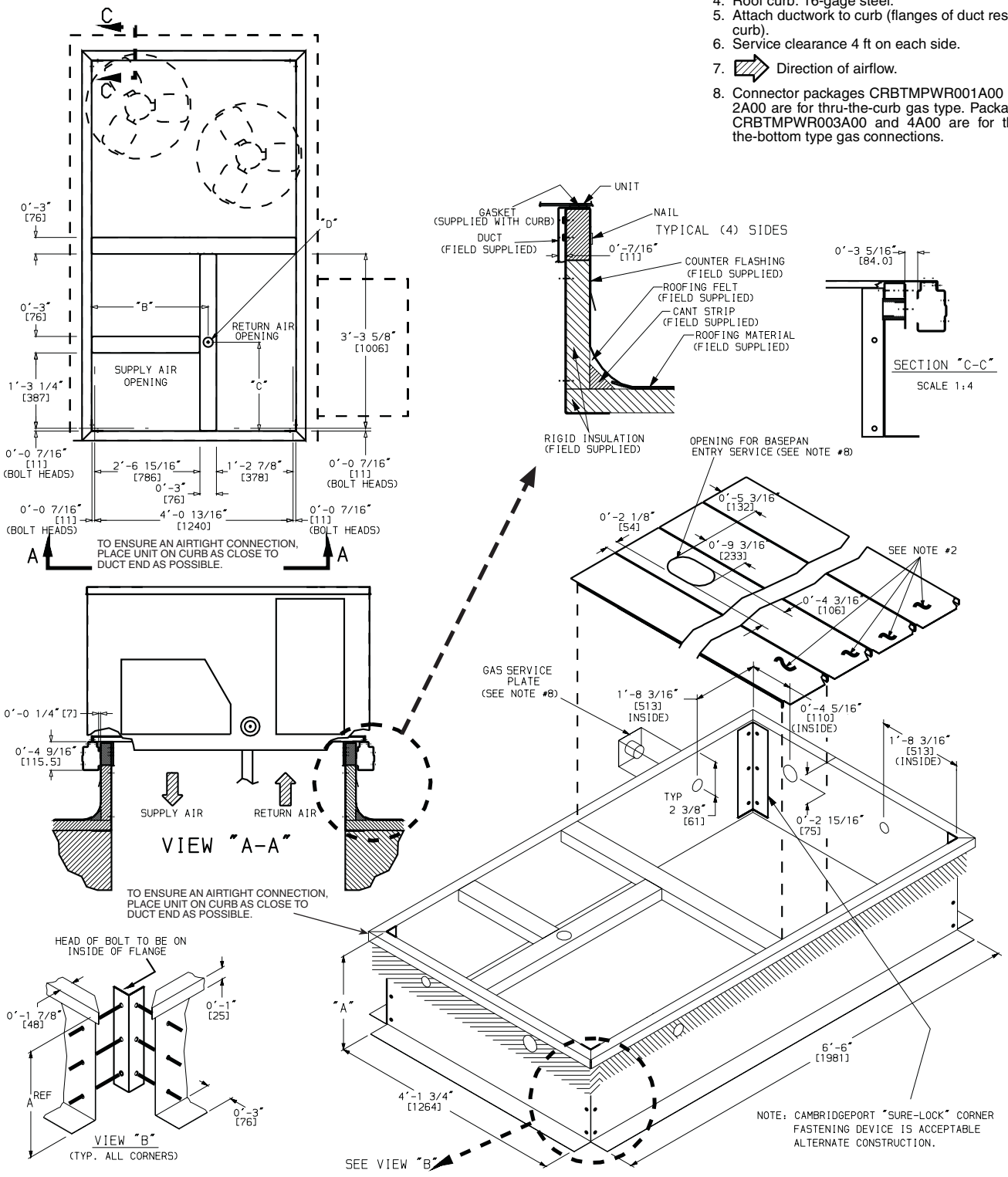
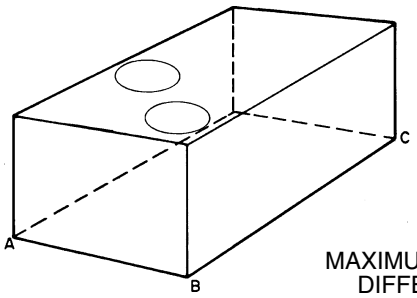


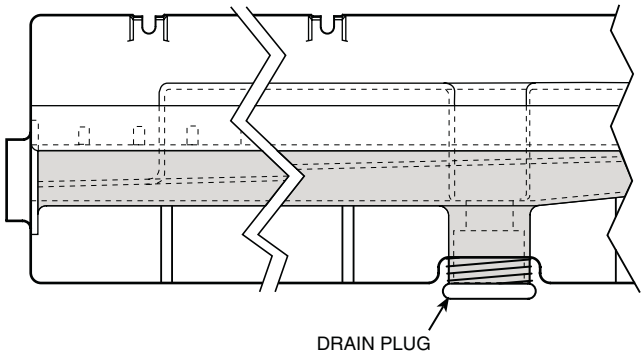
Fig. 2 — Roof Curb Details



MAXIMUM ALLOWABLE DIFFERENCE (in.)

A-B	B-C	A-C
0.5	1.0	1.0

Fig. 3 — Unit Leveling Tolerances



NOTE: Drain plug is shown in factory-installed position.

Fig. 4 — Condensate Drain Pan (Side View)

POSITIONING — Maintain clearance around and above unit to provide minimum distance from combustible materials, proper airflow, and service access. See Fig. 6A and 6B Notes.

Do not install unit in an indoor location. Do not locate unit air inlets near exhaust vents or other sources of contaminated air.

Be sure that unit is installed so that snow will not block the combustion intake or flue outlet.

Unit may be installed directly on wood flooring or on Class A, B, or C roof-covering material when roof curb is used.

Although unit is weatherproof, guard against water from higher level runoff and overhangs.

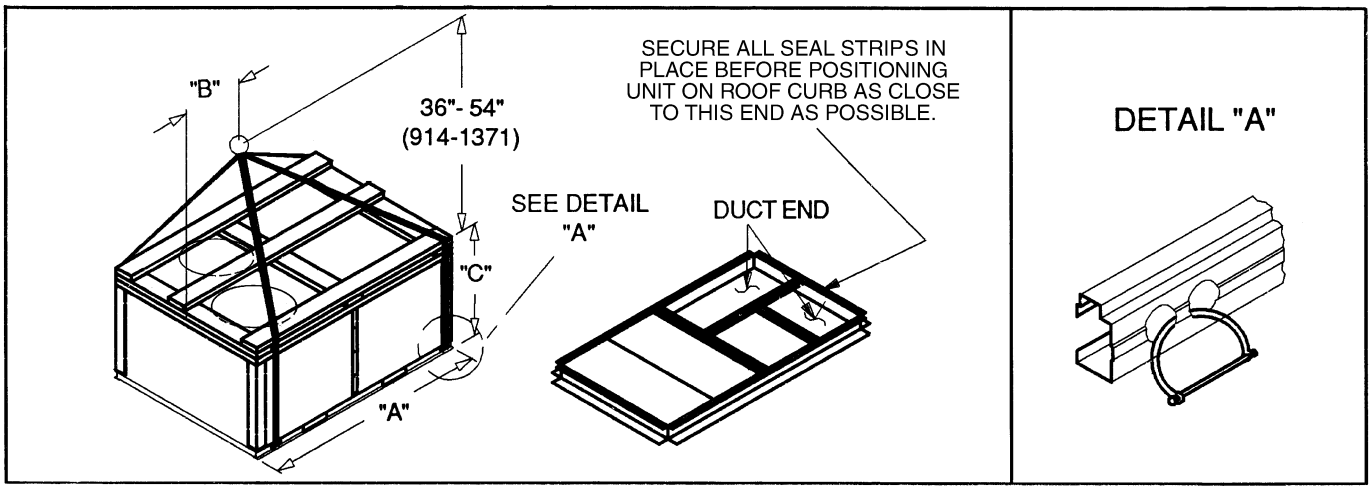
Position unit on roof curb so that the following clearances are maintained: $^1/_4$-in. clearance between roof curb and base rails on each side and duct end of unit; $3^5/_{16}$-in. clearance between roof curb and condenser section end. (See Fig. 2, section C-C.)

Locate mechanical draft system flue assembly at least 48 in. from an adjacent building or combustible material. When unit is located adjacent to public walkways, flue assembly must be at least 7 ft above grade.

Flue vent discharge must have a minimum horizontal clearance of 48 in. from electric and gas meters, gas regulators, and gas relief equipment.

Flue gas can deteriorate building materials. Orient unit so that flue gas will not affect building materials.

Adequate combustion-air space must be provided for proper operation of this equipment. Be sure that installation complies with all local codes and Section 5.3, Air for Combustion and Ventilation, NFGC (National Fuel Gas Code), ANSI (American National Standards Institute) Z223.1-latest year and addendum Z223.1A-latest year. In Canada, installation must be in accordance with the CAN1. B149.1 and CAN1.B149.2 installation codes for gas burning appliances.



NOTES:
1. Dimension in ( ) is in millimeters.
2. Hook rigging shackles through holes in base rail as shown in detail "A." Holes in base rails are centered around the unit center of gravity. Use wooden top skid when rigging to prevent rigging straps from damaging unit.
3. Weights include base unit without economizer. See Tables 1A and 1B for economizer weights.

⚠ CAUTION

All panels must be in place when rigging.

UNIT 48TF, 48TM	OPERATING WEIGHT		DIMENSIONS					
			"A"		"B"		"C"	
	lb	kg	in.	mm	in.	mm	in.	mm
008	870	395	87.38	2219	40.25	1022	41.31	1050
009	880	399	87.38	2219	40.25	1022	41.31	1050
012	1035	469	87.38	2219	40.25	1022	49.31	1253
014	1050	476	87.38	2219	40.25	1022	49.31	1253

Fig. 5 — Rigging Details

Table 1A — Physical Data — 48TF008-014

UNIT SIZE 48TF		D/E/F008	D/E/F009	D/E/F012	D/E014
NOMINAL CAPACITY (tons)		7 1/2	8 1/2	10	12 1/2
OPERATING WEIGHT (lb)					
Unit					
Al/Al*		870	880	1035	1050
Al/Cu*		881	896	1057	1077
Cu/Cu*		893	907	1080	1100
Economizer					
EconoMi$er2		75	75	75	75
Roof Curb†		143	143	143	143
COMPRESSOR		Reciprocating	Reciprocating	Reciprocating	Scroll
Quantity		2	2	2	2
No. Cylinders (per Circuit)		2	2	2	—
Oil (oz)		42 ea	65 ea	54 ea	54 ea
REFRIGERANT TYPE		R-22			
Expansion Device		Fixed Orifice Metering Device			
Operating Charge (lb-oz)					
Circuit 1		4-13	6-14	7- 3	8-10
Circuit 2		4-14	9- 2	7-13	8- 6
CONDENSER COIL		Enhanced Copper Tubes, Aluminum Lanced Fins			
Rows...Fins/in.		1...17	2...17	2...17	2...17
Total Face Area (sq ft)		20.50	18.00	20.47	25.00
CONDENSER FAN		Propeller Type			
Nominal Cfm		6400	6400	7000	7000
Quantity...Diameter (in.)		2...22	2...22	2...22	2...22
Motor Hp...Rpm		1/4...1100	1/4...1100	1/4...1100	1/4...1100
Watts Input (Total)		600	600	600	600
EVAPORATOR COIL		Enhanced Copper Tubes, Aluminum Double-Wavy Fins, Face Split			
Rows...Fins/in.		3...15	3...15	3...15	4...15
Total Face Area (sq ft)		8.0	8.0	10.0	11.1
EVAPORATOR FAN		Centrifugal Type			
Quantity...Size (in.)	**Std**	1...15 x 15	1...15 x 15	1...15 x 15	1...15 x 15
	Alt	1...15 x 15	—	1...15 x 15	1...15 x 15
	High-Static	1...15 x 15	1...15 x 15	1...15 x 15	—
Type Drive	**Std**	Belt	Belt	Belt	Belt
	Alt	Belt	—	Belt	Belt
	High-Static	Belt	Belt	Belt	—
Nominal Cfm		3000	3100	4000	5000
Maximum Continuous Bhp	**Std**	2.40	2.40	2.40	3.70
	Alt	2.40	—	2.90	5.25
	High-Static	3.70	3.70	5.25	—
Motor Frame Size	**Std**	56	56	56	56
	Alt	56	—	56	56
	High-Static	56	56	56	—
	High-Static	1725	1725	1725	1725
Fan Rpm Range	**Std**	590-840	685-935	685-935	860-1080
	Alt	685-935	—	835-1085	900-1260
	High-Static	860-1080	860-1080	830-1130	—
Motor Bearing Type		Ball	Ball	Ball	Ball
Maximum Allowable Rpm		2100	2100	2100	2100
Motor Pulley Pitch Diameter Min/Max (in.)	**Std**	2.4/3.4	2.8/3.8	2.8/3.8	4.0/5.0
	Alt	2.8/3.8	—	3.4/4.4	3.1/4.1
	High-Static	4.0/5.0	4.0/5.0	2.8/3.8	—
Nominal Motor Shaft Diameter (in.)	**Std**	5/8	5/8	5/8	7/8
	Alt	1/2	—	7/8	7/8
	High-Static	7/8	7/8	7/8	—
Fan Pulley Pitch Diameter (in.)	**Std**	7.0	7.0	7.0	8.0
	Alt	7.0	—	7.0	5.9
	High-Static	8.0	8.0	5.8	—
Belt, Quantity...Type...Length (in.)	**Std**	1...A...49	1...A...49	1...A...49	1...A...52
	Alt	1...A...49	—	1...A...49	1...BX...46
	High-Static	1...A...55	1...A...55	1...BX...46	—
Pulley Center Line Distance (in.)	**Std**	16.75-19.25	16.75-19.25	15.85-17.50	15.85-17.50
	Alt	16.75-19.25	—	15.85-17.50	15.85-17.50
	High-Static	16.75-19.25	16.75-19.25	15.85-17.50	—
Speed Change per Full Turn of Movable Pulley Flange (rpm)	**Std**	50	50	50	44
	Alt	50	—	50	50
	High-Static	60	60	60	—
Movable Pulley Maximum Full Turns From Closed Position	**Std**	5	5	5	5
	Alt	5	—	5	6
	High-Static	5	5	6	—
Factory Setting	**Std**	5	5	5	5
	Alt	5	—	5	5
	High-Static	5	5	5	—
Factory Speed Setting (rpm)	**Std**	590	685	685	860
	Alt	685	—	835	960
	High-Static	860	860	887	—
Fan Shaft Diameter at Pulley (in.)		1	1	1	1

LEGEND

Al — Aluminum
Bhp — Brake Horsepower
Cu — Copper

*Evaporator coil fin material/condenser coil fin material. Contact your local representative for details about coated fins.

†Weight of 14-in. roof curb.
**Rollout switch lockout is manually reset by interrupting power to unit or resetting thermostat.

NOTE: High-static motor not available on size 014 units.

Table 1A — Physical Data — 48TF008-014 (cont)

UNIT SIZE 48TF			D/E/F008	D/E/F009	D/E/F012	D/E014
FURNACE SECTION						
Rollout Switch Cutout Temp (F)**			195	195	195	195
Burner Orifice Diameter (in. ...drill size)						
Natural Gas	Std	TFD	.120...31	.120...31	.120...31	.120...31
		TFE	.120...31	.120...31	.120...31	.129...30
		TFF	.120...31	.120...31	.129...30	—
Liquid Propane	Alt	TFD	.096...41	.096...41	.096...41	.096...41
		TFE	.096...41	.096...41	.096...41	.102...38
		TFF	.096...41	.096...41	.102...38	—
Thermostat Heat Anticipator Setting (amps)						
208/230 v and 575 Stage 1			.14	.14	.14	.14
Stage 2			.20	.20	.20	.20
460 v Stage 1			.14	.14	.14	.14
Stage 2			.20	.20	.20	.20
Gas Input (Btuh) Stage 1		TFD	125,000	125,000	120,000	180,000
		TFE	120,000	120,000	180,000	200,000
		TFF	180,000	180,000	200,000	—
Stage 2		TFD	—	—	180,000	220,000
		TFE	180,000	180,000	220,000	250,000
		TFF	220,000	220,000	250,000	—
Efficiency (Steady State) (%)			80	80	80	80
Temperature Rise Range		TFD	20-50	20-50	35-65	35-65
		TFE	35-65	35-65	35-65	40-70
		TFF	45-75	45-75	40-70	—
Manifold Pressure (in. wg)						
Natural Gas	Std		3.5	3.5	3.5	3.5
Liquid Propane	Alt		3.5	3.5	3.5	3.5
Gas Valve Quantity			1	1	1	1
Gas Valve Pressure Range						
Psig			0.180-0.487	0.180-0.487	0.180-0.487	0.180-0.487
in. wg			5.0-13.5	5.0-13.5	5.0-13.5	5.0-13.5
Field Gas Connection Size (in.)			1/2/3/4/3/4	1/2/3/4/3/4	3/4/3/4/3/4	3/4/3/4
HIGH-PRESSURE SWITCH (psig)						
Standard Compressor				450 ± 50		500 ± 50
Internal Relief (Differential)						
Cutout				428		428
Reset (Auto.)				320		320
LOSS-OF-CHARGE (LOW-PRESSURE) SWITCH (psig)						
Cutout				7 ± 3		
Reset (Auto.)				22 ± 7		
FREEZE PROTECTION THERMOSTAT (F)						
Opens				30 ± 5		
Closes				45 ± 5		
OUTDOOR-AIR INLET SCREENS				Cleanable		
Quantity...Size (in.)				1...20 x 24 x 1 1...16 x 25 x 1		
RETURN-AIR FILTERS				Throwaway		
Quantity...Size (in.)			4...16 x 20 x 2	4...16 x 20 x 2	4...20 x 20 x 2	4...20 x 20 x 2

LEGEND

Al — Aluminum
Bhp — Brake Horsepower
Cu — Copper

*Evaporator coil fin material/condenser coil fin material. Contact your local representative for details about coated fins.

†Weight of 14-in. roof curb.
**Rollout switch lockout is manually reset by interrupting power to unit or resetting thermostat.

NOTE: High-static motor not available on size 014 units.

Table 1B — Physical Data — 48TM008-014

UNIT SIZE 48TM		D/E/F008	D/E/F009	D/E/F012	D/E014
NOMINAL CAPACITY (tons)		7 1/2	8 1/2	10	12 1/2
OPERATING WEIGHT (lb)					
Unit					
Al/Al*		870	880	1035	1050
Al/Cu*		891	896	1057	1077
Cu/Cu*		893	907	1080	1100
Economizer					
EconoMi$er2		75	75	75	75
Roof Curb†		143	143	143	143
COMPRESSOR		Reciprocating	Scroll	Scroll	Scroll
Quantity		2	2	2	2
No. Cylinders (per Circuit)		2	2	2	2
Oil (oz) (each compressor)		42	53	50	60
REFRIGERANT TYPE		R-22			
Expansion Device		Acutrol™ Metering Device			
Operating Charge (lb-oz)					
Circuit 1		7-10	7-14	8-10	9-8
Circuit 2		8-2	8-5	8-8	9-5
CONDENSER COIL		Enhanced Copper Tubes, Aluminum Lanced Fins			
Rows...Fins/in.		2...17	2...17	2...17	2...17
Total Face Area (sq ft)		20.50	20.50	25.00	25.00
CONDENSER FAN		Propeller Type			
Nominal Cfm		6500	6500	7000	7000
Quantity...Diameter (in.)		2...22	2...22	2...22	2...22
Motor Hp...Rpm		1/4...1100	1/4...1100	1/4...1100	1/4...1100
Watts Input (Total)		650	650	650	650
EVAPORATOR COIL		Enhanced Copper Tubes, Aluminum Double-Wavy Fins, Face Split			
Rows...Fins/in.		3...15	3...15	3...15	4...15
Total Face Area (sq ft)		8.9	8.9	10.0	11.1
EVAPORATOR FAN		Centrifugal Type			
Quantity...Size (in.)	**Std**	1...15 x 15	1...15 x 15	1...15 x 15	1...15 x 15
	Alt	1...15 x 15	—	1...15 x 15	1...15 x 15
	High-Static	1...15 x 15	1...15 x 15	1...15 x 15	—
Type Drive	**Std**	Belt	Belt	Belt	Belt
	Alt	Belt	—	Belt	Belt
	High-Static	Belt	Belt	Belt	—
Nominal Cfm		2900	3000	3200	5000
Maximum Continuous Bhp	**Std**	2.40	2.40	2.40	3.70
	Alt	2.40	—	2.90	5.25
	High-Static	3.70	3.70	5.25	—
Motor Frame Size	**Std**	56	56	56	56
	Alt	56	—	56	56
	High-Static	56	56	56	—
Fan Rpm Range	**Std**	590-840	685-935	685-935	860-1080
	Alt	685-935	—	835-1085	900-1260
	High-Static	860-1080	860-1080	830-1130	—
Motor Bearing Type		Ball	Ball	Ball	Ball
Maximum Allowable Rpm		2100	2100	2100	2100
Motor Pulley Pitch Diameter Min/Max (in.)	**Std**	2.4/3.4	2.8/3.8	2.8/3.8	4.0/5.0
	Alt	2.8/3.8	—	3.4/4.4	3.1/4.1
	High-Static	4.0/5.0	4.0/5.0	2.8/3.8	—
Nominal Motor Shaft Diameter (in.)	**Std**	5/8	5/8	5/8	7/8
	Alt	5/8	—	7/8	7/8
	High-Static	7/8	7/8	7/8	—
Fan Pulley Pitch Diameter (in.)	**Std**	7.0	7.0	7.0	8.0
	Alt	7.0	—	7.0	5.9
	High-Static	8.0	8.0	5.8	—
Belt, Quantity...Type...Length (in.)	**Std**	1...A...49	1...A...49	1...A...49	1...A...52
	Alt	1...A...49	—	1...A...49	1...BX...46
	High-Static	1...A...55	1...A...55	1...BX...46	—
Pulley Center Line Distance (in.)	**Std**	16.75-19.25	16.75-19.25	15.85-17.50	15.85-17.50
	Alt	16.75-19.25	—	15.85-17.50	15.85-17.50
	High-Static	16.75-19.25	16.75-19.25	15.85-17.50	—
Speed Change per Full Turn of Movable Pulley Flange (rpm)	**Std**	50	50	50	44
	Alt	50	—	50	50
	High-Static	60	60	60	—
Movable Pulley Maximum Full Turns From Closed Position	**Std**	5	5	5	5
	Alt	5	—	5	6
	High-Static	5	5	6	—
Factory Setting	**Std**	5	5	5	5
	Alt	5	—	5	5
	High-Static	5	5	5	—
Factory Speed Setting (rpm)	**Std**	590	685	685	860
	Alt	685	—	835	960
	High-Static	860	860	887	—
Fan Shaft Diameter at Pulley (in.)		1	1	1	1

LEGEND

Al — Aluminum
Bhp — Brake Horsepower
Cu — Copper

*Evaporator coil fin material/condenser coil fin material. Contact your local representative for details about coated fins.

†Weight of 14-in. roof curb.

**Rollout switch lockout is manually reset by interrupting power to unit or resetting thermostat.

NOTE: High-static motor not available on size 014 units.

Table 1B — Physical Data — 48TM008-014 (cont)

UNIT SIZE 48TM			D/E/F008	D/E/F009	D/E/F012	D/E014
FURNACE SECTION						
Rollout Switch Cutout Temp (F)**			195	195	195	195
Burner Orifice Diameter (in. ...drill size)						
Natural Gas	Std	TMD	.120...31	.120...31	.120...31	.120...31
		TME	.120...31	.120...31	.120...31	.129...30
		TMF	.120...31	.120...31	.129...30	—
Liquid Propane	Alt	TMD	.096...41	.096...41	.096...41	.096...41
		TME	.096...41	.096...41	.096...41	.102...38
		TMF	.096...41	.096...41	.102...38	—
Thermostat Heat Anticipator Setting (amps)						
208/230 v and 575 Stage 1			.14	.14	.14	.14
Stage 2			.20	.20	.20	.20
460 v Stage 1			.14	.14	.14	.14
Stage 2			.20	.20	.20	.20
Gas Input (Btuh) Stage 1		TMD	125,000	125,000	120,000	180,000
		TME	120,000	120,000	180,000	200,000
		TMF	180,000	180,000	200,000	—
Stage 2		TMD	—	—	180,000	220,000
		TME	180,000	180,000	220,000	250,000
		TMF	220,000	220,000	250,000	—
Efficiency (Steady State) (%)			80	80	80	80
Temperature Rise Range		TMD	20-50	20-50	35-65	35-65
		TME	35-65	35-65	35-65	40-70
		TMF	45-75	45-75	40-70	—
Manifold Pressure (in. wg)						
Natural Gas	Std		3.5	3.5	3.5	3.5
Liquid Propane	Alt		3.5	3.5	3.5	3.5
Gas Valve Quantity			1	1	1	1
Gas Valve Pressure Range						
Psig			0.180-0.487	0.180-0.487	0.180-0.487	0.180-0.487
in. wg			5.0-13.5	5.0-13.5	5.0-13.5	5.0-13.5
Field Gas Connection Size (in.)		TMD	1/2	1/2	3/4	3/4
		TME	3/4	3/4	3/4	3/4
		TMF	3/4	3/4	3/4	—
HIGH-PRESSURE SWITCH (psig)						
Standard Compressor			450 ± 50			500 ± 50
Internal Relief (Differential)						
Cutout			428			428
Reset (Auto.)			320			320
LOW-PRESSURE SWITCH (psig)			7 ± 3			
Cutout						
Reset (Auto.)			22 ± 7			
FREEZE PROTECTION THERMOSTAT (F)						
Opens			30 ± 5			
Closes			45 ± 5			
OUTDOOR-AIR INLET SCREENS			Cleanable			
Quantity...Size (in.)			1...20 x 24 x 1 1...16 x 25 x 1			
RETURN-AIR FILTERS			Throwaway			
Quantity...Size (in.)			4...16 x 20 x 2	4...16 x 20 x 2	4...20 x 20 x 2	4...20 x 20 x 2

LEGEND

Al — Aluminum
Bhp — Brake Horsepower
Cu — Copper

*Evaporator coil fin material/condenser coil fin material. Contact your local representative for details about coated fins.

†Weight of 14-in. roof curb.
**Rollout switch lockout is manually reset by interrupting power to unit or resetting thermostat.

NOTE: High-static motor not available on size 014 units.

UNIT	STD. UNIT WEIGHT		DURABLADE ECON WEIGHT		ECONOMI$ER2 WEIGHT		ECONOMI$ER2 W/ P.E. WEIGHT		CORNER WEIGHT (A)		CORNER WEIGHT (B)		CORNER WEIGHT (C)		CORNER WEIGHT (D)		"H"		"J"		"K"		"L"	
	LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	FT.-IN.	MM	FT.-IN.	MM	FT.-IN.	MM	FT.-IN.	MM
48TFF/TFE/TFD008	870	395	44	20	75	34.1	145	65.9	189	86	161	73	239	109	280	127	1′-2 7/8″	378	3′-5 5/16″	1050	2′-9 11/16″	856	2′-2 7/16″	672
48TFF/TFE/TFD009	880	399							191	87	163	74	242	110	284	129	3′-3 7/8″	1013	3′-5 5/16″	1050	2′-9 11/16″	856	2′-2 7/16″	672
48TFF/TFE/TFD012	1035	469							225	102	192	87	285	129	333	151	2′-5 7/8″	759	4′-1 5/16″	1253	3′-0 3/8″	924	2′-10 7/16″	875
48TFE/TFD014	1050	476							228	103	195	88	289	131	338	153	1′-2 7/8″	378	4′-1 5/16″	1253	3′-0 3/8″	924	2′-10 7/16″	875

NOTES:
1. Dimensions in [ ] are in millimeters.
2. Center of gravity.
3. Direction of airflow.
4. On vertical discharge units, ductwork to be attached to accessory roof curb only. For horizontal discharge units, field-supplied flanges should be attached to horizontal discharge openings, and all ductwork should be attached to the flanges.
5. Minimum clearance (local codes or jurisdiction may prevail):
   a. Between unit, flue side and combustible surfaces, 48 in., 18 in. when using accessory flue discharge deflector.
   b. Bottom of unit to combustible surfaces (when not using curb), 1 inch. Bottom of base rail to combustible surfaces (when not using curb) 0 inches.
   c. Condenser coil, for proper airflow, 36 in. one side, 12 in. the other. The side getting the greater clearance is optional.
   d. Overhead, 60 in. to assure proper condenser fan operation.
   e. Between units, control box side, 42 in. per NEC (National Electrical Code).
   f. Between unit and ungrounded surfaces, control box side, 36 in. per NEC.
   g. Between unit and block or concrete walls and other grounded surfaces, control box side, 42 in. per NEC.
   h. Horizontal supply and return end, 0 in. when the alternate condensate drain is used.
6. With the exception of the clearance for the condenser coil and combustion side as stated in Notes 5a, b and c, a removable fence or barricade requires no clearance.
7. Units may be installed on combustible floors made from wood or Class A, B, or C roof covering material if set on base rail.
8. The vertical center of gravity is 1′-7″ [483] for 008 and 009, 1′-11″ [485] for 012 and 014 up from the bottom of the base rail.

	CONNECTION SIZES
A	1 3/8″ DIA. [35] FIELD POWER SUPPLY HOLE
B	2 1/2″DIA [64] POWER SUPPLY KNOCK-OUT
C	1 3/4″DIA [44] CHARGING PORT HOLE
D	7/8″ DIA. [22] FIELD CONTROL WIRING HOLE
E	3/4″-14 NPT CONDENSATE DRAIN
F	1/2″-14 NPT GAS CONN.48TFD008 & 009 3/4″-14 NPT GAS CONNECTION 48TFE,F008 & 009 48TFD,E012 & 014, 48TFF012
G	2″ DIA [51] POWER SUPPLY KNOCK-OUT

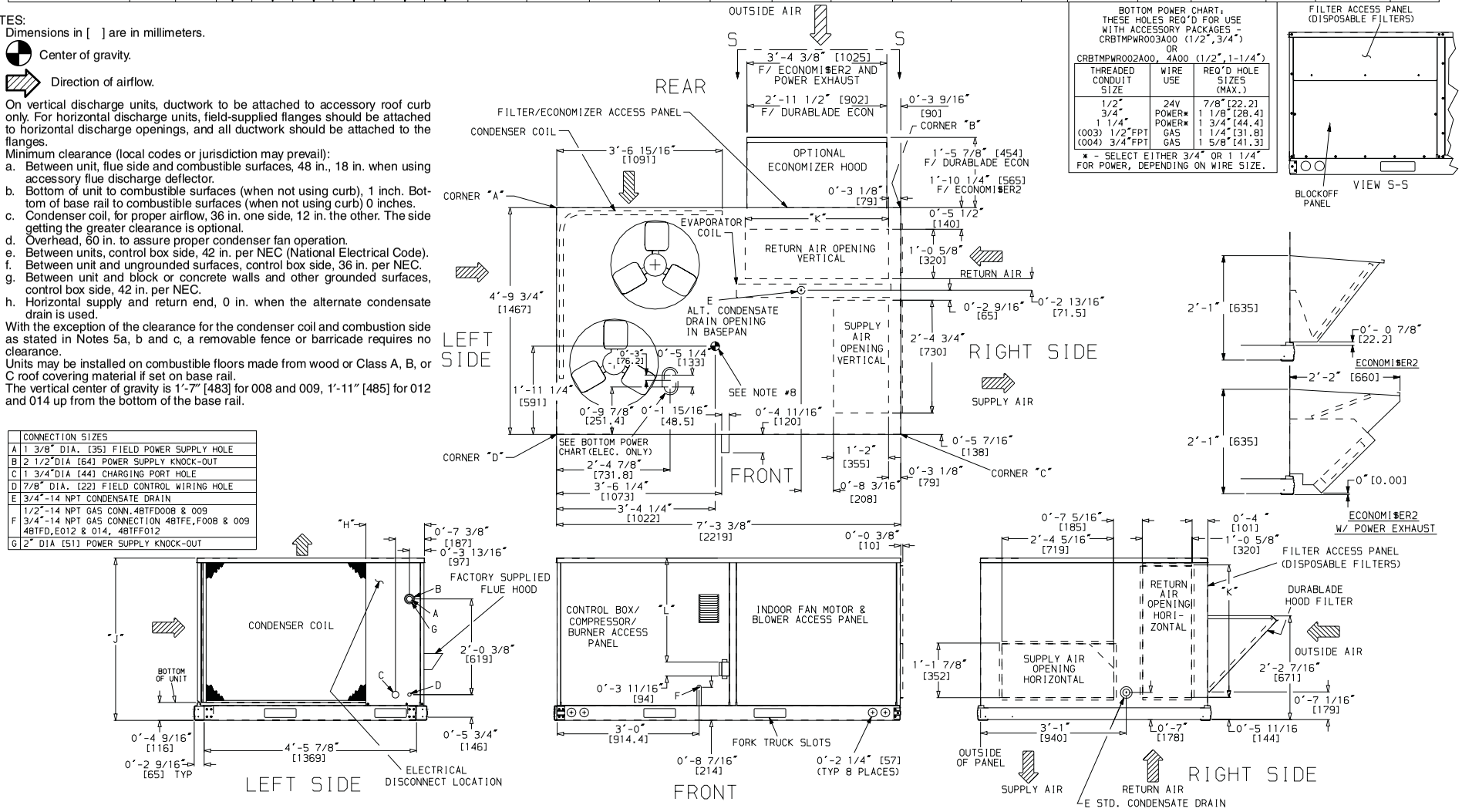


Fig. 6A — Base Unit Dimensions — 48TF008-014

UNIT	STD. UNIT WEIGHT		DURABLADE ECON WEIGHT		ECONOMI$ER2 WEIGHT		ECONOMI$ER2 W/ P.E. WEIGHT		CORNER WEIGHT (A)		CORNER WEIGHT (B)		CORNER WEIGHT (C)		CORNER WEIGHT (D)		"H"		"J"		"K"		"L"	
	LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	FT.-IN.	MM	FT.-IN.	MM	FT.-IN.	MM	FT.-IN.	MM
48TM*008	870	395	44	20	75	34.1	145	65.9	189	86	161	73	239	109	280	127	2'-0 7/8"	632	3'-5 5/16"	1050	2'-9 11/16"	856	2'-2 7/16"	672
48TM*009	880	399	↓	↓	↓	↓	↓	↓	191	87	163	74	242	110	284	129	1' -2 7/8"	378	3'-5 5/16"	1050	2'-9 11/16"	856	2'-2 7/16"	672
48TM*012	1035	469	↓	↓	↓	↓	↓	↓	225	102	192	87	285	129	333	151	1' -2 7/8"	378	4'-1 5/16"	1253	3'-0 3/8"	924	2'-10 7/16"	875
48TM*014	1050	476	↓	↓	↓	↓	↓	↓	228	103	195	88	289	131	338	153	1'-2 7/8"	378	4'-1 5/16"	1253	3'-0 3/8"	924	2'-10 7/16"	875

NOTES:

1. Dimensions in [ ] are in millimeters.
2. Center of gravity.
3. Direction of airflow.
4. On vertical discharge units, ductwork to be attached to accessory roof curb only. For horizontal discharge units, field-supplied flanges should be attached to horizontal discharge openings, and all ductwork should be attached to the flanges.
5. Minimum clearance (local codes or jurisdiction may prevail):
   a. Between unit, flue side and combustible surfaces, 48 in., 18 in. when using accessory flue discharge deflector.
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   c. Condenser coil, for proper airflow, 36 in. one side, 12 in. the other. The side getting the greater clearance is optional.
   d. Overhead, 60 in. to assure proper condenser fan operation.
   e. Between units, control box side, 42 in. per NEC (National Electrical Code).
   f. Between unit and ungrounded surfaces, control box side, 36 in. per NEC.
   g. Between unit and block or concrete walls and other grounded surfaces, control box side, 42 in. per NEC.
   h. Horizontal supply and return end, 0 in. when the alternate condensate drain is used.
6. With the exception of the clearance for the condenser coil and combustion side as stated in Notes 5a, b and c, a removable fence or barricade requires no clearance.
7. Units may be installed on combustible floors made from wood or Class A, B, or C roof covering material if set on base rail.
8. The vertical center of gravity is 1'-7" [483] for 008 and 009, 1'-11" [485] for 012 and 014 up from the bottom of the base rail.

	CONNECTION SIZES
A	1 3/8" DIA. [35] FIELD POWER SUPPLY HOLE
B	2 1/2"DIA [64] POWER SUPPLY KNOCK-OUT
C	1 3/4"DIA [44] CHARGING PORT HOLE
D	7/8" DIA. [22] FIELD CONTROL WIRING HOLE
E	3/4"-14 NPT CONDENSATE DRAIN
F	1/2"-14 NPT GAS CONN; 48HJD008,009; 48TMD008 3/4"-14 NPT GAS CONN.; 48HJE/HJF008,009; 48HJD/HJE/TMD/TME012,014; 48HJF/TMF012; 48TME/TMF008
G	2" DIA [51] POWER SUPPLY KNOCK-OUT

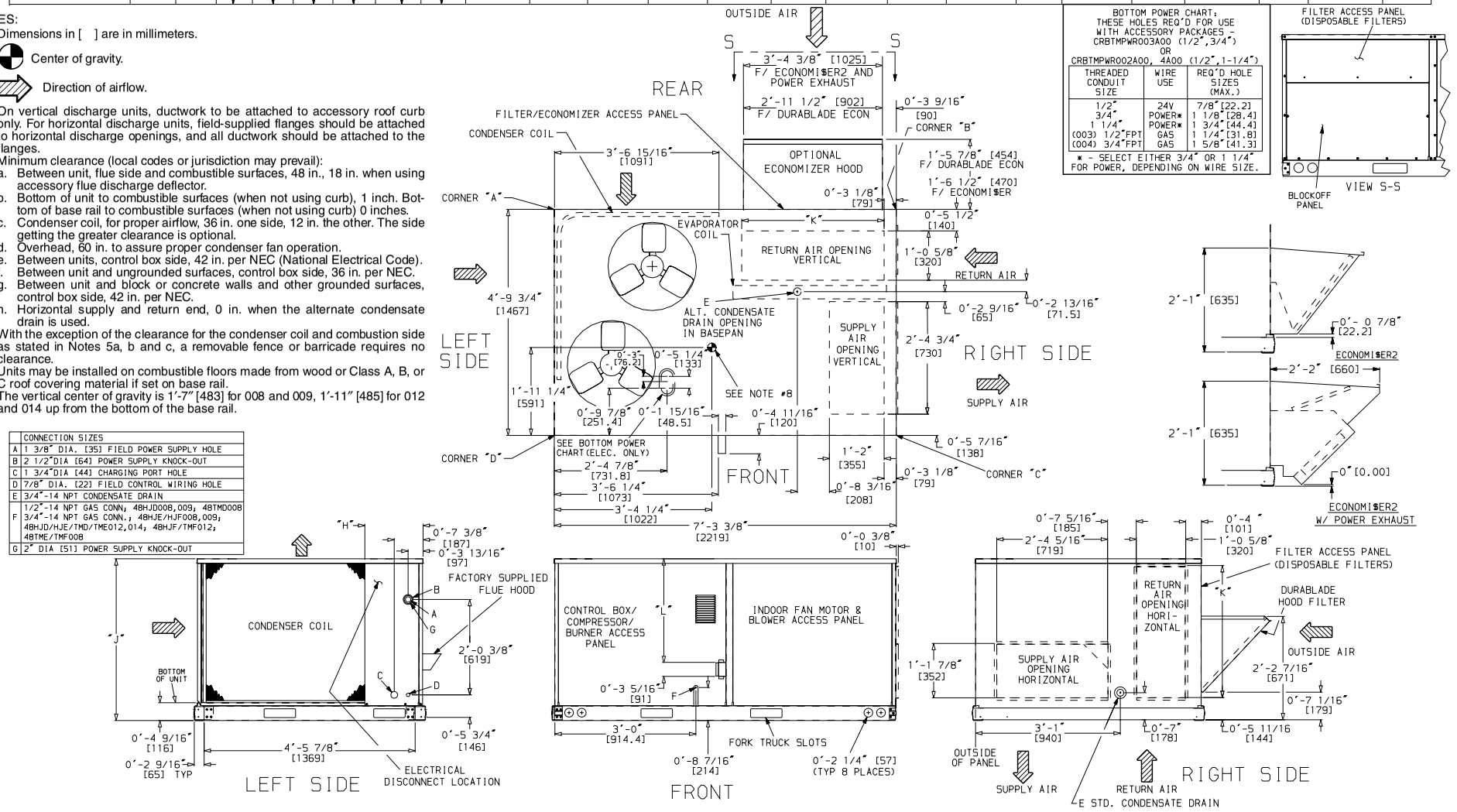


Fig. 6B — Base Unit Dimensions — 48TM008-014

Step 5 — Install Flue Hood — Flue hood is shipped screwed to the burner compartment access panel. Remove from shipping location and, using screws provided, install flue hood and screen in location shown in Fig. 7.

Step 6 — Install Gas Piping — Unit is equipped for use with type of gas shown on nameplate. Refer to local building codes, or in the absence of local codes, to ANSI Z223.1-latest year and addendum Z223.1A-latest year entitled National Fuel Gas Code. In Canada, installation must be in accordance with the CAN1.B149.1 and CAN1.B149.2 installation codes for gas burning appliances when installing gas piping.

For natural gas applications, gas pressure at unit gas connection must not be less than 4.0 in. wg (5.0 in. wg in high heat units) or greater than 13.0 in. wg while unit is operating. For liquid propane applications, the gas pressure must not be less than 5.0 in. wg or greater than 13.0 in. wg at the unit connection.

Size gas supply piping for 0.5 in. wg maximum pressure drop. Do not use supply pipe smaller than unit gas connection.

NOTE: When installing gas piping to gas valve inlet, use properly sized back-up wrench on gas valve inlet flange flats.

Support gas piping as shown in the table in Fig. 8. For example, a 3/4-in. gas pipe must have one field-fabricated support beam every 8 ft.

See Fig. 8 for typical pipe guide and locations of external manual gas shutoff valve.

NOTE: If field-installed thru-the-bottom connections are used, refer to the accessory installation instructions for power wiring and gas connections. Refer to Fig. 6A and 6B for drilling holes in basepan.

Step 7 — Make Electrical Connections

⚠ WARNING

Unit cabinet must have an uninterrupted, unbroken electrical ground to minimize the possibility of personal injury if an electrical fault should occur. This ground may consist of electrical wire connected to unit ground lug in control compartment, or conduit approved for electrical ground when installed in accordance with NEC (National Electrical Code), ANSI/NFPA (National Fire Protection Association), latest edition, and local electrical codes. *Do not use gas piping as an electrical ground.* Failure to follow this warning could result in the installer being liable for personal injury of others.

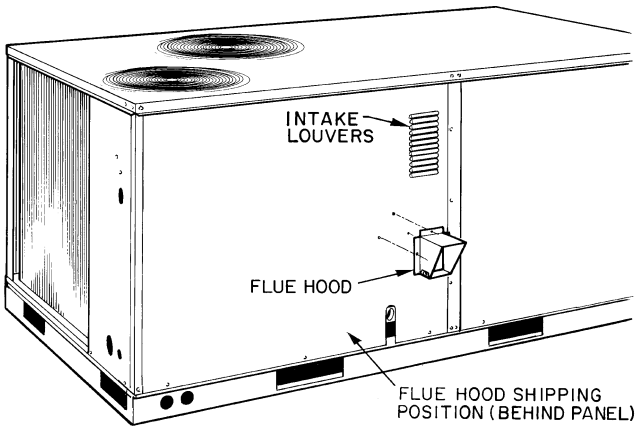
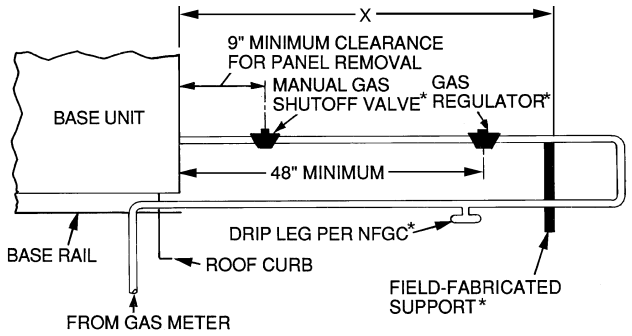


Fig. 7 — Flue Hood Details



LEGEND

NFGC — National Fuel Gas Code

*Field supplied.

NOTE: Follow all local codes.

SPACING OF SUPPORTS

STEEL PIPE NOMINAL DIAMETER (in.)	X DIMENSION (feet)
1/2	6
3/4 or 1	8
1 1/4 or larger	10

Fig. 8 — Gas Piping Guide (With Accessory Thru-the-Curb Service Connections)

FIELD POWER SUPPLY — All units except 208/230-v units are factory wired for the voltage shown on the nameplate. If the 208/230-v unit is to be connected to a 208-v power supply, the transformer *must* be rewired by moving the black wire from the 230-v 1/4-in. spade terminal on the transformer and connecting it to the 208-v 1/4-in. spade terminal from the transformer.

Refer to unit label diagram for additional information. Pigtails are provided for field service.

When installing units, provide a disconnect per NEC. Use copper conductors only when splice connectors are used.

All field wiring must comply with NEC and local requirements. In Canada, electrical connections must be in accordance with CSA (Canadian Standards Association) C22.1 Canadian Electrical Code Part One.

Install conduit through side panel openings indicated in Fig. 6A and 6B. Route power lines through connector to terminal connections as shown in Fig. 9.

On 3-phase units, voltages between phases must be balanced within 2% and the current within 10%. Use the formula shown in Tables 2A-2D, Note 2 to determine the percentage of voltage imbalance. Operation on improper line voltage or excessive phase imbalance constitutes abuse and may cause damage to electrical components. Such operation would invalidate any applicable Carrier warranty.

NOTE: If field-installed thru-the-bottom connections are used, refer to the accessory installation instructions for power wiring and gas connections. Refer to Fig. 6A and 6B for drilling holes in basepan.

FIELD CONTROL WIRING — Install a Carrier-approved accessory thermostat assembly according to installation instructions included with the accessory. Locate thermostat assembly on a solid wall in the conditioned space to sense average temperature in accordance with thermostat installation instructions.

NOTE: For wire runs up to 50 ft, use no. 18 AWG (American Wire Gage) insulated wire (35 C minimum). For 50 to 75 ft, use no. 16 AWG insulated wire (35 C minimum). For over 75 ft, use no. 14 AWG insulated wire (35 C minimum). All wire larger than no. 18 AWG cannot be directly connected to the thermostat and will require a junction box and splice at the thermostat.

Route thermostat cable or equivalent single leads of colored wire from thermostat subbase terminals to low-voltage connections on unit (shown in Fig. 10A and 10B) as described in Steps 1-4 below.

1. If unit is mounted on roof curb and accessory thru-the-curb service plate connection is used, route wire through connection plate.
2. Pass control wires through the hole provided on unit (see connection D in Connection Sizes table in Fig. 6A and 6B).
3. Feed wires through the raceway built into the corner post to the 24-v barrier located on the left side of the control box. See Fig. 11. The raceway provides the UL-required (Underwriters' Laboratories) clearance between high- and low-voltage wiring.
4. Connect thermostat wires to screw terminals on low-voltage connection board.

HEAT ANTICIPATOR SETTINGS — Set heat anticipator settings at .14 amp for the first stage and .20 amp for second-stage heating.

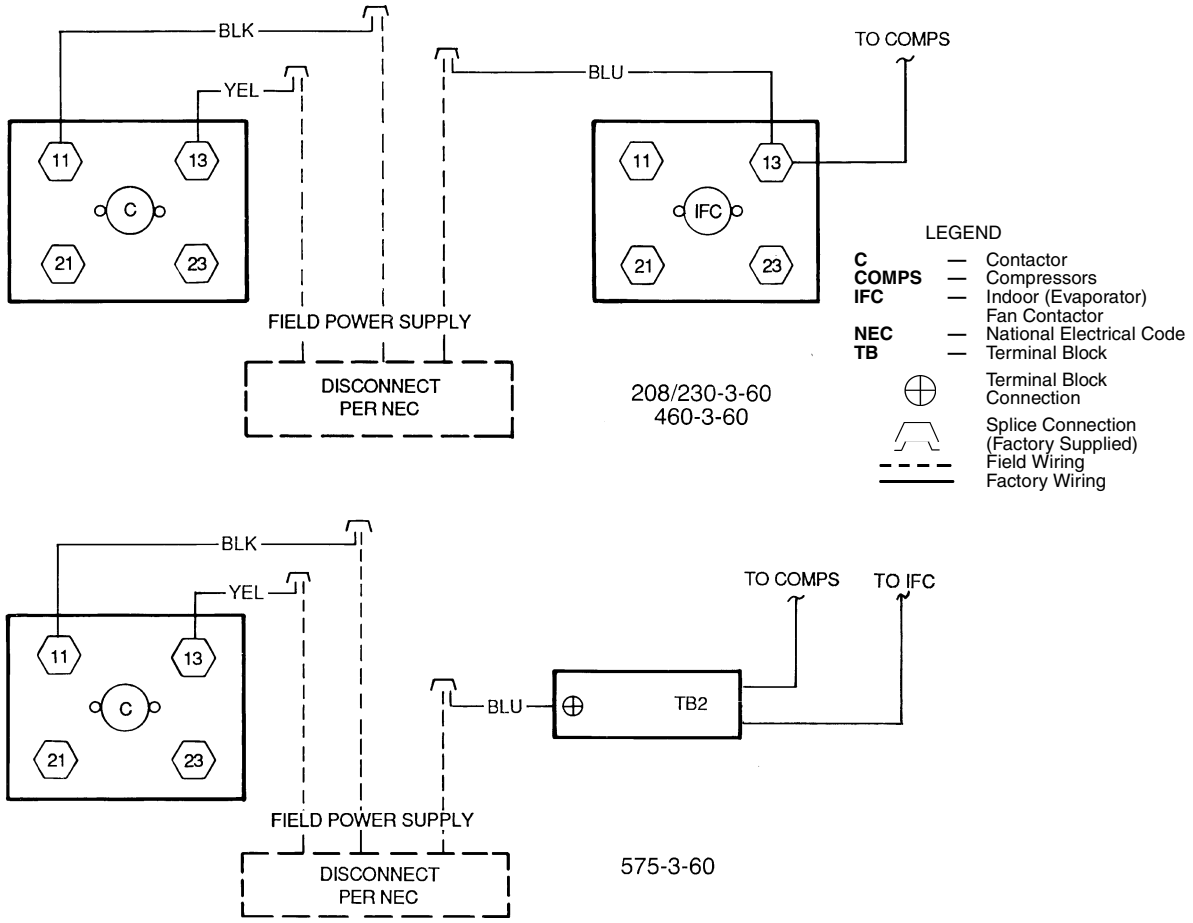


Fig. 9 — Power Wiring Connections

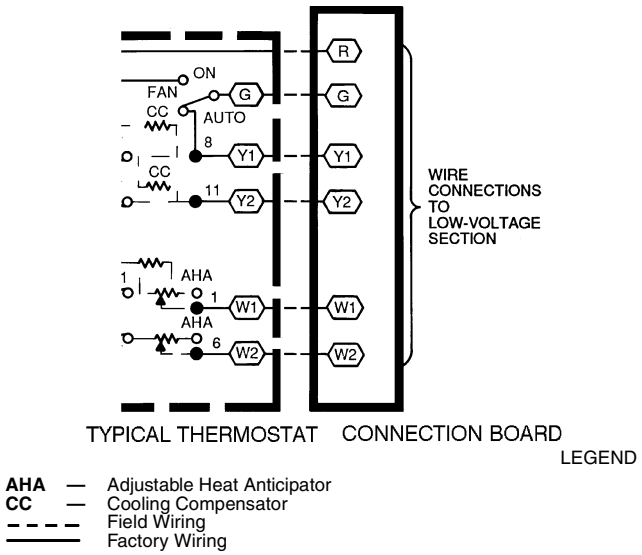


Fig. 10A — Standard Low-Voltage Connections

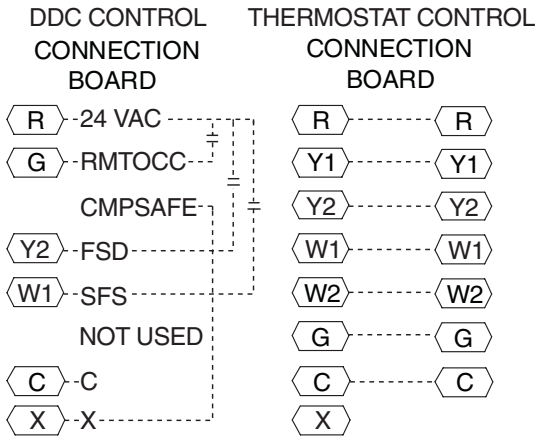


Fig. 10B — Low Voltage Connections (Units with PremierLink™ Controls)

Table 2A — Electrical Data — 48TF Units (Without Convenience Outlet)

UNIT 48TF	NOMINAL VOLTAGE	IFM TYPE	VOLTAGE RANGE		COMPR (ea)		OFM (ea)			IFM	COMBUSTION FAN MOTOR FLA	POWER SUPPLY*		DISCONNECT SIZE†	
			Min	Max	RLA	LRA	Qty	Hp	FLA	FLA		MCA	MOCP**	FLA	LRA
008 (7 1/2 Tons)	208/230-3-60	Std	187	254	14.0	91.0	2	1/4	1.4	5.8	.6	40.1/40.1	45/45	42/42	229/229
		Alt								5.8		40.1/40.1	45/45	42/42	229/229
		High								10.6		44.9/44.9	50/50	48/48	273/273
	460-3-60	Std	414	508	6.4	42.0	2	1/4	0.7	2.6	.3	18.4	20	19	108
		Alt								2.6		18.4	20	19	108
		High								4.8		20.6	25	22	130
	575-3-60	Std	518	632	5.2	39.0	2	1/4	0.7	2.6	.3	14.9	20	16	97
		Alt								2.6		14.9	20	16	97
		High								4.8		16.7	20	18	114
009 (8 1/2 Tons)	208/230-3-60	Std	187	254	16.0	137.0	2	1/4	1.4	5.8	.6	44.6/44.6	50/50	47/47	321/321
		High								10.6		49.4/49.4	60/60	52/52	365/365
	460-3-60	Std	414	508	8.3	69.0	2	1/4	0.7	2.6	.3	22.7	25	24	162
		High								4.8		24.9	30	26	184
	575-3-60	Std	518	632	6.4	58.0	2	1/4	0.7	2.6	.3	17.6	20	18	135
		High								4.8		19.4	25	20	152
012 (10 Tons)	208/230-3-60	Std	187	254	15.8	130.0	2	1/4	1.4	5.8	.6	44.2/44.2	50/50	46/46	307/307
		Alt								7.5		45.9/45.9	50/50	48/48	326/326
		High								15.0		53.4/53.4	60/60	57/57	374/374
	460-3-60	Std	414	508	7.9	64.0	2	1/4	0.7	2.6	.3	21.8	25	23	152
		Alt								3.4		22.6	25	24	191
		High								7.4		26.6	30	28	185
	575-3-60	Std	518	632	6.6	52.0	2	1/4	0.7	2.6	.3	18.1	25	19	123
		Alt								3.4		18.7	25	20	155
		High								7.4		21.9	30	23	150
014 (12 1/2 Tons)	208/230-3-60	Std	187	254	23.0	146.0	2	1/4	1.4	10.6	.6	65.2/65.2	80/80††	68/68	383/383
		Alt								15.0		69.6/69.6	80/80††	73/73	406/406
	460-3-60	Std	414	508	10.4	73.0	2	1/4	0.7	4.8	.3	29.6	40	31	192
		Alt								7.4		32.2	45	34	203
	575-3-60	Std	518	632	8.3	58.4	2	1/4	0.7	4.8	.3	23.6	30	25	154
		Alt								7.4		25.7	30	27	162

LEGEND

FLA — Full Load Amps
HACR — Heating, Air Conditioning and Refrigeration
IFM — Indoor (Evaporator) Fan Motor
LRA — Locked Rotor Amps
MCA — Minimum Circuit Amps
MOCP — Maximum Overcurrent Protection
NEC — National Electrical Code
OFM — Outdoor (Condenser) Fan Motor
RLA — Rated Load Amps

*The values listed in this table do not include power exhaust. See table below for power exhaust requirements.
†Used to determine minimum disconnect per NEC.
**Fuse or HACR circuit breaker.
††Fuse only.
***Data shown in table is for Compressor no. 1.
208/230-3-60: Compressor no. 2 RLA is 14.1 amps and LRA is 105 amps.
460-3-60: Compressor no. 2 RLA is 7.1 amps and LRA is 55 amps.
575-3-60: Compressor no. 2 RLA is 6.4 amps and LRA is 40 amps.

POWER EXHAUST ELECTRICAL DATA

POWER EXHAUST PART NO.	MCA (230 v)	MCA (460 v)	MCA (575 v)	MOCP (for separate power source)
CRPWREXH021A00	N/A	0.9	N/A	15
CRPWREXH022A00	3.3	N/A	1.32	15
CRPWREXH023A00	N/A	1.8	N/A	15
CRPWREXH024A00	1.6	N/A	0.64	15
CRPWREXH025A00	N/A	0.9	N/A	15
CRPWREXH026A00	3.3	N/A	1.32	15
CRPWREXH027A00	N/A	1.8	N/A	15
CRPWREXH028A00	1.7	N/A	0.68	15
CRPWREXH029A00	N/A	1.0	N/A	15
CRPWREXH030A00	1.6	N/A	0.64	15

N/A — Not available

NOTE: If a single power source is to be used, size wire to include power exhaust MCA and MOCP.

Check MCA and MOCP when power exhaust is powered through the unit. Determine the new MCA including the power exhaust using the following formula:

MCA New = MCA unit only + MCA of Power Exhaust

For example, using a 48TF008 unit with MCA = 40.1 and MOCP = 45, with CRPWREXH030A00 power exhaust.

MCA New = 40.1 amps + 1.6 amps = 41.7 amps

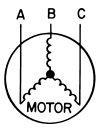
If the new MCA does not exceed the published MOCP, then MOCP would not change. The MOCP in this example is 45 amps and the MCA New is below 45; therefore the MOCP is acceptable. If "MCA New" is larger than the published MOCP, raise the MOCP to the next larger size. For separate power, the MOCP for the power exhaust will be 15 amps per NEC.

NOTES:
1. In compliance with NEC requirements for multimotor and combination load equipment (refer to NEC Articles 430 and 440), the overcurrent protective device for the unit shall be fuse or HACR breaker. Canadian units may be fuse or circuit breaker.
2. **Unbalanced 3-Phase Supply Voltage**
Never operate a motor where a phase imbalance in supply voltage is greater than 2%. Use the following formula to determine the percent of voltage imbalance.

% Voltage Imbalance

$$= 100 \times \frac{\text{max voltage deviation from average voltage}}{\text{average voltage}}$$

Example: Supply voltage is 460-3-60.



AB = 452 v
BC = 464 v
AC = 455 v

$$\text{Average Voltage} = \frac{452 + 464 + 455}{3}$$

$$= \frac{1371}{3}$$

$$= 457$$

Determine maximum deviation from average voltage.
(AB) 457 – 452 = 5 v
(BC) 464 – 457 = 7 v
(AC) 457 – 455 = 2 v

Maximum deviation is 7 v.

Determine percent of voltage imbalance.

$$\text{\% Voltage Imbalance} = 100 \times \frac{7}{457}$$

$$= 1.53\%$$

This amount of phase imbalance is satisfactory as it is below the maximum allowable 2%.

IMPORTANT: If the supply voltage phase imbalance is more than 2%, contact your local electric utility company immediately.

Table 2B — Electrical Data — 48TF Units (With Convenience Outlet)

UNIT 48TF	NOMINAL VOLTAGE	IFM TYPE	VOLTAGE RANGE		COMPR (ea)		OFM (ea)			IFM	COMBUSTION FAN MOTOR FLA	POWER SUPPLY*		DISCONNECT SIZE†	
			Min	Max	RLA	LRA	Qty	Hp	FLA	FLA		MCA	MOCP**	FLA	LRA
008 (7 1/2 Tons)	208/230-3-60	Std	187	254	14.0	91.0	2	1/4	1.4	5.8	.6	46.1/46.1	50/50	48/48	233/233
		Alt								5.8		46.1/46.1	50/50	48/48	233/233
		High								10.6		50.9/50.9	60/60	53/53	273/273
	460-3-60	Std	414	508	6.4	42.0	2	1/4	0.7	2.6	.3	24.1	25	22	110
		Alt								2.6		24.1	25	22	110
		High								4.8		23.3	30	24	132
	575-3-60	Std	518	632	5.2	39.0	2	1/4	0.7	2.6	.3	20.9	25	18	99
		Alt								2.6		20.9	25	18	99
		High								4.8		19.4	25	20	117
009 (8 1/2 Tons)	208/230-3-60	Std	187	254	16.0	137.0	2	1/4	1.4	5.8	.6	50.6/50.6	60/60	52/52	325/325
		High								10.6		55.4/55.4	60/60	58/58	369/369
	460-3-60	Std	414	508	8.3	69.0	2	1/4	0.7	2.6	.3	25.4	30	26	164
		High								4.8		27.6	35	29	138
	575-3-60	Std	518	632	6.4	58.0	2	1/4	0.7	2.6	.3	20.3	25	20	137
		High								4.8		22.1	30	22	155
012 (10 Tons)	208/230-3-60	Std	187	254	15.8	130.0	2	1/4	1.4	5.8	.6	50.2/50.2	60/60	52/52	311/311
		Alt								7.5		51.9/51.9	60/60	54/54	330/330
		High								15.0		59.4/59.4	70/70††	62/62	378/378
	460-3-60	Std	414	508	7.9	64.0	2	1/4	0.7	2.6	.3	27.8	30	25	154
		Alt								3.4		25.3	30	26	193
		High								7.4		29.3	35	31	187
	575-3-60	Std	518	632	6.6	52.0	2	1/4	0.7	2.6	.3	24.1	25	21	109
		Alt								3.4		24.7	25	22	141
		High								7.4		24.6	30	25	136
014 (12 1/2 Tons)	208/230-3-60	Std	187	254	23.0	146.0	2	1/4	1.4	10.6	.6	71.2/71.2	80/80††	74/74	387/387
		Alt								15.0		75.6/75.6	80/80††	79/79	410/410
	460-3-60	Std	414	508	10.4	73.0	2	1/4	0.7	4.8	.3	35.6	40	34	194
		Alt								7.4		34.9	40	37	205
	575-3-60	Std	518	632	8.3	58.4	2	1/4	0.7	4.8	.3	29.6	35	27	156
		Alt								7.4		31.7	35	29	165

LEGEND

FLA — Full Load Amps
HACR — Heating, Air Conditioning and Refrigeration
IFM — Indoor (Evaporator) Fan Motor
LRA — Locked Rotor Amps
MCA — Minimum Circuit Amps
MOCP — Maximum Overcurrent Protection
NEC — National Electrical Code
OFM — Outdoor (Condenser) Fan Motor
RLA — Rated Load Amps

*The values listed in this table do not include power exhaust. See table below for power exhaust requirements.
†Used to determine minimum disconnect per NEC.
**Fuse or HACR circuit breaker.
††Fuse only.
***Data shown in table is for Compressor no. 1.
208/230-3-60: Compressor no. 2 RLA is 14.1 amps and LRA is 105 amps.
460-3-60: Compressor no. 2 RLA is 7.1 amps and LRA is 55 amps.
575-3-60: Compressor no. 2 RLA is 6.4 amps and LRA is 40 amps.

POWER EXHAUST ELECTRICAL DATA

POWER EXHAUST PART NO.	MCA (230 v)	MCA (460 v)	MCA (575 v)	MOCP (for separate power source)
CRPWREXH021A00	N/A	0.9	N/A	15
CRPWREXH022A00	3.3	N/A	1.32	15
CRPWREXH023A00	N/A	1.8	N/A	15
CRPWREXH024A00	1.6	N/A	0.64	15
CRPWREXH025A00	N/A	0.9	N/A	15
CRPWREXH026A00	3.3	N/A	1.32	15
CRPWREXH027A00	N/A	1.8	N/A	15
CRPWREXH028A00	1.7	N/A	0.68	15
CRPWREXH029A00	N/A	1.0	N/A	15
CRPWREXH030A00	1.6	N/A	0.64	15

N/A — Not available

NOTE: If a single power source is to be used, size wire to include power exhaust MCA and MOCP.

Check MCA and MOCP when power exhaust is powered through the unit. Determine the new MCA including the power exhaust using the following formula:

MCA New = MCA unit only + MCA of Power Exhaust

For example, using a 48TF008 unit with MCA = 40.1 and MOCP = 45, with CRPWREXH030A00 power exhaust.

MCA New = 40.1 amps + 1.6 amps = 41.7 amps

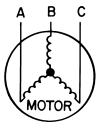
If the new MCA does not exceed the published MOCP, then MOCP would not change. The MOCP in this example is 45 amps and the MCA New is below 45; therefore the MOCP is acceptable. If "MCA New" is larger than the published MOCP, raise the MOCP to the next larger size. For separate power, the MOCP for the power exhaust will be 15 amps per NEC.

NOTES:
1. In compliance with NEC requirements for multimotor and combination load equipment (refer to NEC Articles 430 and 440), the overcurrent protective device for the unit shall be fuse or HACR breaker. Canadian units may be fuse or circuit breaker.
2. **Unbalanced 3-Phase Supply Voltage**
Never operate a motor where a phase imbalance in supply voltage is greater than 2%. Use the following formula to determine the percent of voltage imbalance.

% Voltage Imbalance

$$= 100 \times \frac{\text{max voltage deviation from average voltage}}{\text{average voltage}}$$

Example: Supply voltage is 460-3-60.



AB = 452 v
BC = 464 v
AC = 455 v

$$\text{Average Voltage} = \frac{452 + 464 + 455}{3}$$

$$= \frac{1371}{3}$$

$$= 457$$

Determine maximum deviation from average voltage.
(AB) 457 – 452 = 5 v
(BC) 464 – 457 = 7 v
(AC) 457 – 455 = 2 v

Maximum deviation is 7 v.

Determine percent of voltage imbalance.

$$\text{\% Voltage Imbalance} = 100 \times \frac{7}{457}$$

$$= 1.53\%$$

This amount of phase imbalance is satisfactory as it is below the maximum allowable 2%.

IMPORTANT: If the supply voltage phase imbalance is more than 2%, contact your local electric utility company immediately.

Table 2C — Electrical Data — 48TM Units (Without Convenience Outlet)

UNIT 48TM	NOMINAL VOLTAGE	IFM TYPE	VOLTAGE RANGE		COMPR (ea)			OFM (ea)			IFM	COMBUSTION FAN MOTOR FLA	POWER SUPPLY*		DISCONNECT SIZE†	
			Min	Max	Qty	RLA	LRA	Qty	Hp	FLA	FLA		MCA	MOCP**	FLA	LRA
008 (7 1/2 Tons)	208/230-3-60	Std	187	254	2	14.0	91.0	2	1/4	1.4	5.8	.6	40.1/40.1	45/45	42/42	229/229
		Alt									5.8		40.1/40.1	45/45	42/42	229/229
		High									10.6		44.9/44.9	50/50	48/48	273/273
	460-3-60	Std	414	508	2	6.4	42.0	2	1/4	0.7	2.6	.3	18.4	20	19	108
		Alt									2.6		18.4	20	19	108
		High									4.8		20.6	25	22	130
	575-3-60	Std	518	632	2	5.2	39.0	2	1/4	0.7	2.6	.3	14.9	20	16	97
		Alt									2.6		14.9	20	16	97
		High									4.8		16.7	20	18	114
009 (8 1/2 Tons)	208/230-3-60	Std	187	254	2	17.3***	120.0***	2	1/4	1.4	5.8	.6	44.3/44.3	50/50	46/46	272/272
		High									10.6		49.1/49.1	60/60	52/52	316/316
	460-3-60	Std	414	508	2	7.9***	70.0***	2	1/4	0.7	2.6	.3	21.0	25	22	149
		High									4.8		23.2	30	24	171
	575-3-60	Std	518	632	2	5.5***	50.0***	2	1/4	0.7	2.6	.3	16.7	20	17	109
		High									4.8		18.5	25	19	126
012 (10 Tons)	208/230-3-60	Std	187	254	2	16.0	125.0	2	1/4	1.4	5.8	.6	44.6/44.6	50/50	47/47	297/297
		Alt									7.5		46.3/46.3	60/60	49/49	316/316
		High									15.0		53.4/53.4	60/60	57/57	374/374
	460-3-60	Std	414	508	2	8.0	62.5	2	1/4	0.7	2.6	.3	22.0	25	24	188
		Alt									3.4		22.8	25	24	191
		High									7.4		26.8	30	29	182
	575-3-60	Std	518	632	2	6.3	50.0	2	1/4	0.7	2.6	.3	17.4	20	18	119
		Alt									3.4		18.0	20	19	151
		High									7.4		21.2	25	23	146
014 (12 1/2 Tons)	208/230-3-60	Std	187	254	2	19.0	156.0	2	1/4	1.4	10.6	.6	56.2/56.2	70/70††	59/59	359/359
		Alt									15.0		60.6/60.6	70/70††	64/64	378/378
	460-3-60	Std	414	508	2	9.0	75.0	2	1/4	0.7	4.8	.3	26.5	30	28	174
		Alt									7.4		29.1	35	31	213
	575-3-60	Std	518	632	2	7.4	54.0	2	1/4	0.7	4.8	.3	21.6	25	23	127
		Alt									7.4		23.7	30	25	159

LEGEND

FLA — Full Load Amps
HACR — Heating, Air Conditioning and Refrigeration
IFM — Indoor (Evaporator) Fan Motor
LRA — Locked Rotor Amps
MCA — Minimum Circuit Amps
MOCP — Maximum Overcurrent Protection
NEC — National Electrical Code
OFM — Outdoor (Condenser) Fan Motor
RLA — Rated Load Amps

*The values listed in this table do not include power exhaust. See table below for power exhaust requirements.
†Used to determine minimum disconnect per NEC.
**Fuse or HACR circuit breaker.
††Fuse only.
***Data shown in table is for Compressor no. 1.
208/230-3-60: Compressor no. 2 RLA is 14.1 amps and LRA is 105 amps.
460-3-60: Compressor no. 2 RLA is 7.1 amps and LRA is 55 amps.
575-3-60: Compressor no. 2 RLA is 6.4 amps and LRA is 40 amps.

POWER EXHAUST ELECTRICAL DATA

POWER EXHAUST PART NO.	MCA (230 v)	MCA (460 v)	MCA (575 v)	MOCP (for separate power source)
CRPWREXH021A00	N/A	0.9	N/A	15
CRPWREXH022A00	3.3	N/A	1.32	15
CRPWREXH023A00	N/A	1.8	N/A	15
CRPWREXH024A00	1.6	N/A	0.64	15
CRPWREXH025A00	N/A	0.9	N/A	15
CRPWREXH026A00	3.3	N/A	1.32	15
CRPWREXH027A00	N/A	1.8	N/A	15
CRPWREXH028A00	1.7	N/A	0.68	15
CRPWREXH029A00	N/A	1.0	N/A	15
CRPWREXH030A00	1.6	N/A	0.64	15

N/A — Not available

NOTE: If a single power source is to be used, size wire to include power exhaust MCA and MOCP.

Check MCA and MOCP when power exhaust is powered through the unit. Determine the new MCA including the power exhaust using the following formula:

MCA New = MCA unit only + MCA of Power Exhaust

For example, using a 48TMD006---5 unit with MCA = 28.9 and MOCP = 35, with CRPWREXH030A00 power exhaust.

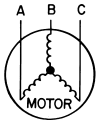
MCA New = 28.9 amps + 1.6 amps = 30.5 amps

If the new MCA does not exceed the published MOCP, then MOCP would not change. The MOCP in this example is 35 amps and the MCA New is below 35; therefore the MOCP is acceptable. If "MCA New" is larger than the published MOCP, raise the MOCP to the next larger size. For separate power, the MOCP for the power exhaust will be 15 amps per NEC.

NOTES:
1. In compliance with NEC requirements for multimotor and combination load equipment (refer to NEC Articles 430 and 440), the overcurrent protective device for the unit shall be fuse or HACR breaker. Canadian units may be fuse or circuit breaker.
2. **Unbalanced 3-Phase Supply Voltage**
Never operate a motor where a phase imbalance in supply voltage is greater than 2%. Use the following formula to determine the percent of voltage imbalance.

$$\% \text{ Voltage Imbalance} = 100 \times \frac{\text{max voltage deviation from average voltage}}{\text{average voltage}}$$

Example: Supply voltage is 460-3-60.



AB = 452 v
BC = 464 v
AC = 455 v

$$\text{Average Voltage} = \frac{452 + 464 + 455}{3}$$

$$= \frac{1371}{3}$$

$$= 457$$

Determine maximum deviation from average voltage.
(AB) 457 – 452 = 5 v
(BC) 464 – 457 = 7 v
(AC) 457 – 455 = 2 v

Maximum deviation is 7 v.

Determine percent of voltage imbalance.

$$\% \text{ Voltage Imbalance} = 100 \times \frac{7}{457}$$

$$= 1.53\%$$

This amount of phase imbalance is satisfactory as it is below the maximum allowable 2%.

IMPORTANT: If the supply voltage phase imbalance is more than 2%, contact your local electric utility company immediately.

Table 2D — Electrical Data — 48TM Units (With Convenience Outlet)

UNIT 48TM	NOMINAL VOLTAGE	IFM TYPE	VOLTAGE RANGE		COMPR (ea)			OFM (ea)			IFM	COMBUSTION FAN MOTOR FLA	POWER SUPPLY*		DISCONNECT SIZE†	
			Min	Max	Qty	RLA	LRA	Qty	Hp	FLA	FLA		MCA	MOCP**	FLA	LRA
008 (7 1/2 Tons)	208/230-3-60	Std	187	254	2	14.0	91.0	2	1/4	1.4	5.8	.6	44.9/44.9	50/50	48/48	234/234
		Alt									5.8		44.9/44.9	50/50	48/48	234/234
		High									10.6		49.7/49.7	60/60	53/53	277/277
	460-3-60	Std	414	508	2	6.4	42.0	2	1/4	0.7	2.6	.3	20.6	25	22	110
		Alt									2.6		20.6	25	22	110
		High									4.8		22.8	30	24	132
	575-3-60	Std	518	632	2	5.2	39.0	2	1/4	0.7	2.6	.3	16.6	20	18	99
		Alt									2.6		16.6	20	18	99
		High									4.8		18.4	20	20	116
009 (8 1/2 Tons)	208/230-3-60	Std	187	254	2	17.3***	120.0***	2	1/4	1.4	5.8	.6	49.1/49.1	60/60	52/52	277/277
		High									10.6		53.9/53.9	60/60	57/57	320/320
	460-3-60	Std	414	508	2	7.9***	70.0***	2	1/4	0.7	2.6	.3	23.2	30	24	151
		High									4.8		25.4	30	27	173
	575-3-60	Std	518	632	2	5.5***	50.0***	2	1/4	0.7	2.6	.3	18.4	25	19	111
		High									4.8		20.2	25	21	128
012 (10 Tons)	208/230-3-60	Std	187	254	2	16.0	125.0	2	1/4	1.4	5.8	.6	49.4/49.4	60/60	52/52	302/302
		Alt									7.5		51.9/51.9	60/60	54/54	321/321
		High									15.0		58.6/58.6	70/70††	63/63	369/369
	460-3-60	Std	414	508	2	8.0	62.5	2	1/4	0.7	2.6	.3	24.2	30	26	151
		Alt									3.4		25.0	30	26	190
		High									7.4		29.0	35	31	184
	575-3-60	Std	518	632	2	6.3	50.0	2	1/4	0.7	2.6	.3	19.1	25	20	121
		Alt									3.4		19.7	25	21	152
		High									7.4		22.9	25	25	148
014 (12 1/2 Tons)	208/230-3-60	Std	187	254	2	19.0	156.0	2	1/4	1.4	10.6	.6	61.0/61.0	70/70††	65/65	364/364
		Alt									15.0		65.4/65.4	80/80††	70/70	383/383
	460-3-60	Std	414	508	2	9.0	75.0	2	1/4	0.7	4.8	.3	28.7	35	30	176
		Alt									7.4		31.3	35	33	215
	575-3-60	Std	518	632	2	8.3	58.4	2	1/4	0.7	4.8	.3	23.3	30	25	129
		Alt									7.4		25.4	30	27	160

LEGEND

FLA — Full Load Amps
HACR — Heating, Air Conditioning and Refrigeration
IFM — Indoor (Evaporator) Fan Motor
LRA — Locked Rotor Amps
MCA — Minimum Circuit Amps
MOCP — Maximum Overcurrent Protection
NEC — National Electrical Code
OFM — Outdoor (Condenser) Fan Motor
RLA — Rated Load Amps

*The values listed in this table do not include power exhaust. See table below for power exhaust requirements.
†Used to determine minimum disconnect per NEC.
**Fuse or HACR circuit breaker.
††Fuse only.
***Data shown in table is for Compressor no. 1.
208/230-3-60: Compressor no. 2 RLA is 14.1 amps and LRA is 105 amps.
460-3-60: Compressor no. 2 RLA is 7.1 amps and LRA is 55 amps.
575-3-60: Compressor no. 2 RLA is 6.4 amps and LRA is 40 amps.

POWER EXHAUST ELECTRICAL DATA

POWER EXHAUST PART NO.	MCA (230 v)	MCA (460 v)	MCA (575 v)	MOCP (for separate power source)
CRPWREXH021A00	N/A	0.9	N/A	15
CRPWREXH022A00	3.3	N/A	1.32	15
CRPWREXH023A00	N/A	1.8	N/A	15
CRPWREXH024A00	1.6	N/A	0.64	15
CRPWREXH025A00	N/A	0.9	N/A	15
CRPWREXH026A00	3.3	N/A	1.32	15
CRPWREXH027A00	N/A	1.8	N/A	15
CRPWREXH028A00	1.7	N/A	0.68	15
CRPWREXH029A00	N/A	1.0	N/A	15
CRPWREXH030A00	1.6	N/A	0.64	15

N/A — Not available

NOTE: If a single power source is to be used, size wire to include power exhaust MCA and MOCP.

Check MCA and MOCP when power exhaust is powered through the unit. Determine the new MCA including the power exhaust using the following formula:

MCA New = MCA unit only + MCA of Power Exhaust

For example, using a 48HJD006---5 unit with MCA = 28.9 and MOCP = 35, with CRPWREXH030A00 power exhaust.

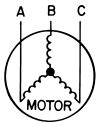
MCA New = 28.9 amps + 1.6 amps = 30.5 amps

If the new MCA does not exceed the published MOCP, then MOCP would not change. The MOCP in this example is 35 amps and the MCA New is below 35; therefore the MOCP is acceptable. If "MCA New" is larger than the published MOCP, raise the MOCP to the next larger size. For separate power, the MOCP for the power exhaust will be 15 amps per NEC.

NOTES:
1. In compliance with NEC requirements for multimotor and combination load equipment (refer to NEC Articles 430 and 440), the overcurrent protective device for the unit shall be fuse or HACR breaker. Canadian units may be fuse or circuit breaker.
2. **Unbalanced 3-Phase Supply Voltage**
Never operate a motor where a phase imbalance in supply voltage is greater than 2%. Use the following formula to determine the percent of voltage imbalance.

$$\%\ \text{Voltage Imbalance} = 100 \times \frac{\text{max voltage deviation from average voltage}}{\text{average voltage}}$$

Example: Supply voltage is 460-3-60.



AB = 452 v
BC = 464 v
AC = 455 v

$$\text{Average Voltage} = \frac{452 + 464 + 455}{3}$$

$$= \frac{1371}{3}$$

$$= 457$$

Determine maximum deviation from average voltage.
(AB) 457 – 452 = 5 v
(BC) 464 – 457 = 7 v
(AC) 457 – 455 = 2 v

Maximum deviation is 7 v.

Determine percent of voltage imbalance.

$$\%\ \text{Voltage Imbalance} = 100 \times \frac{7}{457}$$

$$= 1.53\%$$

This amount of phase imbalance is satisfactory as it is below the maximum allowable 2%.

IMPORTANT: If the supply voltage phase imbalance is more than 2%, contact your local electric utility company immediately.

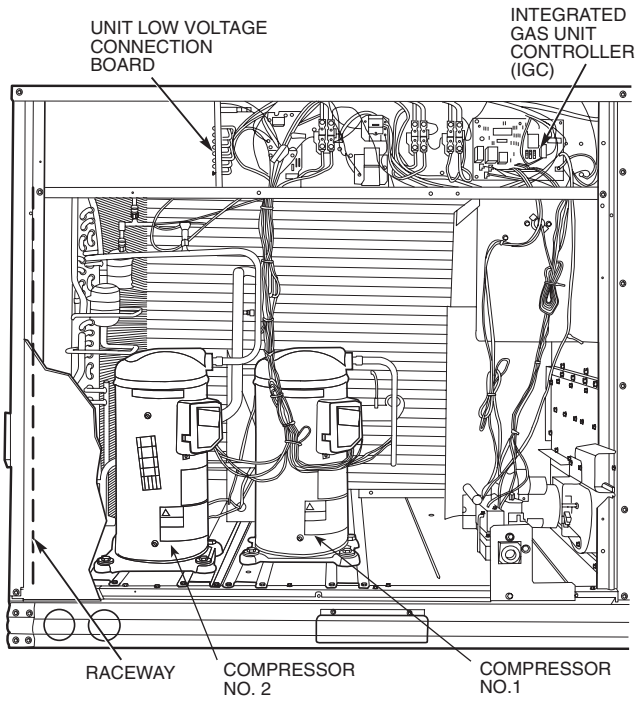


Fig. 11 — Field Control Wiring Raceway and Compressor Location

Step 8 — Adjust Factory-Installed Options

APOLLO CONTROL — The optional Apollo control is used to actively monitor all modes of operation as well as indoor (evaporator) fan status, filter status, and indoor-air quality. The Apollo control is designed to work with Carrier TEMP and VVT® systems.

The thermostat must be wired to the Apollo control before starting the unit. Refer to the Apollo control installation instructions for information on installing the thermostat. See Fig. 12 for Apollo location.

CONVENIENCE OUTLET — An optional convenience outlet provides power for rooftop use. For maintenance personnel safety, the convenience outlet power is off when the unit disconnect is off. Adjacent unit outlets may be used for service tools. An optional "Hot Outlet" is available from the factory as a special order item.

NOVAR CONTROLS — Optional Novar controls (ETM 2024 or ETM3051) are available for replacement or new construction jobs.

MANUAL OUTDOOR-AIR DAMPER — The outdoor-air hood and screen are attached to the basepan at the bottom of the unit for shipping.

Assembly:

1. Determine quantity of ventilation required for building. Record amount for use in Step 8.
2. Remove filter access panel by raising panel and swinging panel outward. Panel is now disengaged from track and can be removed. No tools are required to remove the filter access panel. Remove outdoor-air opening panel. Save panels and screws. See Fig. 13.
3. Separate hood and screen from basepan by removing the screws and brackets securing them. Save all screws and discard brackets.
4. Replace outdoor air opening panel with screws saved from Step 2.
5. Place hood on front of outdoor-air opening panel. See Fig. 14 for hood details. Secure top of hood with the 6 screws removed in Step 3. See Fig. 15.
6. Remove and save 6 screws (3 on each side) from sides of the manual outdoor-air damper.
7. Align screw holes on hood with screw holes on side of manual outdoor-air damper. See Fig. 14 and 15. Secure hood with 6 screws from Step 6.
8. Adjust minimum position setting of the damper blade by adjusting the manual outdoor-air adjustment screws on the front of the damper blade. See Fig. 13. Slide blade vertically until it is in the appropriate position determined by Fig. 16. Tighten screws.
9. Remove and save screws currently on sides of hood. Insert screens. Secure screens to hood using the screws. See Fig. 15.
10. Replace filter access panel. Ensure filter access panel slides along the tracks and is securely engaged.

PREMIERLINK™ CONTROL — The PremierLink controller is compatible with the Carrier Comfort Network (CCN). This control is designed to allow users the access and ability to change factory-defined settings, thus expanding the function of the standard unit control board. Carrier's diagnostic standard tier display tools such as Navigator or Scrolling Marquee can be used with the PremierLink controller.

The PremierLink controller (see Fig. 17) requires the use of a Carrier electronic thermostat or a CCN connection for time broadcast to initiate its internal timeclock. This is necessary for broadcast of time of day functions (occupied/unoccupied). Refer to Fig. 18. No sensors are supplied with the field-mounted PremierLink control. The factory-installed PremierLink control includes only the supply-air sensor (SAT) as standard. The PremierLink control may be mounted in the control panel or an area below the control panel. If used with the EconoMi$er2 the standard OAT (outdoor-air temperature) sensor and Compressor Lockout Sensor supplied with the EconoMi$er2 will need to be changed to the PremierLink compatible sensors. Refer to the PremierLink control and EconoMi$er2 Installation, Start-up and Configuration Instructions for more information.

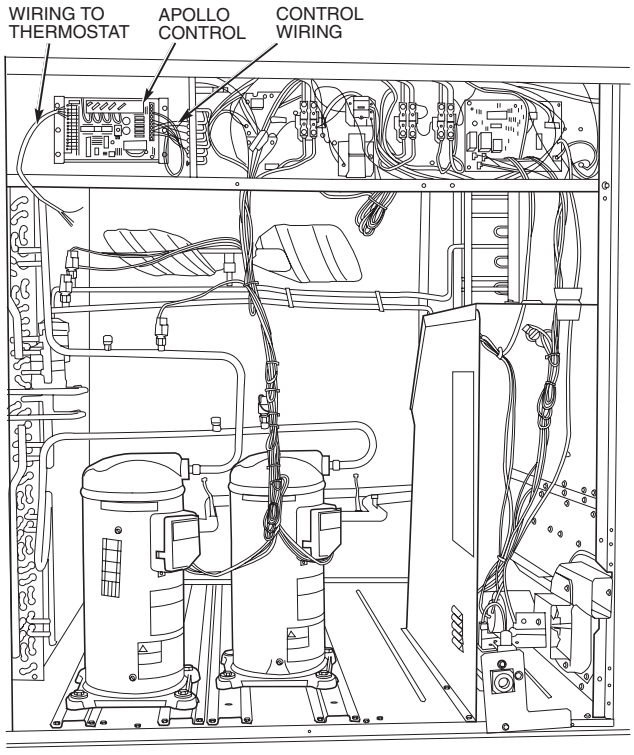


Fig. 12 — Apollo Control Factory-Installed in Typical Unit

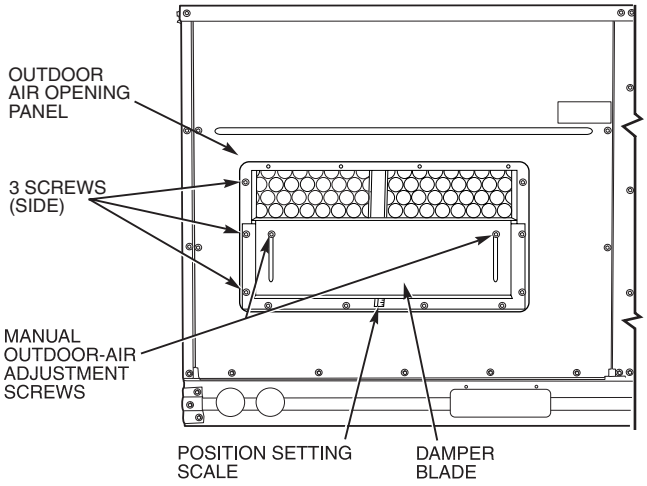


Fig. 13 — Damper Panel with Manual Outdoor-Air Damper Installed

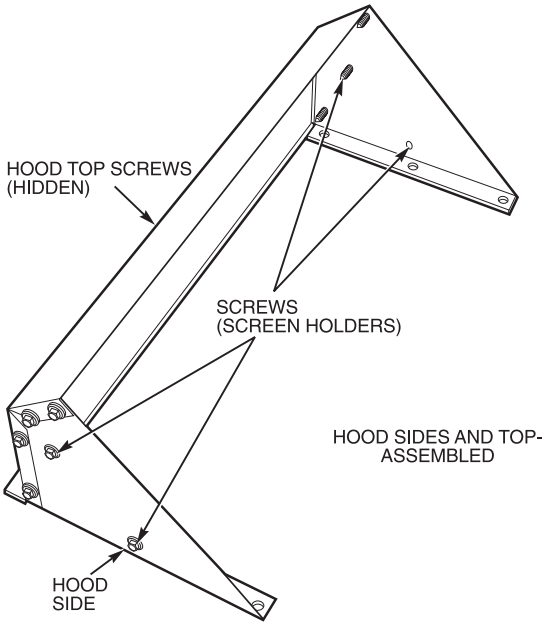


Fig. 14 — Outdoor-Air Hood Details

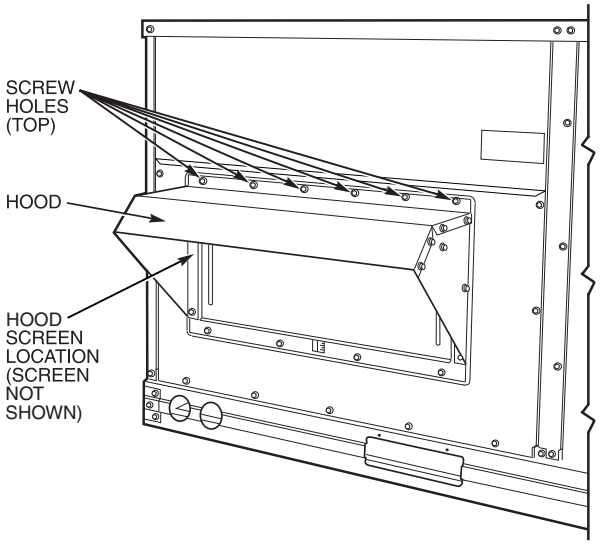


Fig. 15 — Optional Manual Outdoor-Air Damper with Hood Attached

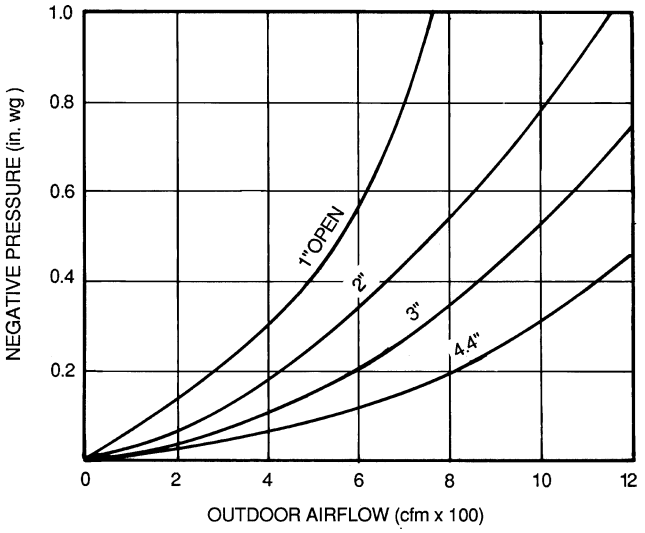


Fig. 16 — Outdoor-Air Damper Position Setting

OPTIONAL ECONOMI$ER2 — See Fig. 19 for Econo-Mi$er2 component locations.

NOTE: These instructions are for installing the optional EconoMi$er2 only. Refer to the accessory EconoMi$er2 installation instructions when field installing an EconoMi$er2 accessory.

1. To remove the existing unit filter access panel, raise the panel and swing the bottom outward. The panel is now disengaged from the track and can be removed. See Fig. 20.
2. The box with the EconoMi$er2 hood components is shipped in the compartment behind the EconoMi$er2. In order to remove the box, the EconoMi$er2 controller must be temporarily moved. This step does not need to be performed on an EconoMi$er2 with 4 to 20 mA control as there is no controller. The entire EconoMi$er2 does not have to be removed to retrieve the hood box. The EconoMi$er2 controller is mounted on top of the EconoMi$er2 in the position shown in Fig. 19.
   a. Remove the screws holding the controller bracket in place. Save screws. Temporarily lower the controller. See Fig. 21. Do not disconnect the controller wiring.
   b. Remove the screw holding the hood box bracket to the top of the EconoMi$er2. Slide the hood box out of the unit. See Fig. 21 and 22.
   c. Reinstall the EconoMi$er2 controller in its original position using the screws saved in Step 2a. Tighten screws.

IMPORTANT: If the power exhaust accessory is to be installed on the unit, the hood shipped with the unit will not be used and must be discarded. Save the aluminum filter for use in the power exhaust.

3. The indoor coil access panel will be used as the top of the hood. Remove the screws along the bottom of the indoor coil access panel. See Fig. 23.
4. Swing out indoor coil access panel and insert the hood sides under the panel (hood top). Use the screws provided to attach the hood sides to the hood top. Use screws provided to attach the hood sides to the unit. See Fig. 24.
5. Remove the shipping tape holding the EconoMi$er2 barometric relief damper in place.

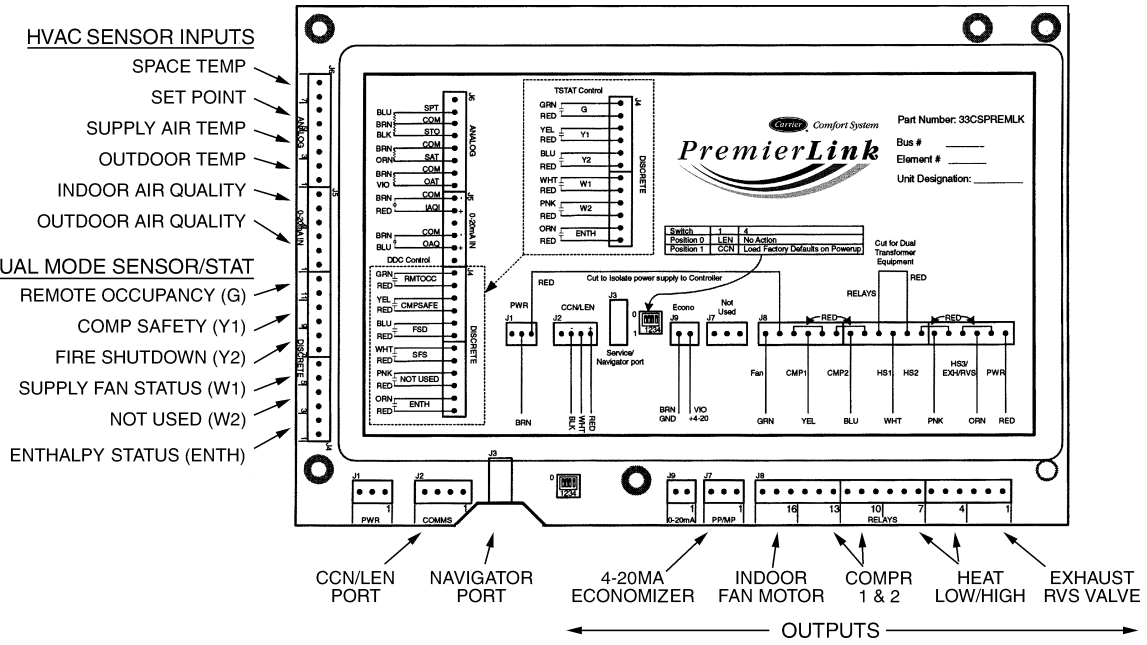
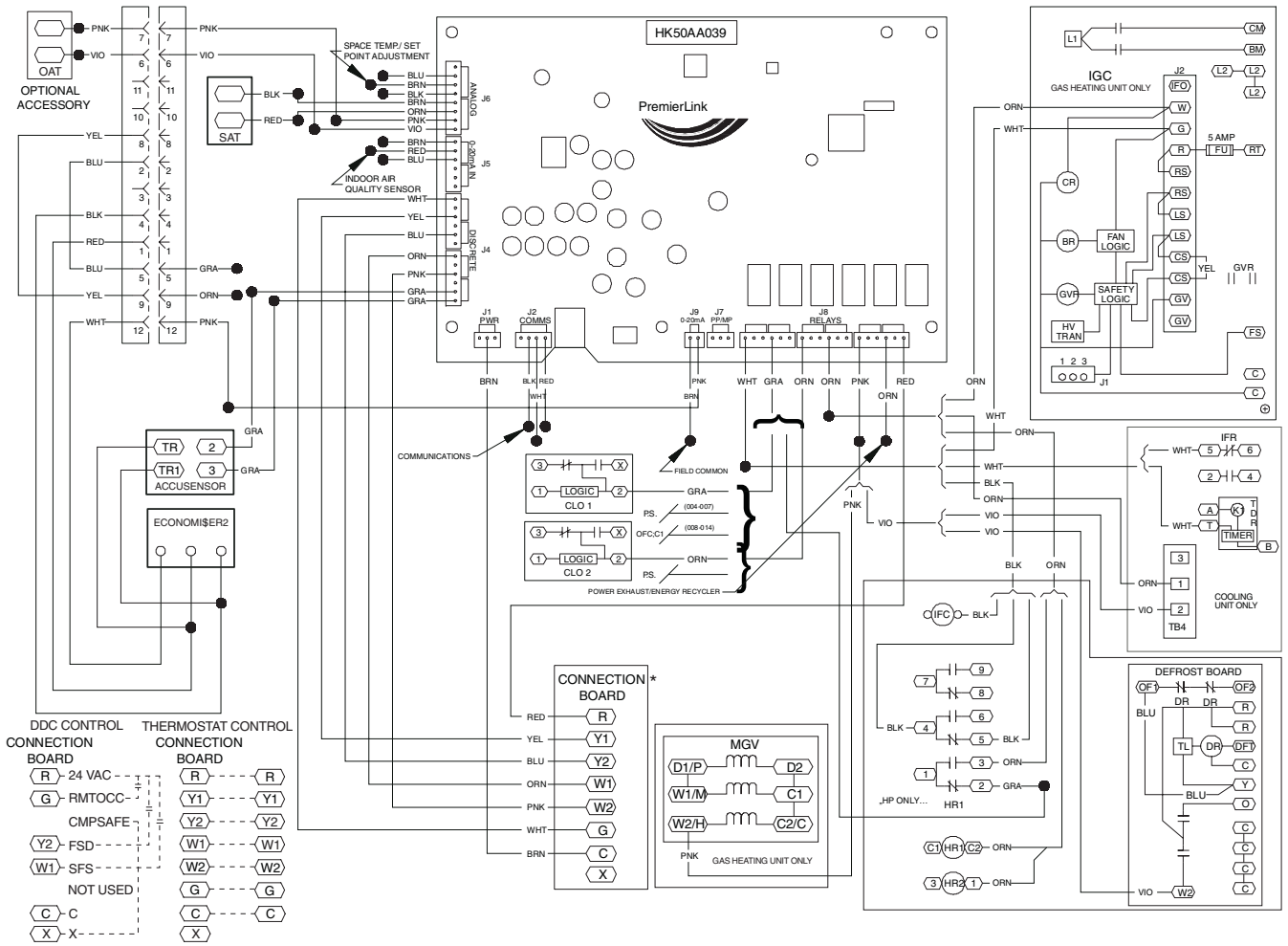


Fig. 17 — PremierLink™ Controller



LEGEND

CLO	—	Compressor Lockout
COMMS	—	Communications
HR	—	Heating Relay
IFC	—	Indoor Fan Contactor
IFR	—	Indoor Fan Relay
IGC	—	Integrated Gas Unit Controller
MGV	—	Main Gas Valve
OAT	—	Outdoor Air Temperature Sensor
PWR	—	Power
SAT	—	Supply Air Temperature Sensor

*When connecting DDC or thermostat controls, wire to connection board. Do not wire directly to PremierLink board.

Fig. 18 — Typical PremierLink Controls Wiring

6. Insert the hood divider between the hood sides. See Fig. 24 and 25. Secure hood divider with 2 screws on each hood side. The hood divider is also used as the bottom filter rack for the aluminum filter.
7. Open the filter clips which are located underneath the hood top. Insert the aluminum filter into the bottom filter rack (hood divider). Push the filter into position past the open filter clips. Close the filter clips to lock the filter into place. See Fig. 25.
8. Caulk the ends of the joint between the unit side panel and the hood top. See Fig. 23.
9. Replace the filter access panel.
10. Install all EconoMi$er2 accessories. EconoMi$er2 wiring is shown in Fig. 26 and 27.

Barometric flow capacity is shown in Fig. 28. Outdoor air leakage is shown in Fig. 29. Return air pressure drop is shown in Fig. 30.

ECONOMI$ER2 CONTROLLER WIRING AND CONFIGURATION

IMPORTANT: The optional EconoMi$er2 with 4 to 20 mA actuator signal control does not include the EconoMi$er2 controller. The EconoMi$er2 actuator is operated by a 4 to 20 mA signal from an existing field-supplied controller (such as PremierLink™ control). See Fig. 27 for wiring information.

Mixed Air Sensor — The mixed air sensor is pre-wired and mounted at the factory. See Fig. 31 for mixed air sensor location.

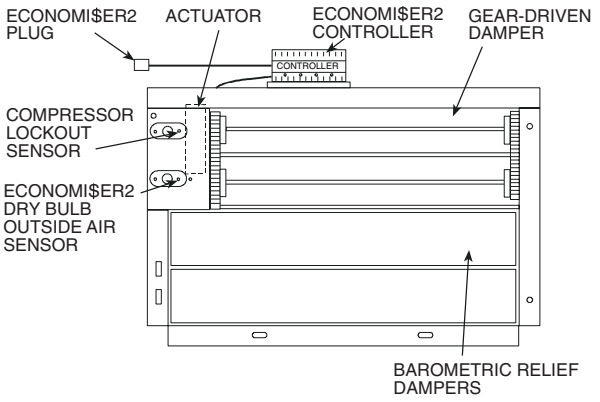


Fig. 19 — EconoMi$er2 Component Locations

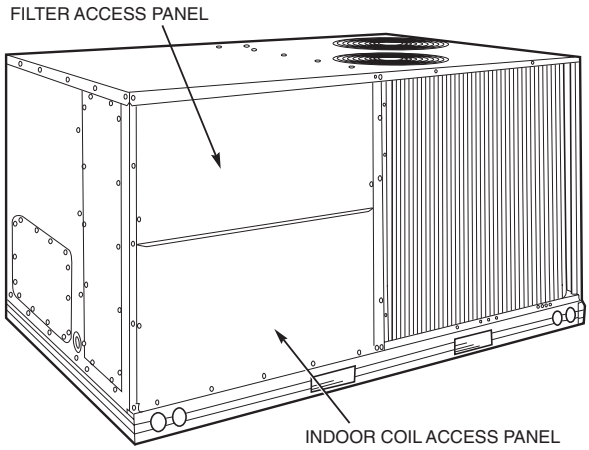


Fig. 20 — Typical Access Panel Locations

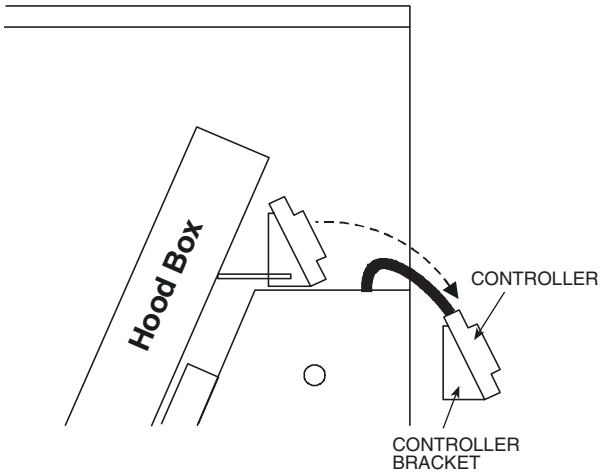


Fig. 21 — Temporary EconoMi$er2 Controller Relocation

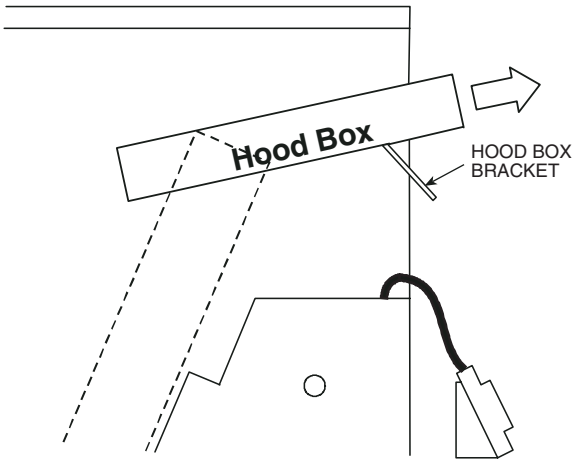


Fig. 22 — Hood Box Removal

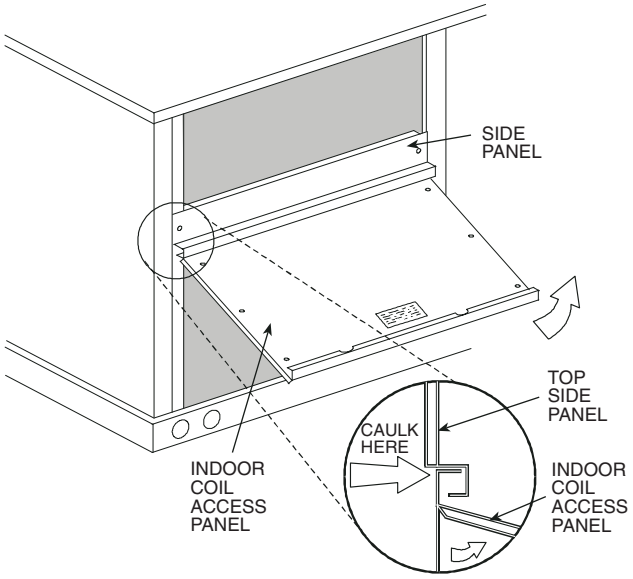


Fig. 23 — Indoor Coil Access Panel Relocation

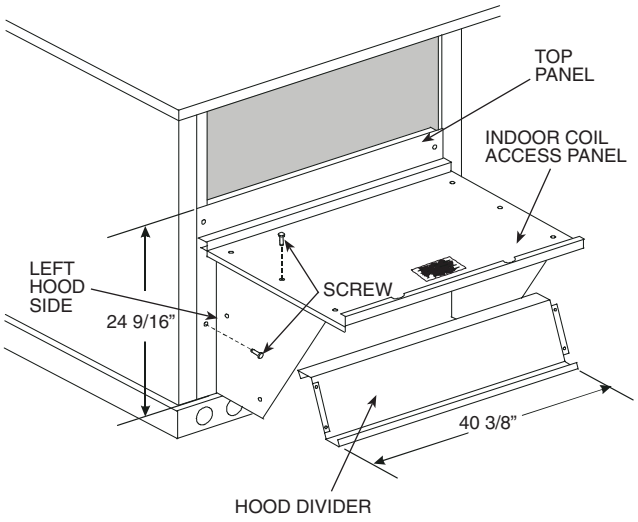


Fig. 24 — Outdoor-Air Hood Construction

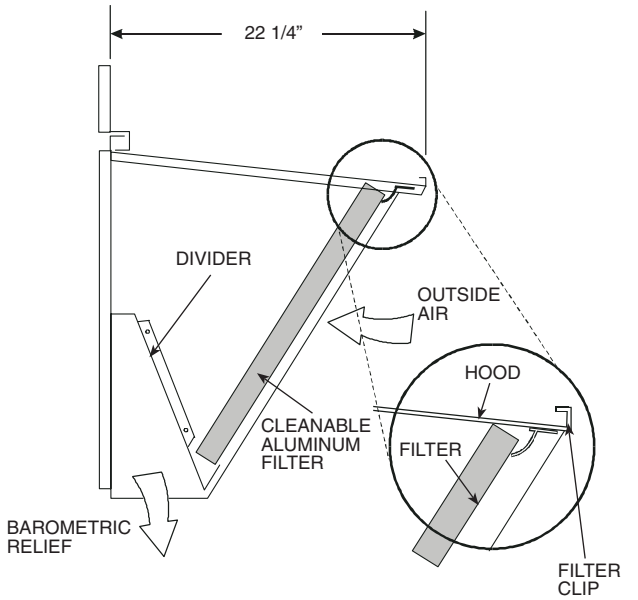


Fig. 25 — Filter Installation

Outdoor Air Enthalpy or Dry Bulb Sensor — The EconoMi$er2 controller accepts input from either an accessory enthalpy sensor or dry bulb sensor. When using differential sensing, both sensors must be the same type. The outdoor air sensor is wired to terminals S_O and + on the EconoMi$er2 controller. See Fig. 26.

When the outdoor air enthalpy rises above the outdoor enthalpy changeover set point, the outdoor-air damper moves to its minimum position. The outdoor enthalpy changeover set point is set with the outdoor enthalpy set point potentiometer on the EconoMi$er2 controller. The set points are A, B, C, and D. See Fig. 32. The factory-installed 620-ohm jumper must be in place across terminals S_R and +. See Fig. 26.

Differential Dry Bulb — For differential dry bulb control the standard outdoor dry bulb sensor (refer to Table 3) must be replaced with two CROASENR001A00 sensors. One must be placed in the return air and the other in the outdoor air stream.

An outdoor-air sensor can be mounted in any orientation that exposes it to freely circulating air and protects it from rain, snow and direct sunlight. Connect it to the S_O terminal and to the + terminal. The second sensor should be mounted in the return airstream. Connect it to the S_R terminal (after removing the 620-ohm resistor) and to the + terminal on the controller. See Fig. 26.

Differential Enthalpy — For differential sensing, the EconoMi$er2 controller uses two enthalpy sensors, one in the outside air and one in the return air duct. The EconoMi$er2 controller compares the outdoor air enthalpy to the return air enthalpy to determine EconoMi$er2 use. The controller selects the lower enthalpy air (return or outdoor) for cooling. For example, when the outdoor air has a lower enthalpy than the return air, the EconoMi$er2 opens to bring in outdoor air for free cooling.

Mount the return air enthalpy sensor in the return air duct. The return air enthalpy sensor is wired to terminals S_R and + on the EconoMi$er2 controller. See Fig. 33. The outdoor enthalpy changeover set point is set with the outdoor enthalpy set point potentiometer on the EconoMi$er2 controller. Turn the potentiometer to the D setting for differential enthalpy. See Fig. 34.

Indoor Air Quality Sensor Input (ISI) — The ISI can be used for any indoor air quality sensor that provides a 2 to 10 vdc output. The controller will modulate the outdoor damper to provide ventilation based on the sensor output and the setting of the ISI potentiometer. The ISI LED will be lit when the ISI signal is above the ISI potentiometer set point.

A separate field-supplied transformer must be used to power the CO_2 sensor. See Fig. 35.

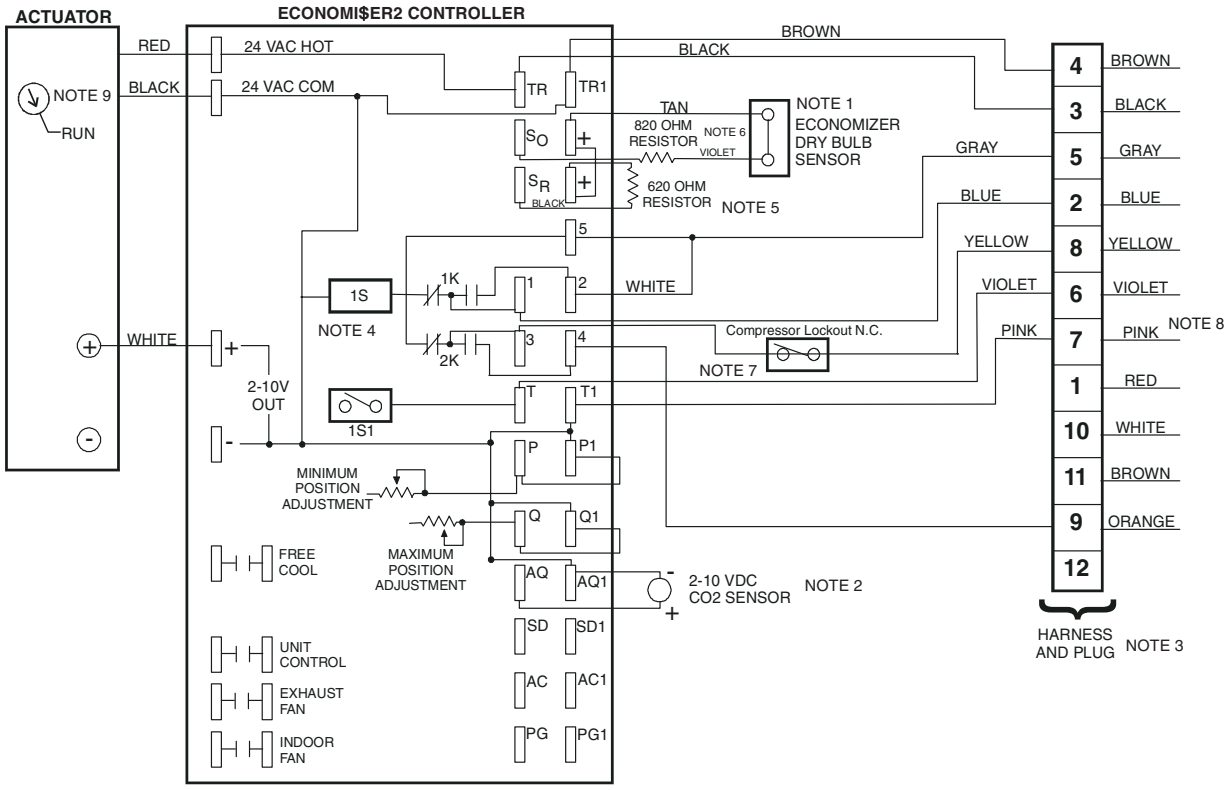
Mount the optional indoor air quality (CO_2) sensor according to manufacturer specifications. The CO_2 sensor should be wired to the AQ and AQ1 terminals of the controller. Adjust the ISI potentiometer to correspond to the ISI voltage output of the indoor air quality sensor at the user-determined set point. See Fig. 26.

Exhaust Set Point Adjustment — Except during purge, shutdown, and air change, the exhaust set point will determine when the exhaust fan runs based on damper position (if accessory power exhaust is installed). When the ISI calls for exhaust, the EconoMi$er2 controller provides a 45 ± 15 second delay before exhaust fan activation. This delay allows the damper to reach the appropriate position to avoid unnecessary fan overload. The set point is modified with the Exhaust Fan Set Point potentiometer. See Fig. 34.

Minimum and Maximum Position Control — There is a minimum damper position potentiometer and a maximum damper position potentiometer on the EconoMi$er2 controller. See Fig. 34. The minimum damper position maintains the minimum airflow into the building during the occupied period.

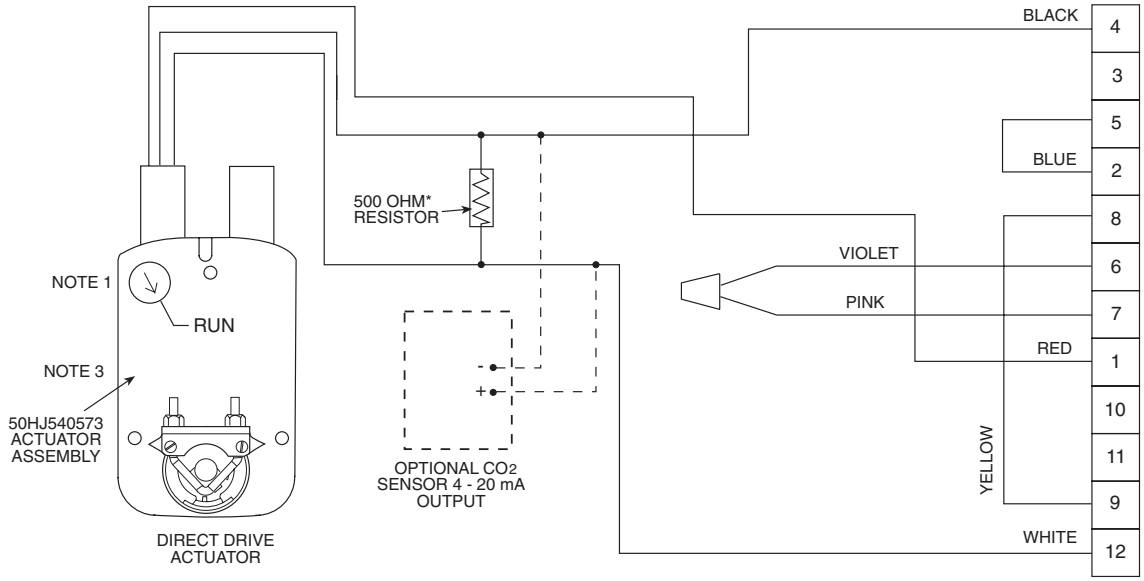
NOTE: The minimum position signal takes priority over the maximum position signal. If the maximum damper position is set below the minimum damper position, the EconoMi$er2 controller will maintain the actuator at minimum position.

Adjust the minimum position potentiometer to allow the minimum amount of outdoor air, as required by local codes, to enter the building. Make minimum position adjustments with at least 10 F (6 C) temperature difference between the outdoor and return-air temperatures.



NOTES:
1. The standard EconoMi$er2 is shipped with a fixed dry bulb sensor. (Open 67 F — Close 52 F.) An adjustable dry bulb or enthalpy sensor can replace the fixed dry bulb. (See Note 6.)
2. CO_2 sensor is optional. Field-installed accessory. See rooftop price pages for ordering data and pricing. Power for CO_2 sensor should be provided by field-supplied transformer.
3. The HVAC unit is shipped with a jumper plug attached to the EconoMi$er2 harness. Remove the jumper plug and save for future use if economizer is removed. Connect the male side of plug (shown above) to the female side in HVAC unit.
4. 1S is an electronic switch that closes when powered by a 24 vac input.
5. Factory-installed 620-ohm, 1 watt, 5% resistor should be removed only when a HH57AC078 enthalpy sensor or CROASENR001A00 adjustable dry bulb is added to SR and + for differential sensing.
6. When replacing the fixed dry bulb sensor with an enthalpy or adjustable dry bulb, remove the 820-ohm resistor.
7. Compressor lockout (Open 35 F — Close 50 F).
8. See EconoMi$er2 Installation Instructions for details on locating and wiring supply air (mixed air) sensor.
9. Switch on actuator must be in run position for economizer to operate.
10. A 2-stage thermostat is recommended.

Fig. 26 — EconoMi$er2 Wiring



*500 ohm ± 5%, 1/2 Watt.

NOTES:
1. Switch on actuator must be in run position for economizer to operate.
2. PremierLink™ control requires that the standard 50HJ540569 outside-air sensor be replaced by either the CROASENR001A00 dry bulb sensor or HH57A077 enthalpy sensor.
3. 50HJ540573 actuator consists of the 50HJ540567 actuator and a harness with 500-ohm resistor.

Fig. 27 — EconoMi$er2 with 4 to 20 mA Control Wiring

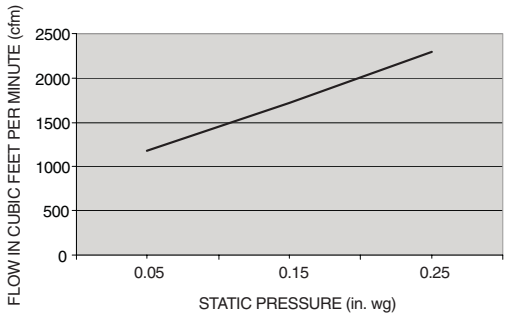


Fig. 28 — Barometric Relief Flow Capacity

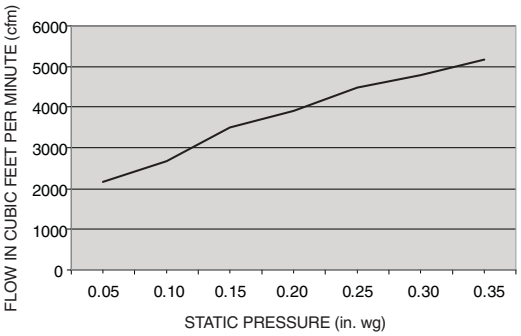


Fig. 30 — Return Air Pressure Drop

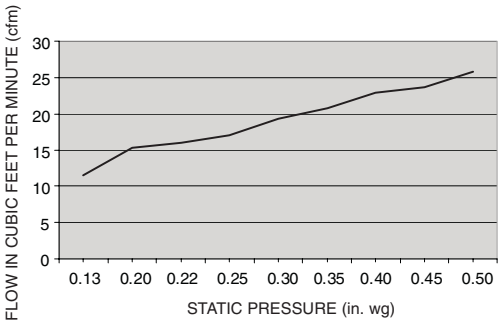


Fig. 29 — Outdoor Air Damper Leakage

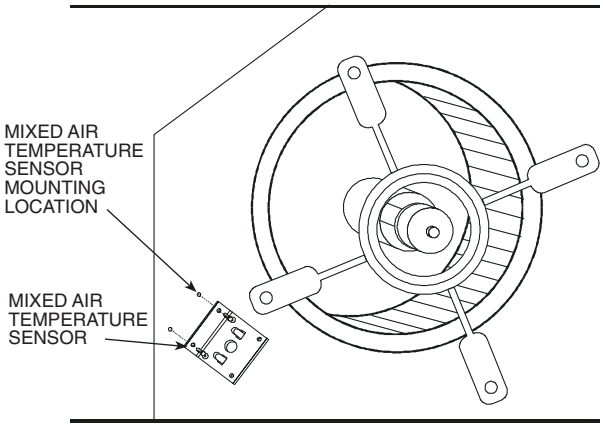


Fig. 31 — Mixed Air Sensor Location

CONTROL CURVE	CONTROL POINT APPROX. °F (°C) AT 50% RH
A	73 (23)
B	70 (21)
C	67 (19)
D	63 (17)

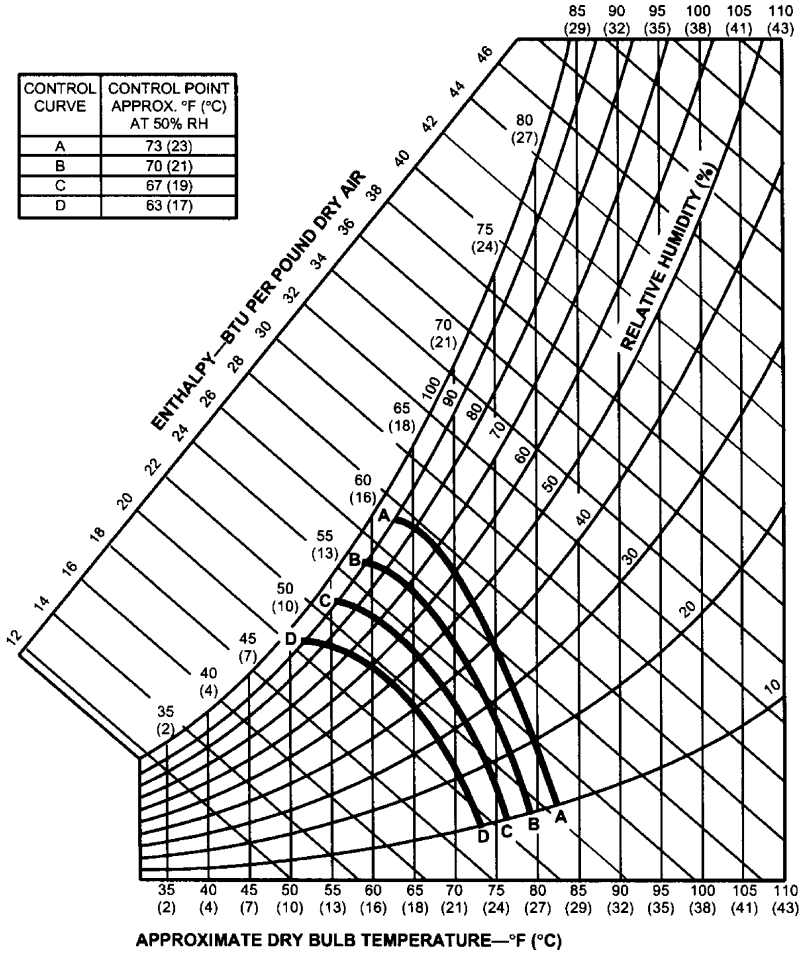
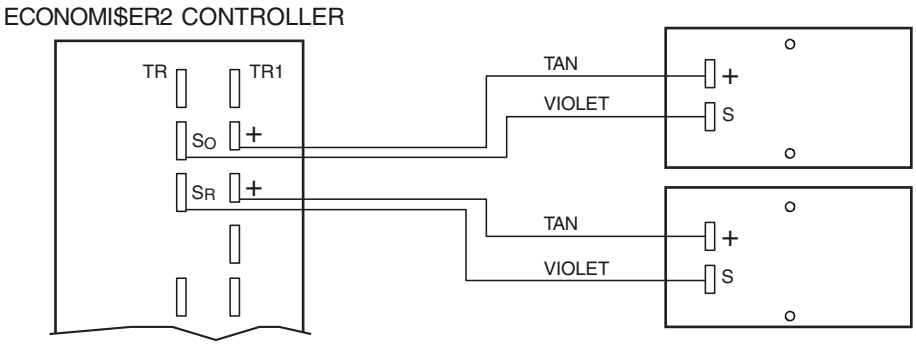


Fig. 32 — Enthalpy Changeover Set Points



NOTE: Remove the 620-ohm resistor between SR and + before installing a differential enthalpy or adjustable dry bulb sensor.

Fig. 33 — Differential Enthalpy Wiring

Table 3 — EconoMi$er2 Sensor Usage

APPLICATION	OUTDOOR AIR TEMPERATURE SENSOR	RETURN AIR TEMPERATURE SENSOR	OUTDOOR AIR ENTHALPY SENSOR	RETURN AIR ENTHALPY SENSOR
Standard Unit*	Included — 50HJ540569	—	—	—
Differential Dry Bulb	Required — CROASENR001A00	Required — CROASENR001A00	—	—
Single Enthalpy	Included — Not Used	—	Required — HH57AC078	—
Differential Enthalpy with Stepper Actuator	Included — Not Used	—	Required — HH57AC078	Required — HH57AC078
Dry Bulb Temperature with PremierLink™† (PremierLink requires 4-20 mA Actuator)	Required — HH79NZ017	—	—	—
Differential Dry Bulb Temperature with PremierLink† (PremierLink requires 4-20 mA Actuator)	Required — HH79NZ017	Required — 33ZCT55SPT or Equivalent	—	—
Single Enthalpy with PremierLink† (PremierLink requires 4-20 mA Actuator)	Included — Not Used	—	Required — HH57AC077	—
Differential Enthalpy with PremierLink† (PremierLink requires 4-20 mA Actuator)	Included — Not Used	—	Required — HH57AC077	Required — HH57AC078

*Optional adjustable Outdoor Air or Return Air Temperature sensor, CROASENR001A00, is available.

†PremierLink Control requires Supply Air Temperature sensor 33ZCSENSAT — Included with factory-installed PremierLink Control; field-supplied and field-installed with field-installed PremierLink Control.

NOTES:

1. CO_2 Sensors (Optional).
   33ZCSENCO2 — Room sensor (adjustable). Aspirator box is required for duct mounting of the sensor.
   33ZCASPCO2 — Aspirator box used for duct-mounted CO_2 room sensor.
   33ZCT55CO2 — Space temperature and CO_2 room sensor with override.
   33ZCT56CO2 — Space temperature and CO_2 room sensor with override and setpoint.
2. All units include the following Standard Sensors:
   Outdoor-Air Sensor — 50HJ540569 — Opens at 67 F, closes at 52 F, not adjustable.
   Mixed-Air Sensor — HH97AZ001 — (PremierLink control requires Supply Air Temperature sensor 33ZCSENSAT)
   Compressor Lockout Sensor — 50HJ540570 — Opens at 35 F, closes at 50 F.

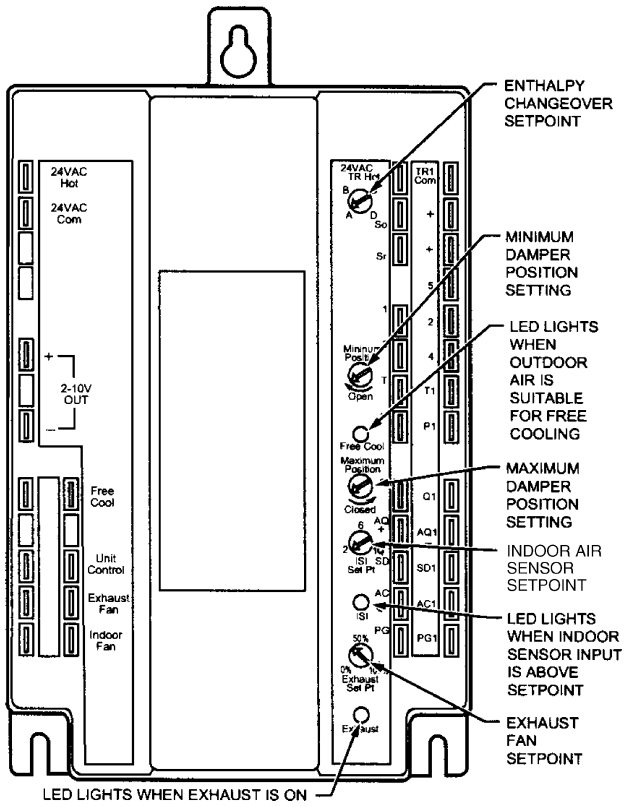


Fig. 34 — EconoMi$er2 Controller Potentiometer and LED Locations

To determine the minimum position setting, perform the following procedure:

1. Calculate the appropriate mixed air temperature using the following formula:
   $(T_O \times OA) + (T_R \times RA) = T_M$
   T_O = Outdoor-Air Temperature
   OA = Percent of Outdoor Air
   T_R = Return-Air Temperature
   RA = Percent of Return Air
   T_M = Mixed-Air Temperature
   As an example, if local codes require 10% outdoor air during occupied conditions, outdoor-air temperature is 60 F, and return-air temperature is 75 F.
   (60 x .10) + (75 x .90) = 73.5 F
2. Disconnect the mixed air sensor from terminals T and T1.
3. Ensure that the factory-installed jumper is in place across terminals P and P1. If remote damper positioning is being used, make sure that the terminals are wired according to Fig. 36 and minimum position potentiometer is turned fully clockwise.
4. Connect 24 vac across terminals TR and TR1.
5. Carefully adjust the minimum position potentiometer until the measured mixed air temperature matches the calculated value.

To adjust the maximum position, perform the following procedure:

1. Disconnect the mixed-air sensor from terminals T and T1 and short terminals T and T1.
2. Ensure that the factory-installed jumper is in place across terminals Q and Q1. If remote damper positioning is being used, make sure that the terminals are wired according to Fig. 36 and minimum position potentiometer is turned fully clockwise.

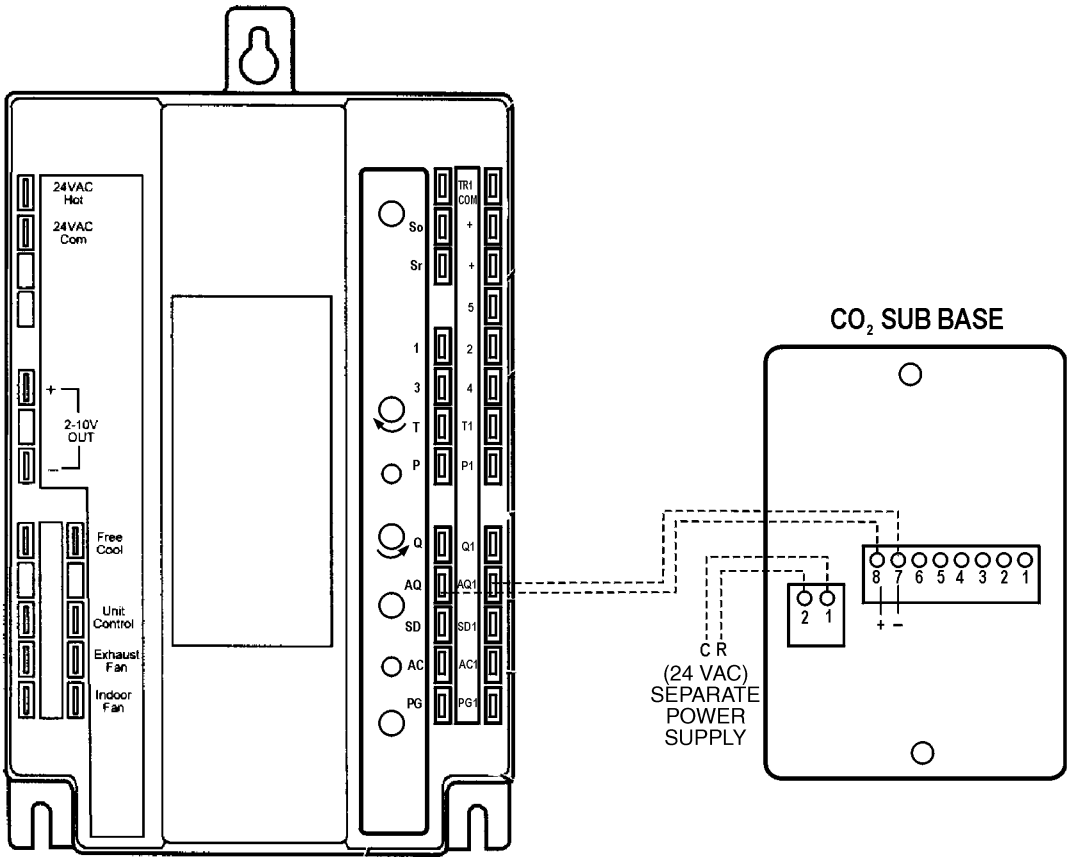


Fig. 35 — Indoor Air Quality Sensor Wiring

3. Connect 24 vac across terminals TR and TR1.
4. Carefully adjust the maximum position potentiometer until the desired position is reached.

Remote control of the EconoMi$er2 damper is desirable when requiring additional temporary ventilation. If a field-supplied remote potentiometer (Honeywell part number S963B1128) is wired to the EconoMi$er2 controller, the minimum and maximum position of the damper can be controlled from a remote location.

To control the minimum damper position remotely, wire the field-supplied potentiometer to the P and P1 terminals on the EconoMi$er2 controller. See Fig. 36.

To control the maximum damper position remotely, wire the field-supplied potentiometer to the Q and Q1 terminals on the EconoMi$er2 controller. See Fig. 36.

EconoMi$er2 Usage Tracking — The Free Cool contacts on the EconoMi$er2 controller close when the EconoMi$er2 is providing free cooling. See Fig. 26. Free cooling usage time can be tracked by connecting a field-supplied timer to the free cool contacts.

Purge — When terminals PG and PG1 on the EconoMi$er2 controller connect through relay contacts or a jumper, the purge function is energized. See Fig. 37. The EconoMi$er2 damper drives fully closed and the exhaust fan turns on.

Shutdown — When terminals SD and SD1 on the EconoMi$er2 controller connect through relay contacts or a jumper, all systems are shut down. See Fig. 37. The EconoMi$er2 damper drives fully closed, the exhaust fan turns off, and the indoor fans turn off.

Air Change (Evacuation) — When terminals AC and AC1 on the EconoMi$er2 controller connect through relay contacts or a jumper, an air change is initiated. See Fig. 37. The EconoMi$er2 damper drives fully opened, the exhaust fan turns on, and the indoor fans turn on.

Demand Control Ventilation — When using the EconoMi$er2 for demand control ventilation, there are some equipment selection criteria which should be considered. When selecting the heat capacity and cool capacity of the equipment, the maximum ventilation rate must be evaluated for design conditions. The maximum damper position must be calculated to provide the desired fresh air.

Typically the maximum ventilation rate will be about 5 to 10% more than the typical CFM required per person, using normal outside air design criteria. The Ventilator software (demand control ventilation software) may be used to determine the CO_2 recovery rate at maximum ventilation.

A proportional anticipatory strategy should be taken with the following conditions: a zone with a large area, varied occupancy, and equipment that cannot exceed the required ventilation rate at design conditions. Exceeding the required ventilation rate means the equipment can condition air at a maximum ventilation rate that is greater than the required ventilation rate for maximum occupancy. A proportional-anticipatory strategy will cause the fresh air supplied to increase as the room CO_2 level increases even though the CO_2 set point has not been reached. By the time the CO_2 level reaches the set point, the damper will be at maximum ventilation and should maintain the set point.

In order to have the CO_2 sensor control the economizer damper in this manner, first determine the damper voltage output for minimum or base ventilation. Base ventilation is the ventilation required to remove contaminants during unoccupied periods. The following equation may be used to determine the percent of outside-air entering the building for a given damper position. For best results there should be at least a 10 degree difference in outside and return-air temperatures.

$(T_O \times OA) + (T_R \times RA) = T_M$

T_O = Outdoor-Air Temperature
OA = Percent of Outdoor Air
T_R = Return-Air Temperature
RA = Percent of Return Air
T_M = Mixed-Air Temperature

The same equation can be used to determine the occupied or maximum ventilation rate to the building. For example, an output of 3.6 volts to the actuator provides a base ventilation rate of 5% and an output of 6.7 volts provides the maximum ventilation rate of 20% (or base plus 15 cfm per person). Use Fig. 38 to determine the maximum setting of the CO_2 sensor. For example, a 1100 ppm set point relates to a 15 cfm per person design. Use the 1100 ppm curve on Fig. 38 to find the point when the CO_2 sensor output will be 6.7 volts. Line up the point on the graph with the left side of the chart to determine that the range configuration for the CO_2 sensor should be 1800 ppm. The EconoMi$er2 controller will output the 6.7 volts from the CO_2 sensor to the actuator when the CO_2 concentration in the space is at 1100 ppm. The ISI set point may be left at 2 volts since the CO_2 sensor voltage will be ignored by the EconoMi$er2 controller until it gets above the 3.6 volt setting of the minimum position potentiometer. The maximum position potentiometer should be left at full open unless the user wants to limit the damper travel for both CO_2 and EconoMi$er2 operation.

CO_2 Sensor Configuration — The CO_2 sensor has preset standard voltage settings that can be selected anytime after the sensor is powered up. See Table 4.

NOTE: Use setting 1 or 2 for Carrier equipment. See Table 4.

1. Press Clear and Mode buttons. Hold at least 5 seconds until the sensor enters the Edit mode.
2. Press Mode twice. The STDSET Menu will appear.
3. Use the Up/Down button to select the preset number. See Table 4.
4. Press Enter to lock in the selection.
5. Press Mode to exit and resume normal operation.

The custom settings of the CO_2 sensor can be changed anytime after the sensor is energized. Follow the steps below to change the non-standard settings:

1. Press Clear and Mode buttons. Hold at least 5 seconds until the sensor enters the Edit Mode.
2. Press Mode twice. The STDSET Menu will appear.

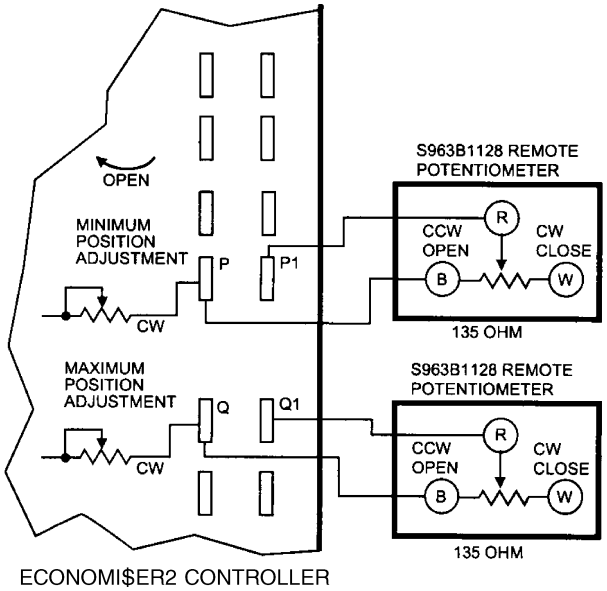


Fig. 36 — Remote Damper Position Potentiometer Wiring

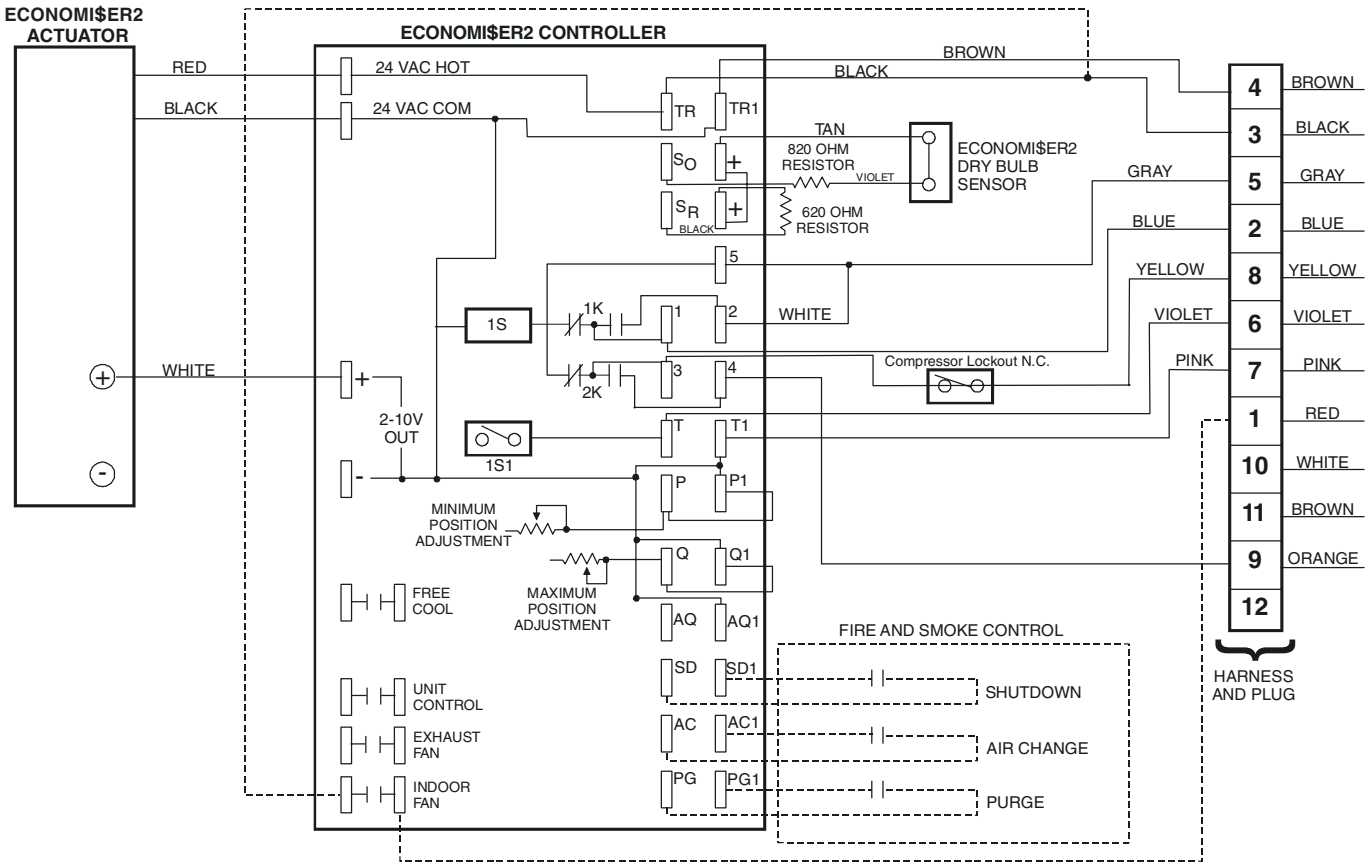


Fig. 37 — Fire and Smoke Control Wiring

Table 4 — CO_2 Sensor Standard Settings

SETTING	EQUIPMENT	OUTPUT	VENTILATION RATE (cfm/Person)	ANALOG OUTPUT	CO_2 CONTROL RANGE (ppm)	OPTIONAL RELAY SETPOINT (ppm)	RELAY HYSTERESIS (ppm)
1	Interface w/Standard Building Control System	Proportional	Any	0-10V 4-20 mA	0-2000	1000	50
2		Proportional	Any	2-10V 7-20 mA	0-2000	1000	50
3		Exponential	Any	0-10V 4-20 mA	0-2000	1100	50
4	Economizer	Proportional	15	0-10V 4-20 mA	0-1100	1100	50
5		Proportional	20	0-10V 4-20 mA	0- 900	900	50
6		Exponential	15	0-10V 4-20 mA	0-1100	1100	50
7		Exponential	20	0-10V 4-20 mA	0- 900	900	50
8	Health & Safety	Proportional	—	0-10V 4-20 mA	0-9999	5000	500
9	Parking/Air Intakes/ Loading Docks	Proportional	—	0-10V 4-20 mA	0-2000	700	50

LEGEND

PPM — Parts Per Million

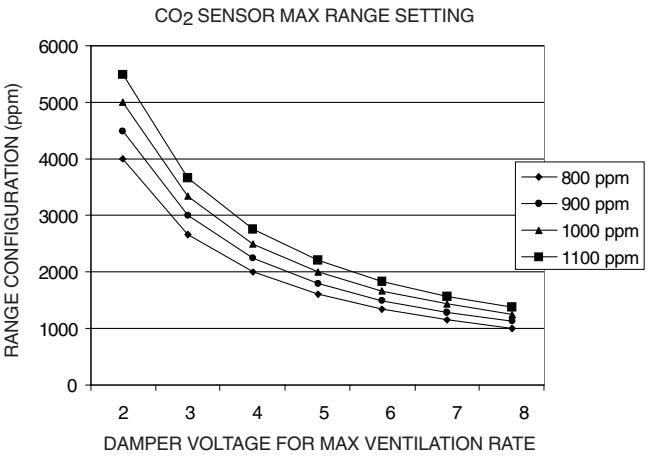


Fig. 38 — CO_2 Sensor Maximum Range Setting

3. Use the Up/Down button to toggle to the NONSTD menu and press Enter.
4. Use the Up/Down button to toggle through each of the nine variables, starting with Altitude, until the desired setting is reached.
5. Press Mode to move through the variables.
6. Press Enter to lock in the selection, then press Mode to continue to the next variable.

Dehumidification of Fresh Air with DCV Control — Information from ASHRAE indicates that the largest humidity load on any zone is the fresh air introduced. For some applications, a device such as a 62AQ energy recovery unit is added to reduce the moisture content of the fresh air being brought into the building when the enthalpy is high. In most cases, the normal heating and cooling processes are more than adequate to remove the humidity loads for most commercial applications.

This makes the control of the of the dehumidification device simple when using the enthalpy or differential enthalpy sensor. The enthalpy sensor or differential enthalpy sensor is installed on the equipment to determine economizer operation. The high enthalpy signal from the enthalpy sensor or differential enthalpy sensor can be used to turn on the outdoor air moisture removal device any time fresh air is required for the space.

The energy recovery device should be sized for maximum latent and sensible conditioning at maximum ventilation on a design day. A calculation for leaving-air temperature on a low ambient, low ventilation day should also be done to determine the mixed-air temperature of the return and pre-conditioned outside air. The design should produce air temperature somewhat near room conditions to prevent reheat of the air mixture. The energy recovery device should be interlocked with the heat to turn off the device when in the heat mode.

Install the Supply Air Temperature Sensor (SAT) — When the unit is supplied with a factory-mounted PremierLink™ control and an EconoMi$er2, the supply-air temperature (SAT) sensor (33ZCSENSAT) is factory-supplied and wired. The wiring is routed from the PremierLink control over the control box, through a grommet, into the fan section, down along the back side of the fan, and along the fan deck over to the supply-air opening.

The SAT probe is wire-tied to the supply-air opening (on the horizontal opening end) in its shipping position. Remove the sensor for installation. Re-position the sensor in the flange of the supply-air opening or in the supply air duct (as required by local codes). Drill or punch a $^1/_2$-in. hole in the flange or duct. Use two field-supplied, self-drilling screws to secure the sensor probe in a horizontal orientation.

NOTE: The sensor must be mounted in the discharge airsteam downstream of the cooling coil and any heating devices. Be sure the probe tip does not come in contact with any of the unit or heat surfaces.

Step 9 — Adjust Evaporator-Fan Speed — Adjust evaporator-fan speed to meet jobsite requirement.

Table 5 shows fan rpm at motor pulley settings, Table 6 shows motor efficiencies and Table 7 gives accessory static pressure drops. Table 8 shows motor performance. Refer to Fan Performance Tables 9-20 to determine fan speed settings. Fan motor pulleys are factory set for speed shown in Tables 1A and 1B.

To change fan speed:

1. Shut off unit power supply and install lockout tag.
2. Loosen belt by loosening fan motor mounting plate nuts (see Fig. 39 and 40).
3. Loosen movable pulley flange setscrew (see Fig. 41).
4. Screw movable flange toward fixed flange to increase fan speed and away from fixed flange to decrease fan speed. Increasing fan speed increases load on motor. Do not exceed maximum speed specified in Tables 1A and 1B.
5. Set movable flange at nearest flat of pulley hub and tighten setscrew (see Tables 1A and 1B for speed change for each full turn of pulley flange).

To align fan and motor pulleys:

1. Loosen fan pulley setscrews.
2. Slide fan pulley along fan shaft.
3. Make angular alignment by loosening motor from mounting plate.

To adjust belt tension (see Fig. 39 and 40):

1. Loosen fan motor mounting plate nuts.
2. *Units 008,009* — Slide motor mounting plate away from fan scroll for proper belt tension ($^1/_2$-in. deflection with 8 to 10 lbs of force) and tighten mounting nuts (see Fig. 39).

   Units 012,014 — Slide motor mounting plate downward to tighten belt tension. Secure motor mounting plate nuts. See Fig. 40. ($^1/_2$-in. deflection with 5 to 10 lbs of force.)
3. Adjust bolt and nut on mounting plate to secure motor in fixed position.

Realign fan and motor pulleys:

1. Loosen fan pulley setscrews.
2. Slide fan pulley along fan shaft.
3. Make angular alignment by loosening motor from mounting plate.

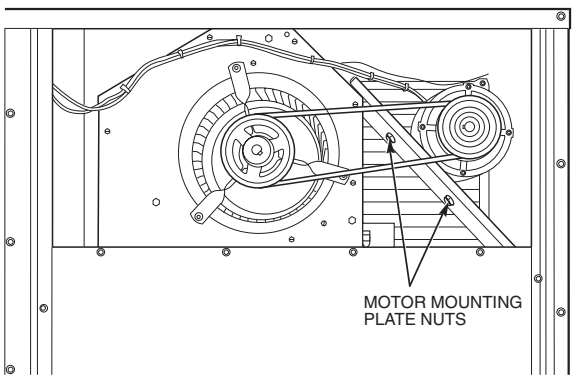


Fig. 39 — Typical Belt-Drive Motor Mounting for Sizes 008,009

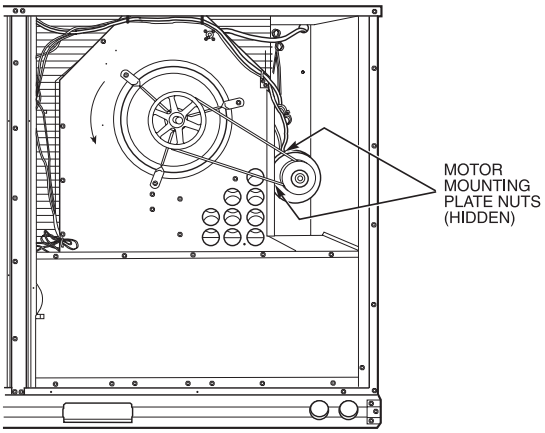


Fig. 40 — Typical Belt-Drive Motor Mounting for Sizes 012,014

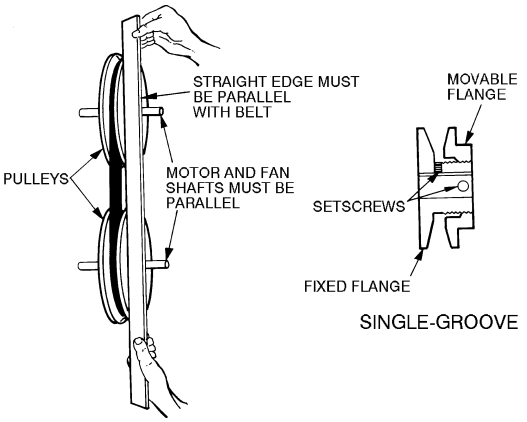


Fig. 41 — Evaporator-Fan Pulley Adjustment

Table 5 — Fan Rpm at Motor Pulley Settings*

UNIT 48TF,TM	MOTOR PULLEY TURNS OPEN												
	0	1/2	1	1 1/2	2	2 1/2	3	3 1/2	4	4 1/2	5	5 1/2	6
008†	840	815	790	765	740	715	690	665	635	615	590	—	—
008**	935	910	885	860	835	810	785	760	735	710	685	—	—
008††	1080	1025	1007	988	970	952	933	915	897	878	860	—	—
009†	935	910	885	860	835	810	785	760	735	710	685	—	—
009††	1080	1025	1007	988	970	952	933	915	897	878	860	—	—
012†	935	910	885	860	835	810	785	760	735	710	685	—	—
012***	1085	1060	1035	1010	985	960	935	910	885	860	835	—	—
012††	1130	1112	1087	1062	1037	1012	987	962	937	912	887	862	830
014†	1080	1060	1035	1015	990	970	950	925	905	880	860	—	—
014***	1260	1220	1185	1155	1130	1100	1075	1045	1015	990	960	930	900

*Approximate fan rpm shown.
†Indicates standard motor and drive package.
**Indicates alternate drive package only.
††Indicates high-static motor and drive package.
***Indicates alternate motor and drive package.

Table 6 — Evaporator Fan Motor Efficiency

UNIT 48TF,TM	MOTOR EFFICIENCY (%)
008-012	80
014	87

Table 7 — Accessory/FIOP Pressure* (in. wg)

COMPONENT	CFM									
	2200	2500	3000	3500	4000	4500	5000	5500	6000	6250
Durablade Economizer	0.02	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09	0.09
Vertical EconoMi$er2	0.06	0.075	0.115	0.15	0.195	0.25	0.325	—	—	—
Horizontal EconoMi$er2	—	0.10	0.15	0.21	0.275	0.34	—	—	—	—

LEGEND

FIOP — Factory-Installed Option

*The static pressure must be added to external static pressure. The sum and the evaporator entering-air cfm should then be used in conjunction with the Fan Performance tables to determine blower rpm and watts.

Table 8 — Evaporator-Fan Motor Performance

UNIT 48TF,TM	EVAPORATOR-FAN MOTOR	UNIT VOLTAGE	MAXIMUM ACCEPTABLE CONTINUOUS BHP*	MAXIMUM ACCEPTABLE OPERATING WATTS	MAXIMUM AMP DRAW
008	Standard, Alternate	208/230	2.40	2120	6.7
		460			3.0
		575			3.0
	High Static	208/230	3.70	3313	12.2
		460			5.5
		575			5.5
009	Standard	208/230	2.40	2120	6.7
		460			3.0
		575			3.0
	High Static	208/230	3.70	3313	12.2
		460			5.5
		575			5.5
012	Standard	208/230	2.40	2120	6.7
		460			3.0
		575			3.0
	Alternate	208/230	2.90	2615	8.6
		460			3.9
		575			3.9
	High Static	208/230	5.25	4400	17.3
		460			8.5
		575			8.5
014	Standard	208/230	3.70	3313	12.2
		460			5.5
		575			5.5
	Alternate	208/230	5.25	4400	17.3
		460			8.5
		575			8.5

LEGEND

BHP — Brake Horsepower

*Extensive motor and electrical testing on these units ensures that the full horsepower range of the motors can be utilized with confidence. Using your fan motors up to the horsepower ratings shown in this table will not result in nuisance tripping or premature motor failure. Unit warranty will not be affected.

†Single phase/three-phase.

NOTES:

1. All indoor-fan motors 5 hp and larger meet the minimum efficiency requirements as established by the Energy Policy Act of 1992 (EPACT) effective October 24, 1997.
2. High-static motor not available on single-phase units.

Table 9 — Fan Performance, 48TF,TM008 Standard Motor and Drive and Alternate Drive — Vertical Discharge Units*

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)														
	0.2			0.4			0.6			0.8			1.0		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2250	**514**	**0.55**	**562**	593	0.76	723	662	0.99	907	724	1.22	1097	781	1.78	1318
2300	**521**	**0.57**	**577**	600	0.79	747	668	1.02	932	730	1.26	1131	786	1.50	1335
2400	**536**	**0.63**	**623**	613	0.85	795	680	1.09	989	741	1.34	1199	796	1.59	1413
2500	**551**	**0.69**	**669**	626	0.93	859	693	1.17	1056	753	1.43	1275	808	1.69	1499
2550	**559**	**0.72**	**692**	634	0.97	891	700	1.21	1089	759	1.48	1318	814	1.74	1543
2600	**567**	**0.75**	**716**	641	1.00	916	706	1.25	1123	764	1.52	1353	819	1.79	1587
2700	**582**	**0.83**	**779**	655	1.08	981	719	1.34	1199	776	1.61	1430	831	1.89	1674
2800	598	0.90	835	670	1.17	1056	732	1.43	1275	789	1.71	1517	**842**	**2.00**	**1770**
2900	614	0.98	899	684	1.25	1123	745	1.53	1361	802	1.81	1604	**854**	**2.11**	**1866**
3000	630	1.07	973	699	1.35	1207	759	1.63	1447	815	1.92	1700	**866**	**2.23**	**1971**
3100	646	1.16	1047	714	1.45	1292	773	1.74	1543	828	2.04	1805	**878**	**2.35**	**2076**
3200	662	1.26	1131	729	1.55	1378	787	1.86	1648	**841**	**2.16**	**1910**	891	2.48	2188
3300	679	1.36	1216	744	1.66	1473	801	1.98	1753	**854**	**2.29**	**2023**	904	2.61	2300
3400	695	1.47	1310	759	1.78	1578	816	2.10	1858	867	2.42	2136	917	2.75	2420
3500	712	1.59	1413	774	1.90	1683	830	2.23	1971	881	2.56	2257	930	2.90	2546
3600	729	1.71	1517	790	2.03	1796	**845**	**2.37**	**2093**	895	2.71	2386	943	3.05	2670
3700	745	1.84	1630	805	2.17	1919	860	2.52	2223	909	2.87	2521	956	3.22	2807
3750	754	1.91	1691	813	2.24	1980	868	2.59	2283	917	2.95	2587	963	3.30	2870

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)								
	1.2			1.4			1.6		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2250	841	1.81	1604	**902**	**2.25**	**1989**	939	2.60	2292
2300	**843**	**1.83**	**1621**	**905**	**2.28**	**2015**	943	2.62	2309
2400	**849**	**1.88**	**1665**	**910**	**2.31**	**2041**	952	2.74	2411
2500	**859**	**1.96**	**1735**	**912**	**2.31**	**2050**	963	2.81	2470
2550	**864**	**2.01**	**1779**	**915**	**2.34**	**2067**	968	2.81	2479
2600	**869**	**2.06**	**1823**	**918**	**2.37**	**2093**	973	2.81	2487
2700	**880**	**2.17**	**1919**	927	2.47	2180	976	2.84	2495
2800	**892**	**2.29**	**2023**	938	2.58	2275	983	2.92	2562
2900	903	2.42	2136	949	2.71	2386	993	3.03	2653
3000	915	2.54	2240	961	2.85	2504	1003	3.17	2767
3100	926	2.67	2352	972	3.00	2629	1015	3.32	2886
3200	938	2.81	2470	983	3.14	2743	1026	3.47	3002
3300	950	2.95	2587	995	3.30	2870	—	—	—
3400	963	3.10	2710	1007	3.45	2987	—	—	—
3500	976	3.25	2831	—	—	—	—	—	—
3600	988	3.41	2956	—	—	—	—	—	—
3700	—	—	—	—	—	—	—	—	—
3750	—	—	—	—	—	—	—	—	—

LEGEND

Bhp — Brake Horsepower Input to Fan
FIOP — Factory-Installed Option
Watts — Input Watts to Motor

*Standard drive range is 590 to 840 rpm. Alternate drive range is 685 to 935 rpm. All other rpms require a field-supplied drive.

NOTES:
1. **Boldface** indicates alternate or field-supplied drive is required.
2. ▒ indicates field-supplied motor and drive are required.
3. Values include losses for filters, unit casing, and wet coils. See Table 7 for accessory/FIOP static pressure information.
4. Maximum continuous bhp is 2.40 and the maximum continuous watts are 2120. Extensive motor and electrical testing on these units ensures that the full range of the motor can be utilized with confidence. Using your fan motors up to the wattage ratings shown will not result in nuisance tripping or premature motor failure. Unit warranty will not be affected. See Evaporator-Fan Motor Performance, Table 8, for additional information.
5. Use of a field-supplied motor may affect wire sizing. Contact your Carrier representative for details.
6. Interpolation is permissible. Do not extrapolate.

Table 10 — Fan Performance, 48TF,TM008 High-Static Motor — Vertical Discharge Units*

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)																	
	0.2			0.4			0.6			0.8			1.0			1.2		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2250	**514**	**0.55**	**562**	**593**	**0.76**	**723**	**662**	**0.99**	**907**	**724**	**1.22**	**1097**	**781**	**1.48**	**1318**	**841**	**1.81**	**1604**
2300	**521**	**0.57**	**577**	**600**	**0.79**	**747**	**668**	**1.02**	**932**	**730**	**1.26**	**1131**	**786**	**1.50**	**1335**	**843**	**1.83**	**1621**
2400	**536**	**0.63**	**623**	**613**	**0.85**	**795**	**680**	**1.09**	**989**	**741**	**1.34**	**1199**	**796**	**1.59**	**1413**	**849**	**1.88**	**1665**
2500	**551**	**0.69**	**689**	**626**	**0.93**	**859**	**693**	**1.17**	**1056**	**753**	**1.43**	**1275**	**808**	**1.69**	**1499**	**859**	**1.96**	**1735**
2550	**559**	**0.72**	**692**	**634**	**0.97**	**891**	**700**	**1.21**	**1089**	**759**	**1.48**	**1318**	**814**	**1.74**	**1543**	864	2.01	1779
2600	**567**	**0.75**	**716**	**641**	**1.00**	**916**	**706**	**1.25**	**1123**	**764**	**1.52**	**1353**	**819**	**1.79**	**1587**	869	2.06	1823
2700	**582**	**0.83**	**779**	**655**	**1.08**	**981**	**719**	**1.34**	**1199**	**776**	**1.61**	**1430**	**831**	**1.89**	**1674**	880	2.17	1919
2800	**598**	**0.90**	**835**	**670**	**1.17**	**1056**	**732**	**1.43**	**1275**	**789**	**1.71**	**1517**	**842**	**2.00**	**1770**	892	2.29	2023
2900	**614**	**0.98**	**899**	**684**	**1.25**	**1123**	**745**	**1.53**	**1361**	**802**	**1.81**	**1604**	**854**	**2.11**	**1866**	903	2.42	2136
3000	**630**	**1.07**	**973**	**699**	**1.35**	**1207**	**759**	**1.63**	**1447**	**815**	**1.92**	**1700**	866	2.23	1971	915	2.54	2240
3100	**646**	**1.16**	**1047**	**714**	**1.45**	**1292**	**773**	**1.74**	**1543**	**828**	**2.04**	**1805**	878	2.35	2076	926	2.67	2352
3200	**662**	**1.26**	**1131**	**729**	**1.55**	**1378**	**787**	**1.86**	**1648**	**841**	**2.16**	**1910**	891	2.48	2188	938	2.81	2470
3300	**697**	**1.36**	**1216**	**744**	**1.66**	**1473**	**801**	**1.98**	**1753**	**854**	**2.29**	**2023**	904	2.61	2300	950	2.95	2587
3400	**695**	**1.47**	**1310**	**759**	**1.79**	**1578**	**816**	**2.10**	**1858**	867	2.42	2136	917	2.75	2420	963	3.10	2710
3500	**712**	**1.59**	**1413**	**774**	**1.90**	**1683**	**830**	**2.23**	**1971**	881	2.56	2257	930	2.90	2546	976	3.25	2831
3600	**729**	**1.71**	**1517**	**780**	**2.03**	**1796**	**845**	**2.37**	**2093**	895	2.71	2386	943	3.05	2670	988	3.41	2956
3700	**745**	**1.84**	**1630**	**805**	**2.17**	**1919**	**860**	**2.52**	**2223**	909	2.87	2521	956	3.22	2807	1019	3.90	3431
3750	**754**	**1.91**	**1691**	**813**	**2.24**	**1980**	868	2.59	2283	917	2.95	2587	963	3.30	2870	1026	4.00	3517

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)											
	1.4			1.6			1.8			2.0		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2250	902	2.25	1989	939	2.60	2292	979	2.94	2589	1015	3.29	2892
2300	905	2.28	2015	943	2.62	2309	983	2.96	2609	1020	3.31	2914
2400	910	2.31	2041	952	2.74	2411	992	3.10	2729	1029	3.46	3048
2500	912	2.31	2050	963	2.81	2470	1004	3.18	2798	1041	3.55	3126
2550	915	2.34	2067	968	2.81	2479	1009	3.18	2798	1047	3.55	3126
2600	918	2.37	2093	973	2.81	2487	1014	3.18	2798	1052	3.55	3126
2700	927	2.47	2180	976	2.84	2495	1017	3.21	2828	1055	3.59	3159
2800	938	2.58	2275	983	2.92	2562	1024	3.30	2908	1063	3.69	3248
2900	949	2.71	2386	993	3.03	2653	1035	3.43	3017	1074	3.83	3370
3000	961	2.85	2504	1003	3.17	2767	1045	3.59	3157	1084	4.01	3526
3100	972	3.00	2629	1015	3.32	2886	1058	3.76	3306	1097	4.20	3693
3200	983	3.14	2743	1026	3.47	3002	1069	3.83	3456	—	—	—
3300	995	3.30	2870	1043	3.80	3341	—	—	—	—	—	—
3400	1007	3.45	2987	1056	3.97	3493	—	—	—	—	—	—
3500	1030	3.82	3362	—	—	—	—	—	—	—	—	—
3600	1043	4.01	3528	—	—	—	—	—	—	—	—	—
3700	—	—	—	—	—	—	—	—	—	—	—	—
3750	—	—	—	—	—	—	—	—	—	—	—	—

LEGEND

Bhp — Brake Horsepower Input to Fan
FIOP — Factory-Installed Option
Watts — Input Watts to Motor

*Motor drive range is 860 to 1080 rpm. All other rpms require a field-supplied drive.

NOTES:
1. **Boldface** indicates field-supplied drive is required.
2. ▭ indicates field-supplied motor and drive are required.
3. Values include losses for filters, unit casing, and wet coils. See Table 7 for accessory/FIOP static pressure information.
4. Maximum continuous bhp is 3.7 and the maximum continuous watts are 3313. Extensive motor and electrical testing on these units ensures that the full range of the motor can be utilized with confidence. Using your fan motors up to the wattage ratings shown will not result in nuisance tripping or premature motor failure. Unit warranty will not be affected. See Evaporator-Fan Motor Performance, Table 8, for additional information.
5. Use of a field-supplied motor may affect wire sizing. Contact your Carrier representative for details.
6. Interpolation is permissible. Do not extrapolate.

Table 11 — Fan Performance, 48TF,TM009 Standard Motor — Vertical Discharge Units*

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)														
	0.2			0.4			0.6			0.8			1.0		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2550	**559**	**0.72**	**692**	**634**	**0.97**	**891**	700	1.21	1089	759	1.48	1318	814	1.74	1543
2600	**567**	**0.75**	**716**	**641**	**1.00**	**916**	706	1.25	1123	764	1.52	1353	819	1.79	1587
2700	**582**	**0.83**	**779**	**655**	**1.08**	**981**	719	1.34	1199	776	1.61	1430	831	1.89	1674
2800	**598**	**0.90**	**835**	**670**	**1.17**	**1056**	732	1.43	1275	789	1.71	1517	842	2.00	1770
2900	**614**	**0.98**	**899**	**684**	**1.25**	**1123**	745	1.53	1361	802	1.81	1604	854	2.11	1866
3000	**630**	**1.07**	**973**	690	1.35	1207	759	1.63	1447	815	1.92	1700	866	2.23	1971
3100	**646**	**1.16**	**1047**	714	1.45	1292	773	1.74	1543	828	2.04	1805	878	2.35	2076
3200	**662**	**1.26**	**1131**	729	1.55	1378	787	1.86	1648	841	2.16	1910	891	2.48	2188
3300	**679**	**1.36**	**1216**	744	1.66	1473	801	1.98	1753	854	2.29	2023	904	2.61	2300
3400	695	1.47	1310	759	1.78	1578	816	2.10	1858	867	2.42	2136	917	2.75	2420
3500	712	1.59	1413	774	1.90	1683	830	2.23	1971	881	2.56	2257	930	2.90	2546
3600	729	1.71	1517	790	2.03	1796	845	2.37	2093	895	2.71	2386	943	3.05	2670
3700	745	1.84	1630	805	2.17	1919	860	2.52	2223	909	2.87	2521	956	3.22	2807
3750	754	1.91	1691	813	2.24	1980	868	2.59	2283	917	2.95	2587	963	3.30	2870
3800	762	1.98	1753	821	2.31	2041	875	2.66	2343	924	3.03	2653	970	3.38	2933
3900	779	2.12	1875	836	2.46	2171	890	2.82	2479	938	3.19	2783	—	—	—
4000	796	2.27	2006	852	2.61	2300	905	2.98	2612	953	3.37	2925	—	—	—
4100	813	2.42	2136	868	2.78	2445	920	3.15	2751	—	—	—	—	—	—
4200	830	2.59	2283	884	2.95	2587	935	3.33	2894	—	—	—	—	—	—
4250	839	2.68	2360	890	3.04	2661	—	—	—	—	—	—	—	—	—

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)								
	1.2			1.4			1.6		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2550	864	2.01	1779	915	2.34	2067	968	2.81	2479
2600	869	2.06	1823	918	2.37	2093	973	2.81	2487
2700	880	2.17	1919	927	2.47	2180	976	2.84	2495
2800	892	2.29	2023	938	2.58	2275	983	2.92	2562
2900	903	2.42	2136	949	2.71	2386	993	3.03	2653
3000	915	2.54	2240	961	2.85	2504	1003	3.17	2767
3100	926	2.67	2352	972	3.00	2629	1016	3.32	2886
3200	938	2.81	2470	983	3.14	2743	1026	3.47	3002
3300	950	2.95	2587	995	3.30	2870	—	—	—
3400	963	3.10	2710	1007	3.45	2987	—	—	—
3500	976	3.25	2831	—	—	—	—	—	—
3600	988	3.41	2956	—	—	—	—	—	—
3700	—	—	—	—	—	—	—	—	—
3750	—	—	—	—	—	—	—	—	—
3800	—	—	—	—	—	—	—	—	—
3900	—	—	—	—	—	—	—	—	—
4000	—	—	—	—	—	—	—	—	—
4100	—	—	—	—	—	—	—	—	—
4200	—	—	—	—	—	—	—	—	—
4250	—	—	—	—	—	—	—	—	—

LEGEND

Bhp — Brake Horsepower Input to Fan
FIOP — Factory-Installed Option
Watts — Input Watts to Motor

*Motor drive range is 685 to 935 rpm. All other rpms require a field-supplied drive.

NOTES:
1. **Boldface** indicates field-supplied drive is required.
2. ▭ indicates field-supplied motor and drive are required.
3. Values include losses for filters, unit casing, and wet coils. See Table 7 for accessory/FIOP static pressure information.
4. Maximum continuous bhp is 2.4 and the maximum continuous watts are 2120. Extensive motor and electrical testing on these units ensures that the full range of the motor can be utilized with confidence. Using your fan motors up to the wattage ratings shown will not result in nuisance tripping or premature motor failure. Unit warranty will not be affected. See Evaporator-Fan Motor Performance, Table 8, for additional information.
5. Use of a field-supplied motor may affect wire sizing. Contact your Carrier representative for details.
6. Interpolation is permissible. Do not extrapolate.

Table 12 — Fan Performance, 48TF,TM009 High-Static Motor — Vertical Discharge Units*

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)														
	0.2			0.4			0.6			0.8			1.0		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2550	**559**	**0.72**	**692**	**634**	**0.97**	**891**	**700**	**1.21**	**1089**	**759**	**1.48**	**1318**	**814**	**1.74**	**1543**
2600	**567**	**0.75**	**716**	**641**	**1.00**	**916**	**706**	**1.25**	**1123**	**764**	**1.52**	**1353**	**819**	**1.79**	**1587**
2700	**582**	**0.83**	**779**	**655**	**1.08**	**981**	**719**	**1.34**	**1199**	**776**	**1.61**	**1430**	**831**	**1.89**	**1674**
2800	**598**	**0.90**	**835**	**670**	**1.17**	**1056**	**732**	**1.43**	**1275**	**789**	**1.71**	**1517**	**842**	**2.00**	**1770**
2900	**614**	**0.98**	**899**	**684**	**1.25**	**1123**	**745**	**1.53**	**1361**	**802**	**1.81**	**1604**	**854**	**2.11**	**1866**
3000	**630**	**1.07**	**973**	**690**	**1.35**	**1207**	**759**	**1.63**	**1447**	**816**	**1.92**	**1700**	866	2.23	1971
3100	**646**	**1.16**	**1047**	**714**	**1.45**	**1292**	**773**	**1.74**	**1543**	**828**	**2.04**	**1805**	878	2.35	2076
3200	**662**	**1.26**	**1131**	**729**	**1.55**	**1378**	**787**	**1.86**	**1648**	**841**	**2.18**	**1910**	891	2.48	2188
3300	**679**	**1.36**	**1216**	**744**	**1.66**	**1473**	**801**	**1.98**	**1753**	**854**	**2.29**	**2023**	904	2.61	2300
3400	**695**	**1.47**	**1310**	**759**	**1.76**	**1578**	**816**	**2.10**	**1858**	867	2.42	2136	917	2.75	2420
3500	**712**	**1.59**	**1413**	**744**	**1.90**	**1683**	**830**	**2.23**	**1971**	881	2.56	2257	930	2.90	2546
3600	**729**	**1.71**	**1517**	**790**	**2.03**	**1796**	**845**	**2.37**	**2093**	895	2.71	2386	943	3.05	2670
3700	**745**	**1.84**	**1630**	**805**	**2.17**	**1919**	**860**	**2.52**	**2223**	909	2.87	2521	956	3.22	2807
3750	**754**	**1.91**	**1691**	**813**	**2.24**	**1980**	868	2.59	2283	917	2.95	2587	963	3.30	2870
3800	**762**	**1.98**	**1753**	**821**	**2.31**	**2041**	875	2.66	2343	924	3.03	2653	970	3.38	2933
3900	**779**	**2.12**	**1875**	**836**	**2.46**	**2171**	890	2.82	2479	938	3.19	2783	981	3.65	3209
4000	**796**	**2.27**	**2006**	**852**	**2.61**	**2300**	905	2.98	2612	953	3.37	2925	996	3.85	3390
4100	**813**	**2.42**	**2136**	868	2.78	2445	920	3.15	2751	974	3.74	3294	—	—	—
4200	**830**	**2.59**	**2283**	884	2.95	2587	935	3.33	2894	990	3.96	3482	—	—	—
4250	**839**	**2.68**	**2360**	890	3.04	2661	965	3.88	3412	—	—	—	—	—	—

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)														
	1.2			1.4			1.6			1.8			2.0		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2550	864	2.01	1779	915	2.34	2067	968	2.81	2479	991	3.02	2654	1012	3.21	2827
2600	869	2.06	1823	918	2.37	2093	973	2.81	2487	996	3.02	2654	1017	3.21	2827
2700	880	2.17	1919	927	2.47	2180	976	2.84	2495	999	3.05	2682	1021	3.25	2857
2800	892	2.29	2023	938	2.58	2275	983	2.92	2562	1006	3.13	2758	1028	3.34	2938
2900	903	2.42	2136	949	2.71	2386	993	3.03	2653	1017	3.25	2862	1038	3.46	3048
3000	915	2.54	2240	961	2.85	2504	1003	3.17	2767	1027	3.40	2994	1049	3.62	3189
3100	926	2.67	2352	972	3.00	2629	1016	3.32	2886	1040	3.65	3136	1062	3.80	3340
3200	938	2.81	2470	983	3.14	2743	1026	3.47	3002	1050	3.72	3277	1073	3.97	3491
3300	950	2.95	2587	995	3.30	2870	1022	3.58	3146	1046	3.84	3377	1069	4.09	3597
3400	963	3.10	2710	1007	3.45	2987	1034	3.74	3289	1059	4.01	3530	—	—	—
3500	976	3.25	2831	1007	3.56	3137	1034	3.86	3399	1058	4.15	3648	—	—	—
3600	988	3.41	2956	1019	3.74	3292	1047	4.05	3566	—	—	—	—	—	—
3700	992	3.59	3161	1023	3.94	3467	—	—	—	—	—	—	—	—	—
3750	999	3.68	3240	1030	4.04	3554	—	—	—	—	—	—	—	—	—
3800	1006	3.77	3318	1038	4.14	3640	—	—	—	—	—	—	—	—	—
3900	1017	4.07	3580	—	—	—	—	—	—	—	—	—	—	—	—
4000	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
4100	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
4200	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
4250	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

LEGEND

Bhp — Brake Horsepower Input to Fan
FIOP — Factory-Installed Option
Watts — Input Watts to Motor

*Motor drive range is 860 to 1080 rpm. All other rpms require a field-supplied drive.

NOTES:
1. **Boldface** indicates field-supplied drive is required.
2. ▭ indicates field-supplied motor and drive are required.
3. Values include losses for filters, unit casing, and wet coils. See Table 7 for accessory/FIOP static pressure information.
4. Maximum continuous bhp is 3.7 and the maximum continuous watts are 3313. Extensive motor and electrical testing on these units ensures that the full range of the motor can be utilized with confidence. Using your fan motors up to the wattage ratings shown will not result in nuisance tripping or premature motor failure. Unit warranty will not be affected. See Evaporator-Fan Motor Performance, Table 8, for additional information.
5. Use of a field-supplied motor may affect wire sizing. Contact your Carrier representative for details.
6. Interpolation is permissible. Do not extrapolate.

Table 13 — Fan Performance, 48TF,TM012 — Vertical Discharge Units*

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)																	
	0.2			0.4			0.6			0.8			1.0			1.2		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
3000	592	0.76	723	661	0.93	859	722	1.09	989	779	1.26	1131	829	1.42	1267	880	1.58	1404
3100	607	0.83	779	676	1.01	924	734	1.17	1056	791	1.34	1199	840	1.51	1344	890	1.68	1491
3200	622	0.90	835	690	1.09	989	746	1.25	1123	803	1.43	1275	852	1.60	1422	900	1.77	1569
3300	638	0.98	899	705	1.17	1056	759	1.33	1190	815	1.52	1353	864	1.70	1508	910	1.88	1665
3400	653	1.06	965	719	1.26	1131	772	1.43	1275	826	1.62	1439	876	1.81	1604	921	1.98	1753
3500	669	1.15	1039	733	1.35	1207	786	1.53	1361	838	1.72	1526	888	1.91	1691	933	2.10	1858
3600	684	1.24	1114	747	1.44	1284	800	1.64	1456	850	1.82	1613	900	2.03	1796	945	2.22	2014
3700	700	1.33	1190	760	1.54	1370	814	1.75	1552	863	1.92	1700	912	2.14	1892	957	2.34	2117
3800	715	1.43	1275	774	1.64	1456	828	1.86	1648	875	2.04	1805	924	2.26	1997	969	2.47	2230
3900	731	1.54	1370	787	1.74	1543	843	1.98	1753	888	2.16	1910	936	2.38	2151	981	2.60	2344
4000	747	1.64	1456	801	1.85	1639	857	2.10	1858	902	2.30	2032	948	2.51	2265	993	2.74	2469
4100	763	1.76	1560	816	1.97	1744	872	2.23	1971	916	2.44	2203	960	2.64	2380	1005	2.88	2596
4200	778	1.88	1665	831	2.10	1884	886	2.36	2084	929	2.58	2326	972	2.78	2505	**1016**	**3.03**	**2735**
4300	794	2.00	1770	846	2.23	1971	900	2.50	2256	943	2.73	2460	**985**	**2.93**	**2642**	**1028**	**3.17**	**2866**
4400	810	2.13	1884	861	2.37	2093	913	2.64	2380	958	2.89	2605	**999**	**3.09**	**2791**	**1040**	**3.32**	**3010**
4500	826	2.27	2006	876	2.52	2273	927	2.78	2505	**973**	**3.04**	**2744**	**1012**	**3.26**	**2952**	—	—	—
4600	842	2.41	2177	892	2.67	2406	**940**	**2.92**	**2633**	**987**	**3.21**	**2904**	—	—	—	—	—	—
4700	858	2.55	2300	907	2.83	2551	**954**	**3.08**	**2782**	**1002**	**3.38**	**3068**	—	—	—	—	—	—
4800	874	2.70	2433	**922**	**2.99**	**2698**	**968**	**3.24**	**2933**	—	—	—	—	—	—	—	—	—
4900	890	2.86	2578	**938**	**3.16**	**2857**	—	—	—	—	—	—	—	—	—	—	—	—
5000	**906**	**3.03**	**2735**	**953**	**3.33**	**3020**	—	—	—	—	—	—	—	—	—	—	—	—

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)											
	1.4			1.6			1.8			2.0		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
3000	924	1.73	1534	970	1.89	1736	1019	2.00	1828	1066	2.30	2082
3100	935	1.84	1695	977	2.00	1828	1026	2.17	1971	1070	2.44	2203
3200	946	1.95	1786	987	2.11	1920	1029	2.28	2065	1075	2.51	2265
3300	957	2.06	1878	998	2.23	2022	1037	2.40	2169	1082	2.58	2326
3400	967	2.17	1971	1009	2.35	2125	1047	2.53	2282	1087	2.70	2433
3500	976	2.29	2074	1020	2.48	2238	1058	2.66	2397	1095	2.84	2560
3600	986	2.41	2177	1030	2.61	2353	1069	2.80	2523	**1106**	**2.98**	**2688**
3700	998	2.54	2291	1039	2.74	2469	**1081**	**2.94**	**2651**	**1117**	**3.13**	**2829**
3800	1010	2.67	2406	1049	2.87	2587	**1091**	**3.08**	**2782**	**1128**	**3.29**	**2981**
3900	1022	2.81	2533	**1060**	**3.02**	**2726**	**1100**	**3.23**	**2923**	—	—	—
4000	**1034**	**2.96**	**2670**	**1072**	**3.17**	**2866**	**1110**	**3.38**	**3068**	—	—	—
4100	**1046**	**3.11**	**2810**	**1084**	**3.32**	**3010**	—	—	—	—	—	—
4200	**1058**	**3.26**	**2952**	—	—	—	—	—	—	—	—	—
4300	—	—	—	—	—	—	—	—	—	—	—	—
4400	—	—	—	—	—	—	—	—	—	—	—	—
4500	—	—	—	—	—	—	—	—	—	—	—	—
4600	—	—	—	—	—	—	—	—	—	—	—	—
4700	—	—	—	—	—	—	—	—	—	—	—	—
4800	—	—	—	—	—	—	—	—	—	—	—	—
4900	—	—	—	—	—	—	—	—	—	—	—	—
5000	—	—	—	—	—	—	—	—	—	—	—	—

LEGEND

Bhp — Brake Horsepower Input to Fan
FIOP — Factory-Installed Option
Watts — Input Watts to Motor

*Standard drive range is 685 to 935 rpm. Alternate motor drive range is 835 to 1085 rpm. High-static motor drive range is 830 to 1130 rpm. All other rpms require a field-supplied drive.

NOTES:
1. **Boldface** indicates optional high-static motor and drive are required.
2. Values include losses for filters, unit casing, and wet coils. See Table 7 for accessory/FIOP static pressure information.
3. Maximum continuous bhp is 2.4 for the standard motor, 2.9 for the alternate motor and 5.25 for the high-static motor. Maximum continuous watts are 2120 for the standard motor, 2615 for the alternate motor and 4400 for the high-static motor. Extensive motor and electrical testing on these units ensures that the full range of the motor can be utilized with confidence. Using your fan motors up to the wattage ratings shown will not result in nuisance tripping or premature motor failure. Unit warranty will not be affected. See Evaporator-Fan Motor Performance, Table 8, for additional information.
4. Use of a field-supplied motor may affect wire sizing. Contact your Carrier representative for details.
5. Interpolation is permissible. Do not extrapolate.

Table 14 — Fan Performance, 48TF,TM014 — Vertical Discharge Units*

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)																	
	0.2			0.4			0.6			0.8			1.0			1.2		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
3750	737	1.41	1300	798	1.64	1486	854	1.84	1658	909	2.12	1893	961	2.35	2097	1014	2.61	2325
3800	745	1.46	1340	805	1.69	1531	861	1.89	1700	915	2.17	1940	967	2.41	2149	1019	2.67	2378
3900	761	1.56	1423	820	1.80	1624	875	2.01	1802	928	2.29	2044	979	2.55	2272	1029	2.80	2494
4000	777	1.67	1514	836	1.92	1725	889	2.14	1914	941	2.40	2140	991	2.68	2387	1040	2.94	2620
4100	793	1.79	1615	851	2.05	1836	904	2.27	2027	955	2.52	2246	1004	2.82	2512	1052	3.08	2746
4200	810	1.91	1717	867	2.18	1948	918	2.41	2149	968	2.65	2361	1017	2.96	2638	1064	3.23	2882
4300	826	2.04	1828	883	2.32	2070	933	2.55	2272	982	2.79	2485	1030	3.11	2773	1076	3.40	3037
4400	842	2.17	1940	898	2.46	2193	948	2.70	2405	996	2.93	2611	1043	3.25	2901	1088	3.56	3184
4500	859	2.31	2061	914	2.60	2316	962	2.85	2539	1010	3.09	2755	1056	3.40	3037	**1101**	**3.73**	**3341**
4600	876	2.45	2184	930	2.76	2459	977	3.01	2683	1024	3.26	2910	1070	3.55	3175	**1114**	**3.90**	**3498**
4700	892	2.60	2316	945	2.91	2593	992	3.18	2837	1039	3.43	3065	**1083**	**3.71**	**3322**	**1126**	**4.07**	**3655**
4800	909	2.77	2468	961	3.07	2737	1008	3.36	3001	1053	3.61	3230	**1097**	**3.88**	**3479**	**1140**	**4.25**	**3822**
4900	926	2.93	2611	977	3.24	2891	1024	3.54	3166	**1068**	**3.80**	**3405**	**1111**	**4.06**	**3646**	**1153**	**4.41**	**3971**
5000	942	3.11	2773	993	3.41	3047	**1039**	**3.73**	**3341**	**1080**	**3.99**	**3581**	**1125**	**4.25**	**3822**	**1166**	**4.59**	**4139**
5100	959	3.29	2937	1009	3.60	3221	**1055**	**3.92**	**3516**	**1097**	**4.19**	**3767**	**1139**	**4.46**	**4018**	**1180**	**4.78**	**4316**
5200	976	3.47	3101	**1025**	**3.78**	**3387**	**1071**	**4.12**	**3702**	**1112**	**4.40**	**3962**	**1153**	**4.67**	**4214**	**1194**	**4.98**	**4503**
5300	993	3.67	3285	**1041**	**3.98**	**3572**	**1086**	**4.33**	**3897**	**1127**	**4.61**	**4158**	**1168**	**4.90**	**4428**	**1208**	**5.19**	**4698**
5400	**1010**	**3.87**	**3470**	**1057**	**4.18**	**3757**	**1102**	**4.54**	**4093**	**1142**	**4.84**	**4372**	**1182**	**5.13**	**4642**	—	—	—
5500	**1027**	**4.07**	**3655**	**1073**	**4.39**	**3953**	**1118**	**4.76**	**4298**	**1157**	**5.07**	**4586**	—	—	—	—	—	—
5600	**1043**	**4.29**	**3860**	**1090**	**4.61**	**4158**	**1133**	**4.99**	**4512**	—	—	—	—	—	—	—	—	—
5700	**1060**	**4.51**	**4065**	**1106**	**4.83**	**4363**	**1149**	**5.22**	**4726**	—	—	—	—	—	—	—	—	—
5800	**1077**	**4.74**	**4279**	**1122**	**5.07**	**4586**	—	—	—	—	—	—	—	—	—	—	—	—
5900	**1094**	**4.98**	**4503**	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
6000	**1111**	**5.22**	**4726**	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
6100	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
6250	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)											
	1.4			1.6			1.8			2.0		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
3750	1065	2.87	2557	1113	3.12	2783	1158	3.36	2997	1197	3.56	3180
3800	1070	2.94	2620	1118	3.19	2846	1163	3.44	3074	1203	3.65	3267
3900	1079	3.07	2737	1128	3.34	2983	1173	3.60	3221	**1214**	**3.83**	**3433**
4000	1089	3.22	2873	1137	3.49	3120	**1183**	**3.76**	**3368**	**1225**	**4.00**	**3590**
4100	1100	3.36	3001	1147	3.65	3267	**1193**	**3.93**	**3525**	**1236**	**4.19**	**3767**
4200	1110	3.51	3138	**1157**	**3.81**	**3414**	**1202**	**4.09**	**3674**	**1245**	**4.38**	**3943**
4300	1121	3.67	3285	**1167**	**3.97**	**3562**	**1212**	**4.27**	**3841**	**1255**	**4.56**	**4111**
4400	**1133**	**3.84**	**3442**	**1178**	**4.14**	**3720**	**1222**	**4.44**	**3999**	**1265**	**4.74**	**4279**
4500	**1144**	**4.00**	**3590**	**1188**	**4.31**	**3878**	**1232**	**4.62**	**4167**	**1274**	**4.93**	**4456**
4600	**1157**	**4.19**	**3767**	**1199**	**4.49**	**4046**	**1242**	**4.81**	**4344**	**1284**	**5.13**	**4642**
4700	**1169**	**4.38**	**3943**	**1210**	**4.68**	**4223**	**1252**	**5.00**	**4521**	—	—	—
4800	**1181**	**4.58**	**4130**	**1222**	**4.87**	**4400**	**1263**	**5.20**	**4707**	—	—	—
4900	**1194**	**4.77**	**4307**	**1234**	**5.09**	**4605**	—	—	—	—	—	—
5000	**1207**	**4.97**	**4493**	—	—	—	—	—	—	—	—	—
5100	**1220**	**5.18**	**4689**	—	—	—	—	—	—	—	—	—
5200	—	—	—	—	—	—	—	—	—	—	—	—
5300	—	—	—	—	—	—	—	—	—	—	—	—
5400	—	—	—	—	—	—	—	—	—	—	—	—
5500	—	—	—	—	—	—	—	—	—	—	—	—
5600	—	—	—	—	—	—	—	—	—	—	—	—
5700	—	—	—	—	—	—	—	—	—	—	—	—
5800	—	—	—	—	—	—	—	—	—	—	—	—
5900	—	—	—	—	—	—	—	—	—	—	—	—
6000	—	—	—	—	—	—	—	—	—	—	—	—
6100	—	—	—	—	—	—	—	—	—	—	—	—
6250	—	—	—	—	—	—	—	—	—	—	—	—

LEGEND

Bhp — Brake Horsepower Input to Fan
FIOP — Factory-Installed Option
Watts — Input Watts to Motor

*Standard drive range is 860 to 1080 rpm. Alternate motor drive range is 900 to 1260 rpm. All other rpms require a field-supplied drive.

NOTES:
1. **Boldface** indicates alternate motor and drive are required.
2. Values include losses for filters, unit casing, and wet coils. See Table 7 for accessory/FIOP static pressure information.
3. Maximum continuous bhp is 3.70 for the standard motor and 5.25 for the alternate motor. Maximum continuous watts are 3313 for the standard motor and 4400 for the alternate motor. Extensive motor and electrical testing on these units ensures that the full range of the motor can be utilized with confidence. Using your fan motors up to the wattage ratings shown will not result in nuisance tripping or premature motor failure. Unit warranty will not be affected. See Evaporator-Fan Motor Performance, Table 8, for additional information.
4. Use of a field-supplied motor may affect wire sizing. Contact your Carrier representative for details.
5. Interpolation is permissible. Do not extrapolate.

Table 15 — Fan Performance, 48TF,TM008 Standard Motor and Drive and Alternate Drive — Horizontal Discharge Units*

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)																	
	0.2			0.4			0.6			0.8			0.9			1.0		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2250	**507**	**0.53**	**547**	**586**	**0.73**	**700**	658	0.97	891	722	1.22	1097	752	1.34	1199	783	1.46	1301
2300	**513**	**0.55**	**562**	592	0.76	723	663	1.00	916	727	1.26	1131	756	1.38	1224	786	1.49	1327
2400	**528**	**0.60**	**600**	606	0.83	779	674	1.06	965	738	1.34	1199	766	1.46	1301	795	1.58	1404
2500	**542**	**0.66**	**646**	619	0.90	835	686	1.13	1022	748	1.41	1258	777	1.55	1370	806	1.68	1491
2550	**550**	**0.69**	**669**	627	0.94	867	692	1.17	1056	754	1.45	1292	783	1.60	1413	812	1.74	1543
2600	**557**	**0.72**	**692**	634	0.97	891	698	1.21	1089	759	1.49	1327	787	1.64	1456	816	1.79	1587
2700	**573**	**0.79**	**747**	648	1.05	956	711	1.29	1156	770	1.58	1404	798	1.73	1534	827	1.88	1665
2800	**588**	**0.86**	**803**	662	1.13	1022	723	1.38	1233	782	1.66	1473	809	1.82	1613	837	1.98	1753
2900	604	0.94	867	676	1.21	1089	737	1.48	1318	794	1.76	1560	821	1.92	1700	848	2.08	1840
3000	620	1.02	932	690	1.30	1165	750	1.58	1404	806	1.86	1648	832	2.02	1788	849	2.18	1927
3100	636	1.11	1006	704	1.39	1241	764	1.69	1499	818	1.97	1744	844	2.13	1884	870	2.29	2023
3200	652	1.21	1089	718	1.49	1327	778	1.80	1595	831	2.09	1849	856	2.25	1980	882	2.40	2119
3300	668	1.31	1173	732	1.59	1413	793	1.92	1700	844	2.21	1954	869	2.37	2093	894	2.53	2232
3400	684	1.41	1258	747	1.70	1508	807	2.04	1805	857	2.35	2076	882	2.51	2206	907	2.66	2343
3500	701	1.53	1361	762	1.82	1613	821	2.16	1910	871	2.48	2188	895	2.64	2326	919	2.80	2462
3600	717	1.65	1465	777	1.94	1718	835	2.29	2023	885	2.63	2317	908	2.79	2453	932	2.95	2587
3700	733	1.77	1569	792	2.07	1831	849	2.42	2136	899	2.78	2445	922	2.95	2579	945	3.11	2718
3750	742	1.84	1630	800	2.14	1892	856	2.49	2197	907	2.86	2512	929	3.03	2653	952	3.20	2719

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)								
	1.2			1.4			1.6		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2250	843	1.81	1604	908	2.25	1989	955	2.59	2283
2300	846	1.84	1630	910	2.25	2015	959	2.61	2300
2400	853	1.88	1665	912	2.31	2041	967	2.68	2360
2500	859	1.94	1718	919	2.37	2093	971	2.73	2403
2550	864	1.99	1761	920	2.39	2110	974	2.76	2428
2600	868	2.04	1805	921	2.41	2136	976	2.78	2445
2700	878	2.16	1910	928	2.45	2162	983	2.88	2529
2800	889	2.29	2023	937	2.57	2266	986	2.91	2554
2900	900	2.41	2128	947	2.70	2377	993	3.01	2637
3000	910	2.52	2223	958	2.85	2504	1002	3.15	2751
3100	920	2.64	2326	968	2.99	2620	1012	3.30	2870
3200	931	2.76	2428	979	3.13	2735	1023	3.47	3002
3300	942	2.89	2537	989	3.26	2839	1034	3.63	3121
3400	954	3.02	2645	1000	3.40	2948	1044	3.79	3237
3500	966	3.15	2751	1011	3.55	3062	1054	3.94	3340
3600	978	3.30	2870	1022	3.69	3165	1065	4.10	3445
3700	990	3.45	2987	1034	3.84	3272	1076	4.26	3544
3750	997	3.54	3055	1040	3.93	3333	—	—	—

LEGEND

Bhp — Brake Horsepower Input to Fan
FIOP — Factory-Installed Option
Watts — Input Watts to Motor

*Standard drive range is 590 to 840 rpm. Alternate drive range is 685 to 935 rpm. All other rpms require a field-supplied drive.

NOTES:
1. **Boldface** indicates field-supplied drive is required.
2. ▒ indicates field-supplied motor and drive are required.
3. Values include losses for filters, unit casing, and wet coils. See Table 7 for accessory/FIOP static pressure information.
4. Maximum continuous bhp is 2.4 and maximum continuous watts are 2120. Extensive motor and electrical testing on these units ensures that the full range of the motor can be utilized with confidence. Using your fan motors up to the wattage ratings shown will not result in nuisance tripping or premature motor failure. Unit warranty will not be affected. See Evaporator-Fan Motor Performance, Table 8, for additional information.
5. Use of a field-supplied motor may affect wire sizing. Contact your Carrier representative for details.
6. Interpolation is permissible. Do not extrapolate.

Table 16 — Fan Performance, 48TF,TM008 High-Static Motor — Horizontal Discharge Units*

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)																	
	0.2			0.4			0.6			0.8			1.0			1.2		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2250	**507**	**0.53**	**547**	**586**	**0.73**	**700**	**658**	**0.97**	**891**	**722**	**1.22**	**1097**	**783**	**1.46**	**1301**	**843**	**1.81**	**1604**
2300	**513**	**0.55**	**562**	**592**	**0.76**	**723**	**663**	**1.00**	**916**	**727**	**1.26**	**1131**	**786**	**1.49**	**1327**	**846**	**1.84**	**1630**
2400	**528**	**0.60**	**600**	**606**	**0.83**	**779**	**674**	**1.06**	**965**	**738**	**1.34**	**1199**	**795**	**1.58**	**1404**	**853**	**1.88**	**1665**
2500	**542**	**0.66**	**646**	**619**	**0.90**	**835**	**686**	**1.13**	**1022**	**748**	**1.41**	**1258**	**806**	**1.68**	**1491**	**859**	**1.94**	**1718**
2550	**550**	**0.69**	**669**	**627**	**0.94**	**867**	**692**	**1.17**	**1056**	**754**	**1.45**	**1292**	**812**	**1.74**	**1543**	864	1.99	1761
2600	**557**	**0.72**	**692**	**634**	**0.97**	**891**	**698**	**1.21**	**1089**	**759**	**1.49**	**1327**	**816**	**1.79**	**1587**	868	2.04	1805
2700	**573**	**0.79**	**747**	**649**	**1.05**	**956**	**711**	**1.29**	**1156**	**770**	**1.58**	**1404**	**827**	**1.88**	**1665**	878	2.16	1910
2800	**588**	**0.86**	**803**	**662**	**1.13**	**1022**	**723**	**1.38**	**1233**	**782**	**1.66**	**1473**	**837**	**1.98**	**1753**	889	2.29	2023
2900	**604**	**0.94**	**867**	**676**	**1.21**	**1089**	**737**	**1.48**	**1318**	**794**	**1.76**	**1560**	**848**	**2.08**	**1840**	900	2.41	2128
3000	**620**	**1.02**	**932**	**690**	**1.30**	**1165**	**750**	**1.58**	**1404**	**806**	**1.86**	**1648**	**849**	**2.18**	**1927**	910	2.52	2223
3100	**636**	**1.11**	**1006**	**704**	**1.39**	**1241**	**764**	**1.69**	**1499**	**818**	**1.97**	**1744**	870	2.29	2023	920	2.64	2326
3200	**652**	**1.21**	**1089**	**718**	**1.49**	**1327**	**778**	**1.80**	**1595**	**831**	**2.09**	**1849**	882	2.40	2119	931	2.76	2428
3300	**668**	**1.31**	**1173**	**732**	**1.59**	**1413**	**793**	**1.92**	**1700**	**844**	**2.21**	**1954**	894	2.53	2232	942	2.89	2537
3400	**684**	**1.41**	**1258**	**747**	**1.70**	**1508**	**807**	**2.04**	**1805**	**857**	**2.35**	**2076**	907	2.66	2343	954	3.02	2645
3500	**701**	**1.53**	**1361**	**762**	**1.82**	**1613**	**821**	**2.16**	**1010**	871	2.48	2188	919	2.80	2462	966	3.15	2751
3600	**717**	**1.65**	**1465**	**777**	**1.94**	**1718**	**835**	**2.29**	**2023**	885	2.63	2317	932	2.95	2587	978	3.30	2870
3700	**733**	**1.77**	**1569**	**792**	**2.07**	**1831**	**849**	**2.42**	**2136**	899	2.78	2445	945	3.11	2718	990	3.45	2987
3750	**742**	**1.84**	**1630**	**800**	**2.14**	**1892**	**856**	**2.49**	**2197**	907	2.86	2512	952	3.20	2719	997	3.54	3055

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)											
	1.4			1.6			1.8			2.0		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2250	908	2.25	1989	955	2.59	2283	991	2.89	2543	1023	3.19	2805
2300	910	2.25	2015	959	2.61	2300	995	2.91	2563	1028	3.21	2827
2400	912	2.31	2041	967	2.68	2360	1003	2.99	2631	1036	3.30	2902
2500	919	2.37	2093	971	2.73	2403	1007	3.05	2681	1041	3.36	2956
2550	920	2.39	2110	974	2.76	2428	1010	3.08	2710	1044	3.40	2989
2600	921	2.41	2136	976	2.78	2445	1012	3.10	2730	1046	3.42	3011
2700	928	2.45	2162	983	2.88	2529	1020	3.21	2828	1053	3.54	3119
2800	937	2.57	2266	986	2.91	2554	1023	3.25	2857	1057	3.58	3151
2900	947	2.70	2377	993	3.01	2637	1030	3.36	2955	1064	3.70	3260
3000	958	2.85	2504	1002	3.15	2751	1039	3.51	3093	1074	3.88	3411
3100	968	2.99	2620	1012	3.30	2870	1050	3.68	3240	1084	4.06	3574
3200	979	3.13	2735	1023	3.47	3002	1061	3.87	3407	—	—	—
3300	989	3.26	2839	1034	3.63	3121	1072	4.05	3564	—	—	—
3400	1000	3.40	2948	1044	3.79	3237	—	—	—	—	—	—
3500	1011	3.55	3062	1054	3.94	3340	—	—	—	—	—	—
3600	1022	3.69	3165	1065	4.10	3445	—	—	—	—	—	—
3700	1034	3.84	3272	—	—	—	—	—	—	—	—	—
3750	1040	3.93	3333	—	—	—	—	—	—	—	—	—

LEGEND

Bhp — Brake Horsepower Input to Fan
FIOP — Factory-Installed Option
Watts — Input Watts to Motor

*Motor drive range is 860 to 1080 rpm. All other rpms require a field-supplied drive.

NOTES:
1. **Boldface** indicates field-supplied drive is required.
2. ▭ indicates field-supplied motor and drive are required.
3. Values include losses for filters, unit casing, and wet coils. See Table 7 for accessory/FIOP static pressure information.
4. Maximum continuous bhp is 3.7 and the maximum continuous watts are 3313. Extensive motor and electrical testing on these units ensures that the full range of the motor can be utilized with confidence. Using your fan motors up to the wattage ratings shown will not result in nuisance tripping or premature motor failure. Unit warranty will not be affected. See Evaporator-Fan Motor Performance, Table 8, for additional information.
5. Use of a field-supplied motor may affect wire sizing. Contact your Carrier representative for details.
6. Interpolation is permissible. Do not extrapolate.

Table 17 — Fan Performance, 48TF,TM009 Standard Motor — Horizontal Discharge Units*

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)																	
	0.2			0.4			0.6			0.8			0.9			1.0		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2550	**550**	**0.69**	**669**	**627**	**0.94**	**867**	692	1.17	1056	754	1.45	1292	783	1.60	1413	812	1.74	1543
2600	**557**	**0.72**	**692**	**634**	**0.97**	**891**	698	1.21	1089	759	1.49	1327	787	1.64	1456	816	1.79	1587
2700	**573**	**0.79**	**747**	**648**	**1.05**	**956**	711	1.29	1156	770	1.58	1404	798	1.73	1534	827	1.88	1665
2800	**588**	**0.86**	**803**	**662**	**1.13**	**1022**	723	1.38	1233	782	1.66	1473	809	1.82	1613	837	1.98	1753
2900	**604**	**0.94**	**867**	**676**	**1.21**	**1089**	737	1.48	1318	794	1.76	1560	821	1.92	1700	848	2.08	1840
3000	**620**	**1.02**	**932**	690	1.30	1165	750	1.58	1404	806	1.86	1648	832	2.02	1788	849	2.18	1927
3100	**636**	**1.11**	**1006**	704	1.39	1241	764	1.69	1499	818	1.97	1744	844	2.13	1884	870	2.29	2023
3200	**652**	**1.21**	**1089**	718	1.49	1327	778	1.80	1595	831	2.09	1849	856	2.25	1980	882	2.40	2119
3300	**668**	**1.31**	**1173**	732	1.59	1413	793	1.92	1700	844	2.21	1954	869	2.37	2093	894	2.53	2232
3400	**684**	**1.41**	**1258**	747	1.70	1508	807	2.04	1805	857	2.35	2076	882	2.51	2206	907	2.66	2343
3500	701	1.53	1361	762	1.82	1613	821	2.16	1910	871	2.48	2188	895	2.64	2326	919	2.80	2462
3600	717	1.65	1465	777	1.94	1718	835	2.29	2023	885	2.63	2317	908	2.79	2453	932	2.95	2587
3700	733	1.77	1569	792	2.07	1831	849	2.42	2136	899	2.78	2445	922	2.95	2579	945	3.11	2718
3750	742	1.84	1630	800	2.14	1892	856	2.49	2197	907	2.86	2512	929	3.03	2653	952	3.20	2719
3800	750	1.90	1683	807	2.21	1954	863	2.56	2257	914	2.93	2571	936	3.11	2847	958	3.28	2854
3900	767	2.04	1805	822	2.35	2076	877	2.71	2386	928	3.09	2702	950	3.27	2979	972	3.45	2987
4000	783	2.18	1927	838	2.50	2206	891	2.86	2512	942	3.26	2839	964	3.45	3187	986	3.63	3121
4100	800	2.34	2067	854	2.66	2343	905	3.02	2645	956	2.43	2971	978	3.62	3244	1000	3.81	3251
4200	817	2.49	2197	869	2.82	2479	920	3.19	2783	970	3.60	3099	992	3.80	3258	1015	4.00	3380
4250	826	2.58	2275	877	2.91	2554	928	3.28	2854	977	3.69	3165	999	3.90	3306	1022	4.10	3445

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)								
	1.2			1.4			1.6		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2550	864	1.99	1761	920	2.39	2110	974	2.76	2428
2600	868	2.04	1805	921	2.41	2136	976	2.78	2445
2700	878	2.16	1910	928	2.45	2162	983	2.88	2529
2800	889	2.29	2023	937	2.57	2266	986	2.91	2554
2900	900	2.41	2128	947	2.70	2377	993	3.01	2637
3000	910	2.52	2223	958	2.85	2504	1002	3.15	2751
3100	920	2.64	2326	968	2.99	2620	1012	3.30	2870
3200	931	2.76	2428	979	3.13	2735	1023	3.47	3002
3300	942	2.89	2537	989	3.26	2839	1034	3.63	3121
3400	954	3.02	2645	1000	3.40	2948	1044	3.79	3237
3500	966	3.15	2751	1011	3.55	3062	1054	3.94	3340
3600	978	3.30	2870	1022	3.69	3165	1065	4.10	3445
3700	990	3.45	2987	1034	3.84	3272	1076	4.26	3544
3750	997	3.54	3055	1040	3.93	3333	1082	5.27	3609
3800	1003	3.62	3114	1045	4.01	3387	1087	4.43	3643
3900	1015	3.80	3244	1057	4.18	3495	1098	4.60	3733
4000	1028	3.99	3373	1070	4.36	3603	1110	4.78	3820
4100	1042	4.18	3495	1082	4.56	3713	1122	4.97	3902
4200	1055	4.38	3614	1095	4.76	3811	1134	5.16	3971
4250	1062	4.49	3676	1102	4.87	3860	—	—	—

LEGEND

Bhp — Brake Horsepower Input to Fan
FIOP — Factory-Installed Option
Watts — Input Watts to Motor

*Motor drive range is 685 to 935 rpm. All other rpms require a field-supplied drive.

NOTES:
1. **Boldface** indicates field-supplied drive is required.
2. ▒ indicates field-supplied motor and drive are required.
3. Values include losses for filters, unit casing, and wet coils. See Table 7 for accessory/FIOP static pressure information.
4. Maximum continuous bhp is 2.4 and the maximum continuous watts are 2120. Extensive motor and electrical testing on these units ensures that the full range of the motor can be utilized with confidence. Using your fan motors up to the wattage ratings shown will not result in nuisance tripping or premature motor failure. Unit warranty will not be affected. See Evaporator-Fan Motor Performance, Table 8, for additional information.
5. Use of a field-supplied motor may affect wire sizing. Contact your Carrier representative for details.
6. Interpolation is permissible. Do not extrapolate.

Table 18 — Fan Performance, 48TF,TM009 High-Static Motor — Horizontal Discharge Units*

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)														
	0.2			0.4			0.6			0.8			1.0		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2550	**550**	**0.69**	**669**	**627**	**0.94**	**867**	**692**	**1.17**	**1056**	**754**	**1.45**	**1292**	**812**	**1.74**	**1543**
2600	**557**	**0.72**	**692**	**634**	**0.97**	**891**	**698**	**1.21**	**1089**	**759**	**1.49**	**1327**	**816**	**1.79**	**1587**
2700	**573**	**0.79**	**747**	**648**	**1.05**	**956**	**711**	**1.29**	**1156**	**770**	**1.59**	**1404**	**827**	**1.88**	**1665**
2800	**688**	**0.86**	**803**	**662**	**1.13**	**1022**	**723**	**1.38**	**1233**	**782**	**1.68**	**1473**	**837**	**1.98**	**1753**
2900	**604**	**0.94**	**867**	**676**	**1.21**	**1089**	**737**	**1.48**	**1318**	**794**	**1.76**	**1560**	**848**	**2.08**	**1840**
3000	**620**	**1.02**	**932**	**690**	**1.30**	**1165**	**750**	**1.58**	**1404**	**806**	**1.86**	**1648**	**849**	**2.18**	**1927**
3100	**636**	**1.11**	**1006**	**704**	**1.39**	**1241**	**764**	**1.69**	**1499**	**818**	**1.97**	**1744**	870	2.29	2023
3200	**652**	**1.21**	**1089**	**718**	**1.49**	**1327**	**778**	**1.80**	**1595**	**831**	**2.09**	**1849**	882	2.40	2119
3300	**668**	**1.31**	**1173**	**732**	**1.59**	**1413**	**793**	**1.92**	**1700**	**844**	**2.21**	**1954**	894	2.53	2232
3400	**684**	**1.41**	**1258**	**747**	**1.70**	**1508**	**807**	**2.04**	**1805**	**857**	**2.35**	**2076**	907	2.66	2343
3500	**701**	**1.53**	**1361**	**762**	**1.82**	**1613**	**821**	**2.16**	**1910**	871	2.48	2188	919	2.80	2462
3600	**717**	**1.65**	**1465**	**777**	**1.94**	**1718**	**835**	**2.29**	**2023**	885	2.36	2317	932	2.95	2587
3700	**733**	**1.77**	**1569**	**792**	**2.07**	**1831**	**849**	**2.42**	**2136**	899	2.78	2445	945	3.11	2718
3750	**742**	**1.84**	**1630**	**800**	**2.14**	**1892**	**856**	**2.49**	**2197**	907	2.86	2512	952	3.20	2719
3800	**750**	**1.90**	**1683**	**807**	**2.21**	**1954**	863	2.56	2257	914	2.93	2571	958	3.28	2854
3900	**767**	**2.04**	**1805**	**822**	**2.35**	**2076**	877	2.71	2386	928	3.09	2702	972	3.45	2987
4000	**783**	**2.18**	**1927**	**838**	**2.50**	**2206**	891	2.86	2512	942	3.26	2839	986	3.63	3121
4100	**800**	**2.34**	**2067**	**854**	**2.66**	**2343**	905	3.02	2645	956	3.43	2971	1000	3.81	3251
4200	**817**	**2.49**	**2197**	869	2.82	2479	920	3.19	2783	970	3.60	3099	1015	4.00	3380
4250	**826**	**2.58**	**2275**	877	2.91	2554	928	3.28	2854	977	3.69	3165	1022	4.10	3445

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)														
	1.2			1.4			1.6			1.8			2.0		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
2550	864	1.99	1761	920	2.39	2110	974	2.76	2428	1005	3.04	2672	1034	3.31	2910
2600	868	2.04	1805	921	2.41	2136	976	2.78	2445	1008	3.06	2691	1037	3.33	2931
2700	878	2.16	1910	928	2.45	2162	893	2.88	2529	1015	3.17	2788	1044	3.45	3036
2800	889	2.92	2023	937	2.57	2266	986	2.91	2554	1018	3.20	2817	1047	3.49	3068
2900	900	2.41	2128	947	2.70	2377	993	3.01	2637	1025	3.31	2914	1055	3.61	3174
3000	910	2.52	2223	958	2.85	2504	1002	3.15	2751	1034	3.47	3049	1064	3.77	3321
3100	920	2.64	2326	968	2.99	2620	1012	3.30	2870	1045	3.63	3195	1075	3.95	3479
3200	931	2.76	2428	979	3.13	2735	1023	3.47	3002	1056	3.82	3359	1087	4.16	3659
3300	942	2.89	2537	989	3.26	2839	1034	3.63	3121	1067	3.99	3514	—	—	—
3400	954	3.02	2645	1000	3.40	2948	1044	3.79	3237	1078	4.17	3669	—	—	—
3500	966	3.15	2751	1011	3.55	3062	1054	3.94	3340	—	—	—	—	—	—
3600	978	3.30	2870	1022	3.69	3165	1065	4.10	3445	—	—	—	—	—	—
3700	990	3.45	2987	1034	3.84	3272	—	—	—	—	—	—	—	—	—
3750	997	3.54	3055	1040	3.93	3333	—	—	—	—	—	—	—	—	—
3800	1003	3.62	3114	1045	4.01	3387	—	—	—	—	—	—	—	—	—
3900	1015	3.80	3244	1057	4.18	3495	—	—	—	—	—	—	—	—	—
4000	1028	3.99	3373	—	—	—	—	—	—	—	—	—	—	—	—
4100	1042	4.28	3495	—	—	—	—	—	—	—	—	—	—	—	—
4200	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
4250	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

LEGEND

Bhp — Brake Horsepower Input to Fan
FIOP — Factory-Installed Option
Watts — Input Watts to Motor

*Motor drive range is 860 to 1080 rpm. All other rpms require a field-supplied drive.

NOTES:
1. **Boldface** indicates field-supplied drive is required.
2. ▭ indicates field-supplied motor and drive are required.
3. Values include losses for filters, unit casing, and wet coils. See Table 7 for accessory/FIOP static pressure information.
4. Maximum continuous bhp is 3.7 and the maximum continuous watts are 3313. Extensive motor and electrical testing on these units ensures that the full range of the motor can be utilized with confidence. Using your fan motors up to the wattage ratings shown will not result in nuisance tripping or premature motor failure. Unit warranty will not be affected. See Evaporator-Fan Motor Performance, Table 8, for additional information.
5. Use of a field-supplied motor may affect wire sizing. Contact your Carrier representative for details.
6. Interpolation is permissible. Do not extrapolate.

Table 19 — Fan Performance, 48TF,TM012 — Horizontal Discharge Units*

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)																	
	0.2			0.4			0.6			0.8			1.0			1.2		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
3000	**552**	**0.68**	**661**	**632**	**0.87**	**810**	701	1.05	956	761	1.22	1097	816	1.36	1216	871	1.54	1370
3100	**565**	**0.74**	**708**	**644**	**0.93**	**859**	711	1.12	1014	772	1.31	1173	825	1.45	1292	879	1.63	1447
3200	**578**	**0.81**	**763**	**656**	**1.00**	**916**	723	1.20	1081	782	1.39	1241	835	1.55	1378	887	1.71	1517
3300	**591**	**0.88**	**818**	**668**	**1.08**	**973**	734	1.28	1148	793	1.47	1310	845	1.65	1465	895	1.80	1595
3400	**605**	**0.96**	**883**	**680**	**1.16**	**1047**	745	1.36	1216	803	1.56	1387	856	1.75	1552	904	1.91	1691
3500	**619**	**1.04**	**948**	**691**	**1.23**	**1106**	755	1.44	1284	813	1.65	1465	867	1.86	1648	914	2.03	1796
3600	**633**	**1.13**	**1022**	703	1.31	1173	766	1.52	1353	824	1.74	1543	877	1.97	1744	924	2.15	1901
3700	**648**	**1.23**	**1106**	714	1.39	1241	777	1.61	1430	835	1.85	1639	887	2.07	1831	935	2.28	2015
3800	**662**	**1.33**	**1190**	726	1.51	1310	789	1.72	1526	846	1.95	1726	897	2.18	1927	946	2.40	2169
3900	**677**	**1.44**	**1284**	738	1.61	1387	801	1.82	1613	857	2.06	1823	908	2.29	2023	956	2.53	2282
4000	**692**	**1.55**	**1378**	750	1.73	1473	813	1.94	1718	868	2.17	1919	918	2.40	2119	967	2.66	2397
4100	707	1.67	1482	762	1.84	1560	825	2.05	1814	878	2.28	2015	929	2.53	2282	977	2.78	2505
4200	722	1.80	1595	775	1.97	1656	837	2.16	1910	889	2.40	2119	941	2.66	2397	987	2.91	2624
4300	737	1.94	1718	787	2.09	1761	848	2.27	2006	900	2.52	2273	952	2.80	2523	999	3.04	2744
4400	752	2.08	1840	800	2.21	1875	860	2.39	2110	912	2.66	2397	962	2.93	2642	1008	3.19	2885
4500	768	2.24	1980	814	2.35	1989	871	2.51	2265	924	2.80	2523	973	3.07	2772	1019	3.34	3029
4600	783	2.40	2119	827	2.50	2121	883	2.64	2380	937	2.95	2661	983	3.21	2904	—	—	—
4700	799	2.56	2309	841	2.64	2291	894	2.77	2496	949	3.10	2800	994	3.36	3049	—	—	—
4800	814	2.74	2469	855	2.80	2424	906	2.91	2624	961	3.26	2952	—	—	—	—	—	—
4900	—	—	—	868	2.90	2578	918	3.05	2754	972	3.40	3088	—	—	—	—	—	—
5000	—	—	—	883	3.10	2735	931	3.21	2904	—	—	—	—	—	—	—	—	—

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)											
	1.4			1.6			1.8			2.0		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
3000	918	1.67	1482	967	1.89	1736	1010	2.09	1903	1063	2.46	2221
3100	928	1.78	1478	973	1.94	1778	1018	2.17	1971	1070	2.51	2265
3200	937	1.90	1745	981	2.04	1861	1026	2.26	2048	1075	2.57	2318
3300	946	2.00	1828	991	2.16	1963	1032	2.32	2099	1080	2.64	2380
3400	953	2.10	1912	1000	2.29	2074	1041	2.44	2203	1083	2.65	2389
3500	961	2.20	1997	1009	2.41	2177	1051	2.57	2318	1090	2.74	2469
3600	970	2.32	2099	1017	2.53	2282	1061	2.72	2451	1099	2.88	2596
3700	980	2.45	2212	1024	2.64	2380	1069	2.87	2587	1109	3.03	2735
3800	989	2.58	2326	1033	2.76	2487	1077	2.99	2698	1118	3.20	2895
3900	1000	2.73	2460	1042	2.91	2624	1085	3.12	2819	1127	3.36	3049
4000	1010	2.87	2587	1052	3.06	2763	1093	3.24	2933	—	—	—
4100	1021	3.02	2726	1062	3.22	2914	1102	3.41	3097	—	—	—
4200	1032	3.17	2866	1072	3.38	2971	—	—	—	—	—	—
4300	1042	3.32	3010	—	—	—	—	—	—	—	—	—
4400	—	—	—	—	—	—	—	—	—	—	—	—
4500	—	—	—	—	—	—	—	—	—	—	—	—
4600	—	—	—	—	—	—	—	—	—	—	—	—
4700	—	—	—	—	—	—	—	—	—	—	—	—
4800	—	—	—	—	—	—	—	—	—	—	—	—
4900	—	—	—	—	—	—	—	—	—	—	—	—
5000	—	—	—	—	—	—	—	—	—	—	—	—

LEGEND

Bhp — Brake Horsepower Input to Fan
FIOP — Factory-Installed Option
Watts — Input Watts to Motor

*Standard motor drive range is 695 to 935 rpm. Alternate motor drive range is 835 to 1085 rpm. High-static motor drive range is 830 to 1130 rpm. All other rpms require a field-supplied drive.

NOTES:
1. **Boldface** indicates field-supplied drive is required.
2. ▒▒▒ indicates optional high-static motor and drive are required.
3. Values include losses for filters, unit casing, and wet coils. See Table 7 for accessory/FIOP static pressure information.
4. Maximum continuous bhp is 2.4 for the standard motor, 2.9 for the alternate motor and 5.25 for the high-static motor. Maximum continuous watts are 2120 for the standard motor, 2615 for the alternate motor and 4400 for the high-static motor. Extensive motor and electrical testing on these units ensures that the full range of the motor can be utilized with confidence. Using your fan motors up to the wattage ratings shown will not result in nuisance tripping or premature motor failure. Unit warranty will not be affected. See Evaporator-Fan Motor Performance, Table 8, for additional information.
5. Use of a field-supplied motor may affect wire sizing. Contact your Carrier representative for details.
6. Interpolation is permissible. Do not extrapolate.

Table 20 — Fan Performance, 48TF,TM014 — Horizontal Discharge Units*

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)																	
	0.2			0.4			0.6			0.8			1.0			1.2		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
3750	**684**	**1.24**	**1162**	**755**	**1.48**	**1353**	**816**	**1.70**	**1540**	875	1.94	1738	933	2.23	1988	989	2.49	2219
3800	**691**	**1.28**	**1194**	**761**	**1.52**	**1390**	**822**	**1.75**	**1582**	880	1.98	1776	937	2.28	2035	993	2.55	2272
3900	**705**	**1.37**	**1267**	**773**	**1.62**	**1473**	**834**	**1.86**	**1674**	891	2.08	1862	947	2.39	2131	1002	2.66	2370
4000	**720**	**1.47**	**1349**	**786**	**1.71**	**1548**	**847**	**1.97**	**1768**	902	2.19	1957	957	2.50	2228	1011	2.79	2485
4100	**734**	**1.56**	**1423**	**800**	**1.82**	**1641**	860	2.09	1871	914	2.31	2061	967	2.60	2316	1021	2.91	2593
4200	**749**	**1.66**	**1506**	**813**	**1.92**	**1725**	873	2.21	1974	926	2.44	2175	978	2.71	2414	1030	3.04	2710
4300	**764**	**1.77**	**1598**	**826**	**2.04**	**1828**	886	2.33	2079	938	2.57	2290	989	2.83	2521	1040	3.18	2837
4400	**779**	**1.88**	**1691**	**840**	**2.16**	**1931**	899	2.46	2193	951	2.71	2414	1000	2.96	2638	1050	3.31	2955
4500	**793**	**1.99**	**1785**	**854**	**2.28**	**2035**	912	2.59	2307	963	2.86	2548	1012	3.09	2755	1061	3.43	3065
4600	**808**	**2.11**	**1888**	868	2.42	2158	925	2.73	2459	975	3.00	2674	1024	3.25	2901	1071	3.56	3184
4700	**822**	**2.24**	**2000**	882	2.56	2281	937	2.86	2548	988	3.16	2819	1036	3.42	3056	1082	3.70	3313
4800	**837**	**2.37**	**2114**	896	2.71	2414	950	3.00	2674	1001	3.32	2964	1048	3.59	3212	1093	3.86	3461
4900	**852**	**2.51**	**2237**	910	2.86	2548	963	3.15	2810	1014	3.48	3111	1060	3.76	3368	1105	4.02	3609
5000	867	2.65	2361	924	3.01	2683	977	3.30	2946	1027	3.65	3267	1073	3.94	3535	1117	4.20	3776
5100	882	2.79	2485	938	3.17	2828	990	3.46	3092	1040	3.82	3424	1085	4.12	3702	1129	4.40	3962
5200	896	2.95	2629	952	3.33	2973	1003	3.63	3248	1053	4.00	3590	1098	4.30	3869	1141	4.60	4148
5300	911	3.11	2773	967	3.50	3129	1017	3.80	3405	1066	4.18	3757	1111	4.50	4055	1153	4.80	4335
5400	926	3.27	2919	981	3.68	3294	1030	3.98	3572	1079	4.35	3916	1124	4.70	4270	1166	5.01	4531
5500	940	3.44	3074	995	3.86	3461	1044	4.17	3748	1092	4.54	4093	1137	4.91	4437	1178	5.22	4726
5600	955	3.62	3239	1010	4.04	3627	1058	4.38	3943	1105	4.73	4270	1150	5.12	4633	—	—	—
5700	970	3.80	3405	1024	4.23	3804	1072	4.59	4139	1118	4.93	4456	—	—	—	—	—	—
5800	985	3.99	3581	1039	4.42	3981	1086	4.80	4335	1131	5.14	4652	—	—	—	—	—	—
5900	1000	4.18	3757	1053	4.62	4167	1100	5.02	4540	—	—	—	—	—	—	—	—	—
6000	1015	4.39	3953	1068	4.83	4363	—	—	—	—	—	—	—	—	—	—	—	—
6100	1030	4.59	4139	1083	5.04	4558	—	—	—	—	—	—	—	—	—	—	—	—
6250	1062	5.02	4560	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

AIRFLOW (Cfm)	EXTERNAL STATIC PRESSURE (in. wg)											
	1.4			1.6			1.8			2.0		
	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts	Rpm	Bhp	Watts
3750	1041	2.75	2445	1086	2.98	2634	1122	3.16	2819	1146	3.28	2928
3800	1046	2.81	2503	1092	3.05	2719	1129	3.25	2901	1156	3.39	3028
3900	1055	2.94	2620	1102	3.20	2855	1143	3.42	3056	1174	3.59	3212
4000	1064	3.07	2737	1112	3.34	2983	1155	3.59	3212	1190	3.80	3405
4100	1072	3.20	2855	1121	3.49	3120	1165	3.76	3368	1203	3.99	3581
4200	1081	3.34	2983	1130	3.64	3258	1175	3.92	3516	1215	4.18	3757
4300	1090	3.48	3111	1139	3.79	3396	1185	4.08	3664	1226	4.36	3925
4400	1100	3.63	3248	1148	3.94	3535	1194	4.25	3822	1236	4.54	4093
4500	1109	3.78	3387	1157	4.09	3674	1203	4.42	3981	1246	4.72	4260
4600	1119	3.93	3525	1166	4.26	3832	1212	4.58	4130	1255	4.91	4437
4700	1129	4.09	3674	1175	4.43	3990	1221	4.76	4298	1264	5.09	4605
4800	1139	4.24	3813	1185	4.60	4148	1230	4.93	4456	—	—	—
4900	1150	4.38	3943	1194	4.77	4307	1239	5.12	4633	—	—	—
5000	1161	4.54	4093	1204	4.95	4475	—	—	—	—	—	—
5100	1172	4.71	4251	1214	5.13	4642	—	—	—	—	—	—
5200	1183	4.91	4419	—	—	—	—	—	—	—	—	—
5300	1194	5.08	4596	—	—	—	—	—	—	—	—	—
5400	—	—	—	—	—	—	—	—	—	—	—	—
5500	—	—	—	—	—	—	—	—	—	—	—	—
5600	—	—	—	—	—	—	—	—	—	—	—	—
5700	—	—	—	—	—	—	—	—	—	—	—	—
5800	—	—	—	—	—	—	—	—	—	—	—	—
5900	—	—	—	—	—	—	—	—	—	—	—	—
6000	—	—	—	—	—	—	—	—	—	—	—	—
6100	—	—	—	—	—	—	—	—	—	—	—	—
6250	—	—	—	—	—	—	—	—	—	—	—	—

LEGEND

Bhp — Brake Horsepower Input to Fan
FIOP — Factory-Installed Option
Watts — Input Watts to Motor

*Standard motor drive range is 860 to 1080 rpm. Alternate motor drive range is 900 to 1260 rpm. All other rpms require a field-supplied drive.

NOTES:
1. **Boldface** indicates field-supplied drive is required.
2. ▭ indicates alternate motor and drive are required.
3. Values include losses for filters, unit casing, and wet coils. See Table 7 for accessory/FIOP static pressure information.
4. Maximum continuous bhp is 3.7 for the standard motor, and 5.25 for the alternate motor. The maximum continuous watts are 3313 for the standard motor and 4400 for the alternate motor. Extensive motor and electrical testing on these units ensures that the full range of the motor can be utilized with confidence. Using your fan motors up to the wattage ratings shown will not result in nuisance tripping or premature motor failure. Unit warranty will not be affected. See Evaporator-Fan Motor Performance, Table 8, for additional information.
5. Use of a field-supplied motor may affect wire sizing. Contact your Carrier representative for details.
6. Interpolation is permissible. Do not extrapolate.

START-UP

Unit Preparation — Make sure that unit has been installed in accordance with these installation instructions and applicable codes. Make sure that Start-Up Checklist, located on back page of this booklet, has been completed and filled out.

Return-Air Filters — Make sure correct filters are installed in filter tracks (see Tables 1A and 1B). Do not operate unit without return-air filters.

Outdoor-Air Inlet Screens — Outdoor-air inlet screens must be in place before operating unit.

Compressor Mounting — Compressors are internally spring mounted. Do not loosen or remove compressor hold-down bolts. On 48TF,TM014 units, remove the tiedown bands that hold the compressors together.

Internal Wiring — Check all electrical connections in unit control boxes. Tighten as required. Ensure wiring does not come into direct contact with refrigerant tubing.

Gas Piping — Check gas piping for leaks.

⚠ WARNING



Disconnect gas piping from unit when leak testing at pressure greater than 1/2 psig. Pressures greater than 1/2 psig will cause gas valve damage resulting in hazardous condition. If gas valve is subjected to pressure greater than 1/2 psig, it *must* be replaced before use. When pressure testing field-supplied gas piping at pressures of 1/2 psig or less, a unit connected to such piping must be isolated by manually closing the gas valve.

Refrigerant Service Ports — To service refrigerant service ports, remove compressor access panel. Each unit system has 3 Schrader-type service gage ports: one on the suction line, one on the liquid line, and one on the compressor discharge line. Be sure that caps on the ports are tight. The Schrader-type valve on the discharge line is located under the low-pressure switch. Another valve is located on the discharge line underneath the high-pressure switch. It is screwed on a Schrader fitting but there is no Schrader core.

High Flow Valves — Located on the compressor hot gas and suction tubes are High Flow Valves. Large black plastic caps distinguish these valves with O-rings located inside the caps. These valves can not be accessed for service in the field. Ensure the plastic caps are in place and tight or the possibility of refrigerant leakage could occur.

Compressor Rotation — On 3-phase units with scroll compressors, it is important to be certain compressor is rotating in the proper direction. To determine whether or not compressor is rotating in the proper direction:

1. Connect service gages to suction and discharge pressure fittings.
2. Energize the compressor.
3. The suction pressure should drop and the discharge pressure should rise, as is normal on any start-up.

If the suction pressure does not drop and the discharge pressure does not rise to normal levels:

1. Note that the evaporator fan is probably also rotating in the wrong direction.
2. Turn off power to the unit and install lockout tag.
3. Reverse any two of the unit power leads.
4. Reapply power to unit. Reenergize compressor.

The suction and discharge pressure levels should now move to their normal start-up levels.

NOTE: When the compressor is rotating in the wrong direction, the unit will make an elevated level of noise and will not provide cooling.

Cooling — To start unit, turn on main power supply. Set system selector switch at COOL position and fan switch at AUTO. position. Adjust thermostat to a setting below room temperature. Compressor, indoor and outdoor fans start on closure of contactors.

Check unit charge. Refer to Checking and Adjustment Refrigerant Charge section, page 46. Unit must operate a minimum of 10 minutes before adjusting charge.

Reset thermostat at a position above room temperature. Compressor and outdoor fans will shut off. Evaporator fan will shut off after 30-second delay.

TO SHUT OFF UNIT — Set system selector switch at OFF position. Resetting thermostat at a position above room temperature shuts unit off temporarily until space temperature exceeds thermostat setting.

Main Burners — Main burners are factory set and should require no adjustment.

TO CHECK ignition of main burners and heating controls, move thermostat set point above room temperature and verify that the burners light and evaporator fan is energized. After ensuring that the unit continues to heat the building, lower the thermostat setting below the room temperature and verify that the burners and evaporator fan turn off (fan will turn off only if fan selector switch is in the AUTO. position). Refer to Table 21 for the correct orifice to use at high altitudes.

NOTE: Upon a call for heat, the main burners will remain on for a minimum of 60 seconds.

ADJUST GAS INPUT — The gas input to the unit is determined by measuring the gas flow at the meter or by measuring the manifold pressure. Manifold pressure should be 3.5 in. wg.

NOTE: On units that use a 2 stage gas valve there is no need to adjust the "Low Fire" manifold pressure.

Measure Gas Flow (Natural Gas Units) — Minor adjustment to the gas flow can be made by changing the manifold pressure. The manifold pressure must be 3.5 in wg. Normal manifold pressure is 3.5 in. wg in high fire (W1 and W2 inputs to gas valve).

Proceed as follows:

1. Turn off gas supply to unit.
2. Remove pipe plug on manifold then connect manometer at this point. Turn on gas to unit. Ensure gas valve is in high fire operation.

Observe manifold pressure and proceed as follows to adjust gas input:

1. Remove cover screw over regulator adjustment screw on gas valve. Ensure gas valve is operating in high fire mode.
2. Turn regulator adjustment screw clockwise to increase gas input, or turn regulator adjustment screw counterclockwise to decrease input. High fire manifold pressure must be 3.5 in. wg.

⚠ WARNING

Unsafe operation of the unit may result if manifold pressure is outside 3.4 to 3.6 in. wg range. Personal injury or unit damage may result.

3. Replace cover screw cap on gas valve.

4. Turn off gas supply to unit. Remove manometer from pressure tap and replace pipe plug on gas valve. Turn on gas to unit and check for leaks.

Heating

1. Purge gas supply line of air by opening union ahead of gas valve. When gas odor is detected, tighten union and wait 5 minutes before proceeding.
2. Turn on electrical supply and open manual gas valve.
3. Set system switch selector at HEAT position and fan switch at AUTO. or ON position. Set heating temperature lever above room temperature.
4. The induced-draft motor will start, purging heat exchangers.
5. After a call for heating, the main burners should light within 5 seconds. If the burners do not light, then there is a 22-second delay before another 5-second ignition try. If the burners still do not light, the time delay is repeated. If the burners do not light within 15 minutes, there is a lockout. To reset the control, break the 24 v power to W1.
6. The evaporator-fan motor will turn on 45 seconds after the burners are ignited.
7. The evaporator-fan motor will turn off 45 seconds after the thermostat temperature is satisfied.
8. Adjust airflow to obtain a temperature rise within the range specified on the unit nameplate and in Tables 1A and 1B.

Table 21 — Altitude Compensation*

ELEVATION (Ft)	125,000, 180,000, AND 220,000 BTUH NOMINAL INPUT		250,000 BTUH NOMINAL INPUT	
	Natural Gas Orifice Size†	Liquid Propane Orifice Size†	Natural Gas Orifice Size†	Liquid Propane Orifice Size†
0-2,000	31	41	30	38
2,000	32	42	30	39
3,000	32	42	31	40
4,000	32	42	32	41
5,000	33	43	33	42
6,000	34	43	34	43
7,000	35	44	35	43
8,000	36	44	36	44
9,000	37	45	37	44
10,000	38	46	38	45
11,000	39	47	39	45
12,000	40	47	40	46
13,000	41	48	41	47
14,000	42	48	42	47

*As the height above sea level increases, there is less oxygen per cubic foot of air. Therefore, heat input rate should be reduced at higher altitudes.
†Orifice available through your local distributor.

Integrated Gas Controller (IGC) Operation

NOTE: The default value for the evaporator-fan motor ON and OFF delay is 45 seconds. The Integrated Gas Controller (IGC) modifies this value when abnormal limit switch cycles occur. Based upon unit operating conditions, the ON delay can be reduced to 0 seconds and the OFF delay can be extended to 180 seconds.

When one flash of the LED (light-emitting diode) is observed, the evaporator-fan ON/OFF delay has been modified. If the limit switch trips at the start of the heating cycle during the evaporator ON delay, the time period of the ON delay for the next cycle will be 5 seconds less than the time at which the switch tripped. (Example: If the limit switch trips at 30 seconds, the evaporator-fan ON delay for the next cycle will occur at 25 seconds.) To prevent short-cycling, a 5-second reduction will only occur if a minimum of 10 minutes has elapsed since the last call for heating.

The evaporator-fan OFF delay can also be modified. Once the call for heating has ended, there is a 10-minute period during which the modification can occur. If the limit switch trips during this period, the evaporator-fan OFF delay will increase by 15 seconds on the next cycle. A maximum of 9 trips can occur, extending the evaporator-fan OFF delay to 180 seconds. To restore the original default value, reset the power to the unit.

TO SHUT OFF UNIT — Set system selector switch at OFF position. Resetting heating selector lever below room temperature will shut unit off temporarily until space temperature falls below thermostat setting.

Safety Relief — A soft-solder joint at the suction line Schrader port provides pressure relief under abnormal temperature and pressure conditions.

Ventilation (Continuous Fan) — Set fan and system selector switches at ON and OFF positions, respectively. Evaporator fan operates continuously to provide constant air circulation. When the evaporator-fan selector switch is turned to the OFF position, there is a 30-second delay before the fan turns off.

Operating Sequence

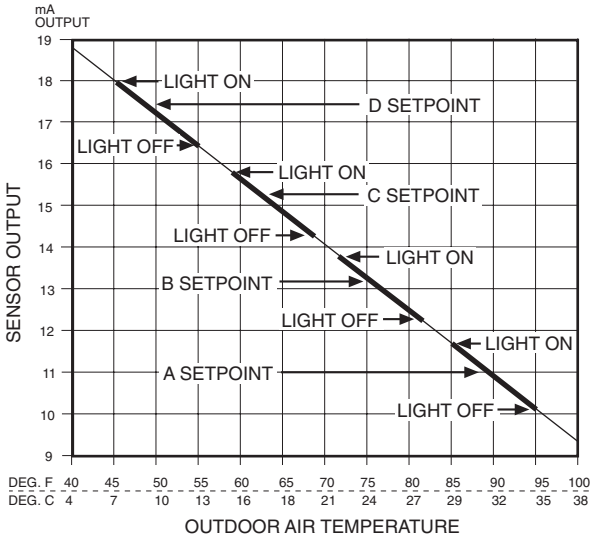
COOLING, UNITS WITHOUT ECONOMIZER — When thermostat calls for cooling, terminals G and Y1 are energized. The indoor (evaporator) fan contactor (IFC), compressor contactor no. 1 (C1) and outdoor-fan contactor (OFC) are energized, and evaporator-fan motor, compressor no. 1, and both condenser fans start. The condenser-fan motors run continuously while unit is cooling. If the thermostat calls for a second stage of cooling by energizing Y2, compressor contactor no. 2 (C2) is energized and compressor no. 2 starts.

When the thermostat is satisfied, C1 and C2 are deenergized and the compressors and outdoor (condenser) fan motors (OFM) shut off. After a 30-second delay, the indoor (evaporator) fan motor (IFM) shuts off. If the thermostat fan selector switch is in the ON position, the evaporator-fan motor will run continuously.

HEATING, UNITS WITHOUT ECONOMI$ER2 — When the thermostat calls for heating, terminal W1 is energized. In order to prevent thermostat short-cycling, the unit is locked into the Heating mode for at least 1 minute when W1 is energized. The induced-draft motor (IDM) is then energized and the burner ignition sequence begins. The indoor (evaporator) fan motor (IFM) is energized 45 seconds after a flame is ignited. On units equipped for two stages of heat, when additional heat is needed, W2 is energized and the high-fire solenoid on the main gas valve (MGV) is energized. When the thermostat is satisfied and W1 and W2 are deenergized, the IFM stops after a 45-second time-off delay.

COOLING, UNITS WITH ECONOMI$ER2 — When free cooling is not available, the compressors will be controlled by the zone thermostat. When free cooling is available, the outdoor-air damper is positioned through the use of a Proportional-Integral (PI) control process to provide a 55 F supply-air temperature into the zone. As the supply-air temperature fluctuates above or below 55 F, the dampers will be modulated (open or close) to bring the supply-air temperature back within control.

If mechanical cooling is utilized with free cooling, the outdoor-air damper will be locked into its current position at the time the compressor is started. If the increase in cooling capacity causes the supply-air temperature to drop below 45 F, then the outdoor-air damper position will be decreased to the minimum position. If the supply-air temperature continues to fall, the outdoor-air damper will close. See Fig. 42 for EconoMi$er2 outdoor-air damper operation.



NOTES:
1. Light ON: Outside-air damper will open to allow free cooling below this temperature.
2. Light OFF: Outside-air damper will close to its minimum position and compressor will be energized above this temperature.

Fig. 42 — EconoMi$er2 Outdoor-Air Damper Operation

If optional power exhaust is installed, as the outdoor-air damper opens and closes, the power exhaust fans will be energized and deenergized.

If field-installed accessory CO_2 sensors are connected to the EconoMi$er2 control, a PI-controlled demand ventilation strategy will begin to operate. As the CO_2 level in the zone increases above the CO_2 set point, the minimum position of the damper will be increased proportionally. As the CO_2 level decreases because of the increase in fresh air, the outdoor-air damper will be proportionally closed.

HEATING, UNITS WITH ECONOMI$ER2 — When the thermostat calls for heating, terminal W1 is energized. To prevent thermostat short-cycling, the unit is locked into the Heating mode for at least 1 minute when W1 is energized. The induced-draft motor is energized and the burner ignition sequence begins. The indoor (evaporator) fan motor (IFM) is energized 45 seconds after a flame is ignited and the damper moves to the minimum position. On units equipped for two stages of heat, when additional heat is needed, W2 energized and the high-fire solenoid on the main gas valve (MGV) is energized. When the thermostat is satisfied and W1 is deenergized, the IFM stops after a 45-second time-off delay. The economizer damper then moves to the fully closed position. When using continuous fan, the damper will remain in the minimum position.

SERVICE

⚠ CAUTION

When servicing unit, shut off all electrical power to unit and tag disconnect to avoid shock hazard or injury from rotating parts.

Cleaning — Inspect unit interior at the beginning of each heating and cooling season or more frequently as operating conditions require.

EVAPORATOR COIL — Clean coil as required. Inspect coil at beginning of heating and cooling seasons.

1. Turn unit power off and install lockout tag. Remove evaporator coil access panel.
2. If economizer is installed, remove economizer by disconnecting Molex plug and removing economizer mounting screws. Refer to Accessory Economizer Installation Instructions or Optional EconoMi$er2 section on page 18 for more details.
3. Remove filters from unit.
4. Clean coil using a commercial coil cleaner or dishwasher detergent in a pressurized spray canister. Wash both sides of coil and flush with clean water. For best results, backflush toward return-air section to remove foreign material. Caution should be taken as to not overflow the evaporator drain condensate pan.
5. Flush condensate pan after completion.
6. Reinstall economizer and filters.
7. Reconnect wiring.
8. Replace access panels.

CONDENSER COIL — Inspect coil monthly. Clean condenser coil annually, and as required by location and outdoor-air conditions.

One-Row Coils — Wash coil with commercial cleaner. Clean outer surfaces with a stiff brush in the normal manner. It is not necessary to remove top panel.

2-Row Coils — Clean coil as follows:

1. Turn off unit power and tag disconnect.
2. Remove top panel screws on condenser end of unit.
3. Remove condenser coil corner post. See Fig. 43. To hold top panel open, place coil corner post between top panel and center post. See Fig. 44.
4. Remove screws securing coil to center post.
5. Remove fastener holding coil sections together at return end of condenser coil. Carefully separate the outer coil section 3 to 4 in. from the inner coil section. See Fig. 45.
6. Use a water hose or other suitable equipment to flush down between the 2 coil sections to remove dirt and debris. Clean the outer surfaces with a stiff brush in the normal manner.
7. Secure inner and outer coil rows together with a field-supplied fastener.
8. Reposition the outer coil section and remove the coil corner post from between the top panel and center post.
9. Reinstall the coil corner post and replace all screws.

CONDENSATE DRAIN — Check and clean each year at start of cooling season. In winter, keep drain dry or protect against freeze-up.

FILTERS — Clean or replace at start of each heating and cooling season, or more often if operating conditions require it. Replacement filters must be same dimensions as original filters.

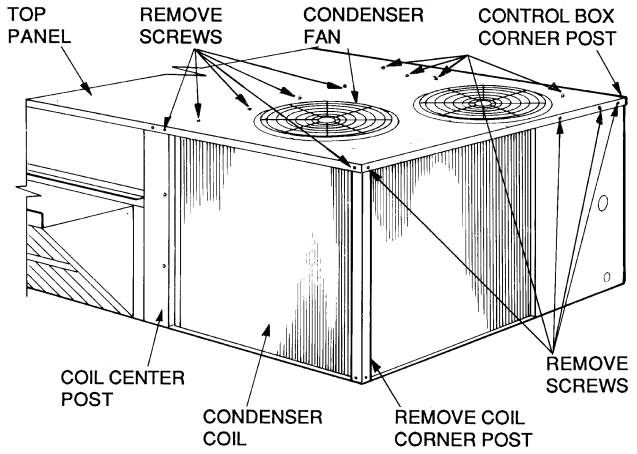


Fig. 43 — Cleaning Condenser Coil

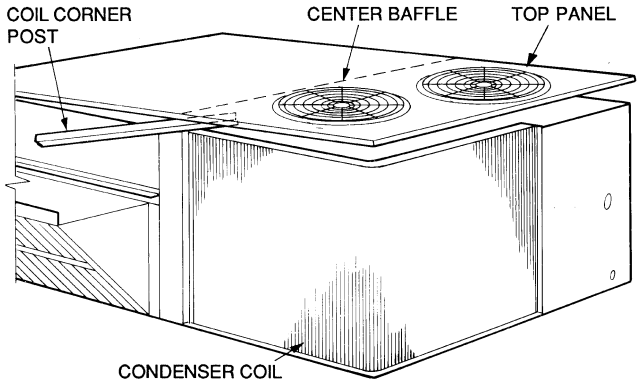


Fig. 44 — Propping Up Top Panel

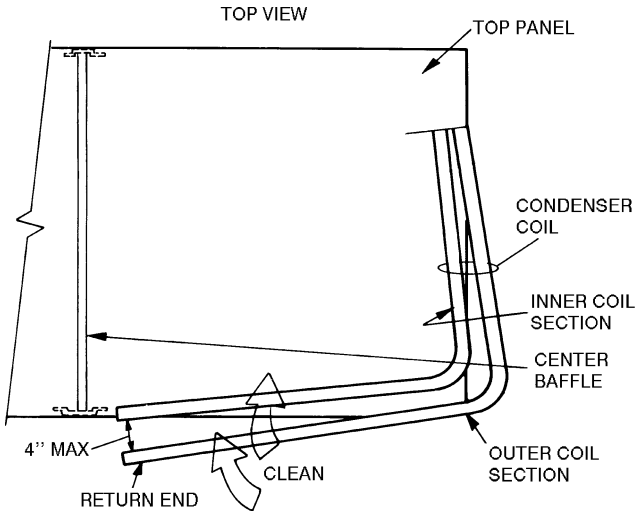


Fig. 45 — Separating Coil Sections

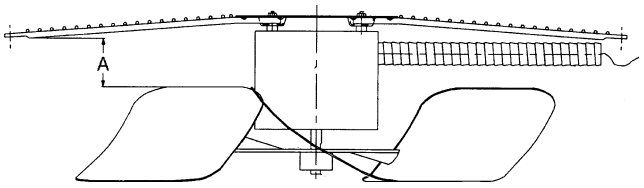
Lubrication

COMPRESSORS — Each compressor is charged with the correct amount of oil at the factory.

FAN-MOTOR BEARINGS — *Fan-motor bearings are of the permanently lubricated type. No further lubrication is required.* No lubrication of condenser- or evaporator-fan motors is required.

Condenser-Fan Adjustment (Fig. 46)

1. Shut off unit power supply and tag disconnect.
2. Remove condenser-fan assembly (grille, motor, motor cover, and fan) and loosen fan hub setscrews.
3. Adjust fan height as shown in Fig. 46.
4. Tighten setscrews and replace condenser-fan assembly.



48TF,TM UNIT VOLTAGE	FAN HEIGHT "A" (in.)
208/230 V	2.75
460 V and 575 V	3.50

Fig. 46 — Condenser-Fan Adjustment

Blower Belt Adjustment — Inspect blower belt for wear, proper belt tension, and pulley alignment as conditions require or at the beginning of each heating and air conditioning season.

Refer to Step 9 — Adjust Evaporator Fan Speed on page 28 for adjustment and alignment procedures.

Manual Outdoor-Air Damper — If outdoor-air damper blade is required, see Manual Outdoor-Air Damper section on page 17.

Economizer Adjustment — Refer to Optional Econo-Mi$er section on page 18.

Condenser Coil Grille — Condenser coil grille is shipped factory-installed. No adjustments are required.

High-Pressure Switch — Located on the compressor hot gas line is a high-pressure switch. This switch opens at 428 psig and closes at 320 psig. No adjustment is necessary. Refer to Tables 1A and 1B.

Loss-of-Charge Switch — Located on the condenser liquid line is a low-pressure switch which functions as a loss-of-charge switch. This switch contains a Schrader core depressor. This switch opens at 7 psig and closes at 22 psig. No adjustment is necessary. Refer to Tables 1A and 1B.

Freezestat — Located on the "hair pin" end of the evaporator coil is a bimetal temperature sensing switch. This switch protects the evaporator coil from freeze-up due to lack of airflow. The switch opens at 30 F and closes at 45 F. No adjustment is necessary. Refer to Tables 1A and 1B.

Checking and Adjustment Refrigerant Charge — The refrigerant system is fully charged with R-22 refrigerant, tested, and factory-sealed. Unit must operate in Cooling mode a minimum of 10 minutes before checking charge.

NOTE: Adjustment of the refrigerant charge is not required unless the unit is suspected of not having the proper R-22 charge.

A superheat charging chart is attached to the outside of the service access panel. The chart includes the required suction line temperature at given suction line pressures and outdoor ambient temperatures.

An accurate superheat, thermocouple- or thermistor-type thermometer, and a gage manifold are required when using the superheat charging method for evaluating the unit charge. *Do not use mercury or small dial-type thermometers because they are not adequate for this type of measurement.*

⚠ CAUTION

When evaluating the refrigerant charge, an indicated adjustment to the specified factory charge must always be very minimal. If a substantial adjustment is indicated, an abnormal condition exists somewhere in the cooling system, such as insufficient airflow across either coil or both coils.

Proceed as follows:

1. Remove caps from low- and high-pressure Schrader valve fittings.
2. Using hoses with valve core depressors, attach low- and high-pressure gage hoses to low- and high-pressure service fittings, respectively.
3. Start unit in Cooling mode and let unit run until system pressure stabilize.
4. Measure and record the following:
   a. Outdoor ambient-air temperature (F db).
   b. Evaporator inlet-air temperature (F wb).
   c. Suction-tube temperature (F) at low-side service fitting.
   d. Suction (low-side) pressure (psig).
5. Using "Cooling Charging Charts" compare outdoor-air temperature (F db) with the suction line pressure (psig) to determine desired system operating suction line temperature. See Fig. 47-54.
6. Compare measured suction-tube temperature with desired suction-tube temperature. Using a tolerance of ± 3° F, add refrigerant if measured temperature is more than 3° F higher than proper suction-tube temperature, or remove refrigerant if actual temperature is more than 3° F lower than required suction-tube temperature.

TO USE COOLING CHARGING CHART — This method is to be used in Cooling mode only. Take the outdoor ambient temperature and read the suction pressure gage. Refer to appropriate chart to determine what suction temperature should be. If suction temperature is high, add refrigerant. If suction temperature is low, carefully recover some of the charge. Recheck the suction pressure as charge is adjusted. Example (Fig. 50, Circuit 2):

Outdoor Temperature. . . . . .85 F
Suction Pressure. . . . . 74 psig
Suction Temperature should be. . . . . .56 F
(Suction temperature may vary ± 3 F.)

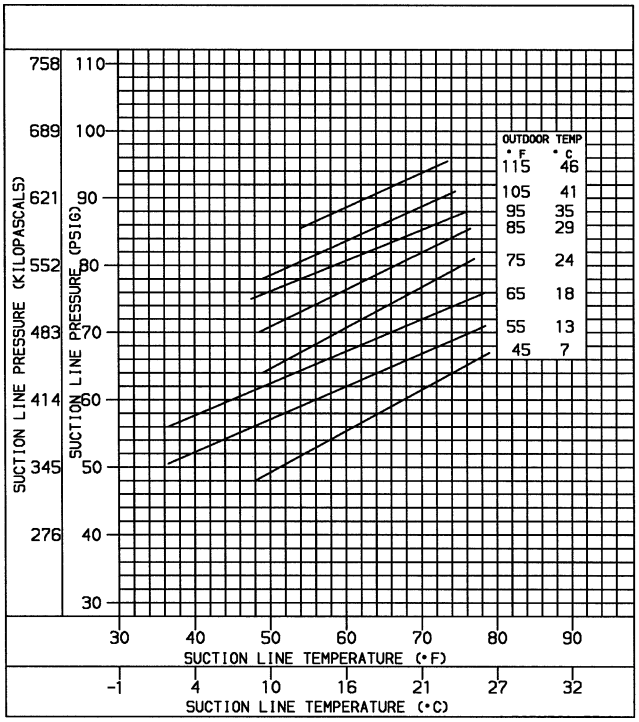


Fig. 47 — Cooling Charging Chart; 48TF008

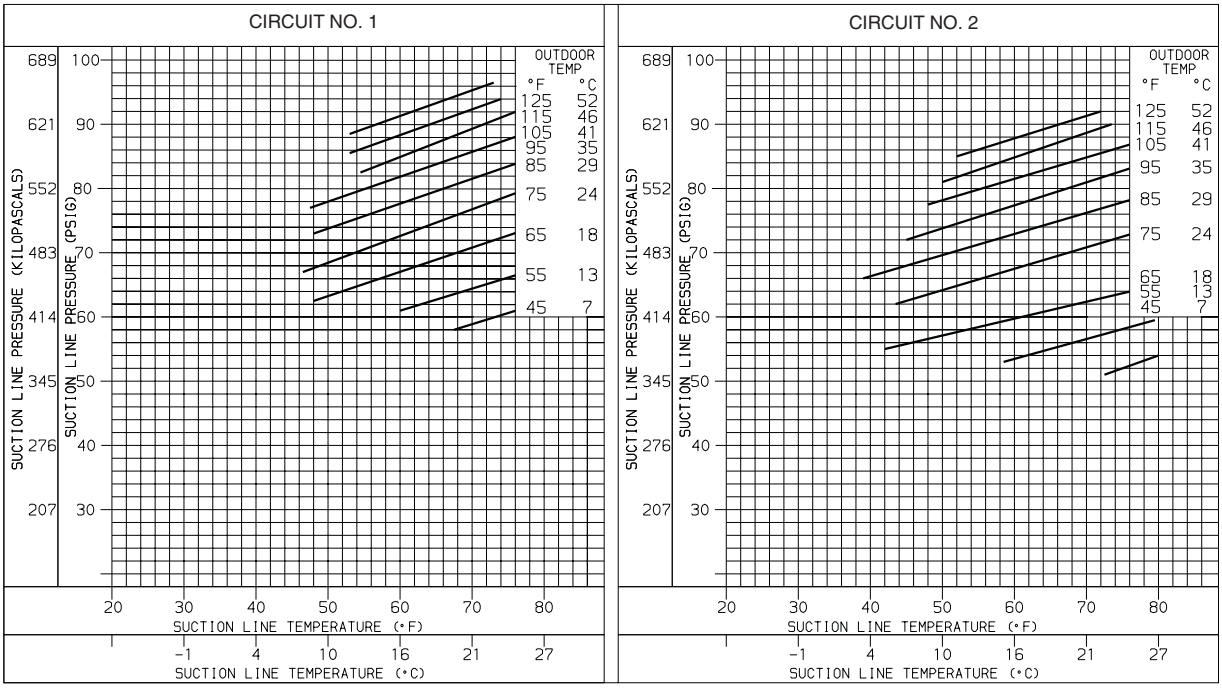


Fig. 48 — Cooling Charging Chart; 48TF009

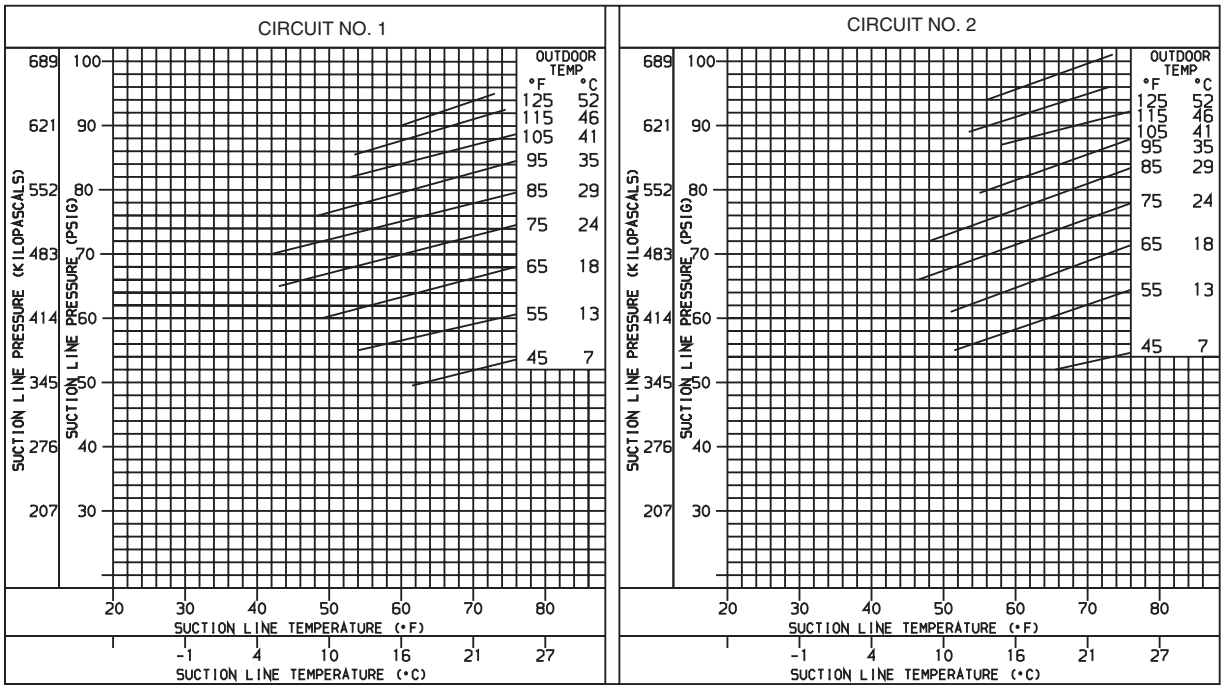


Fig. 49 — Cooling Charging Chart; 48TF012

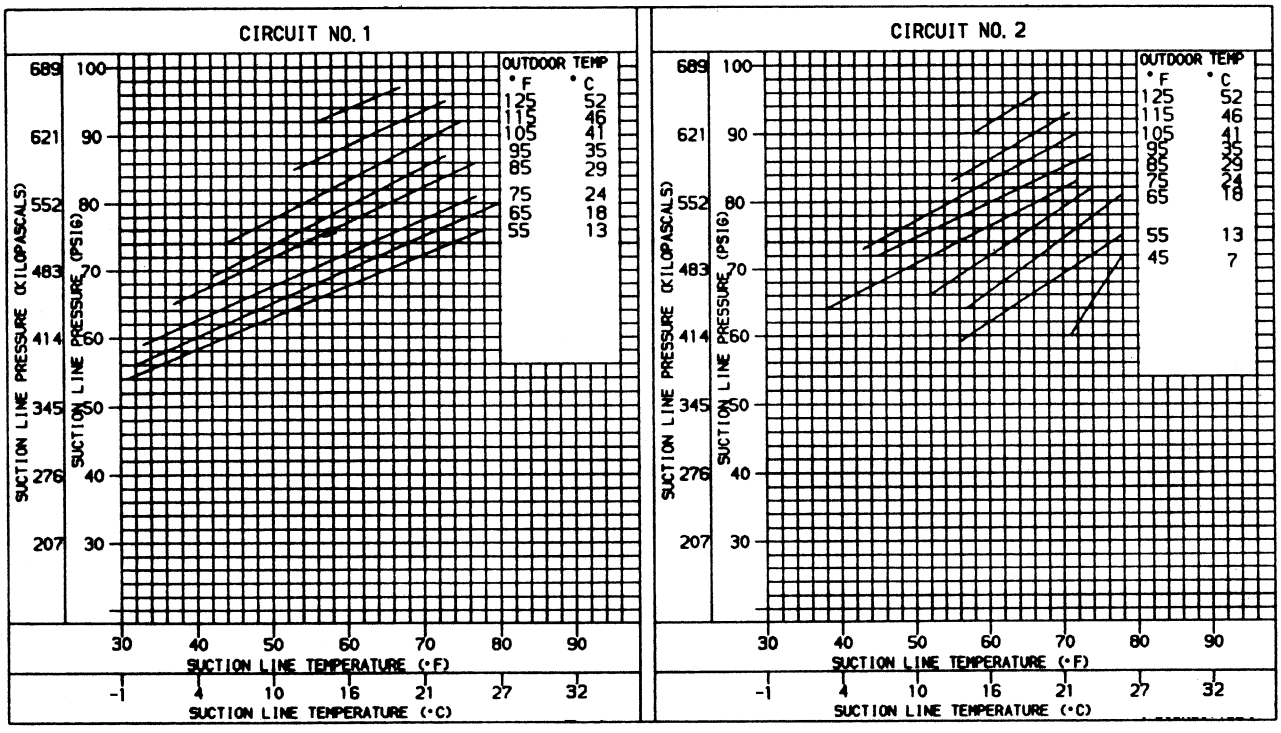


Fig. 50 — Cooling Charging Chart; 48TF014

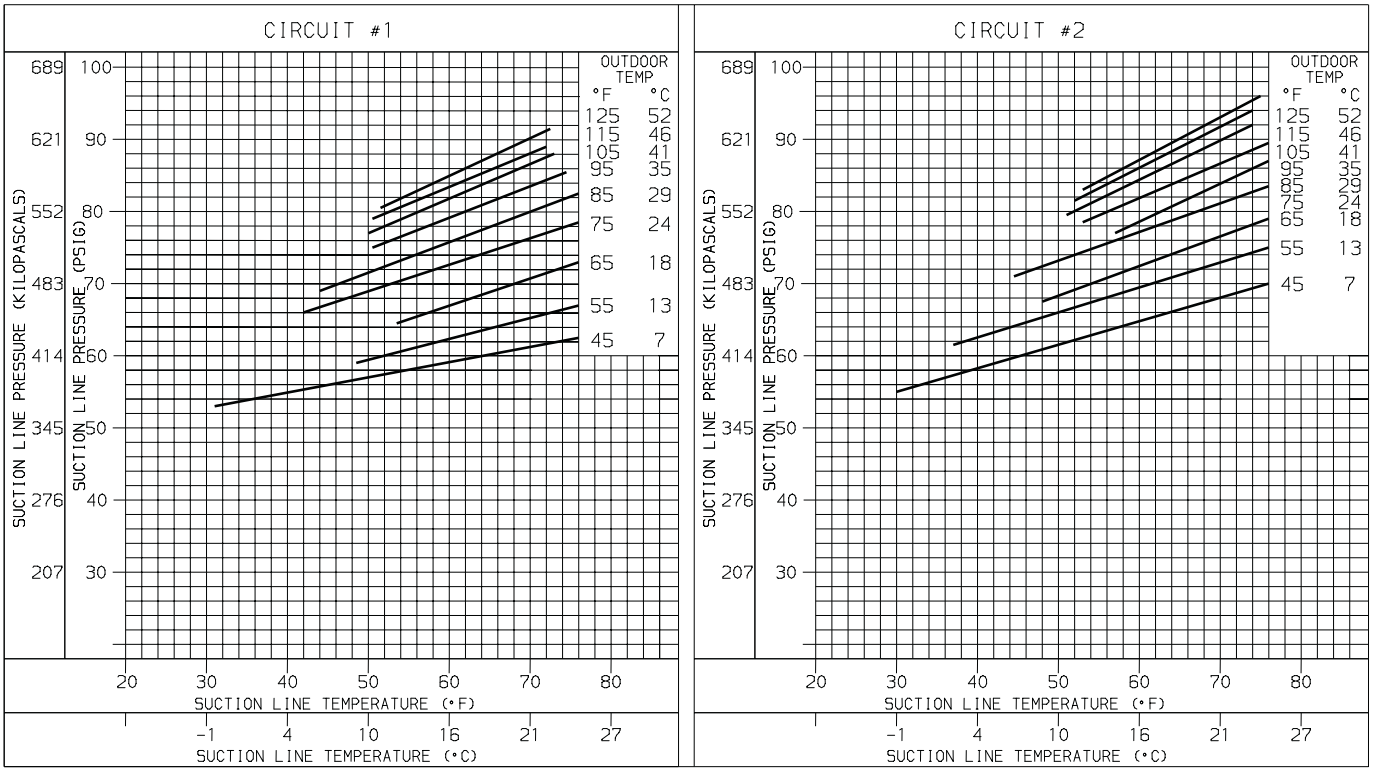


Fig. 51 — Cooling Charging Chart; 48TM008

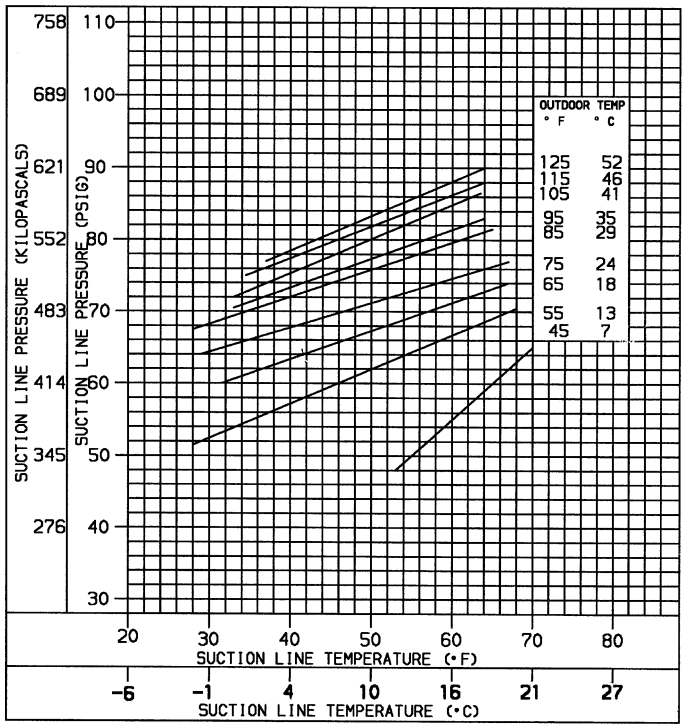


Fig. 52 — Cooling Charging Chart; 48TM009

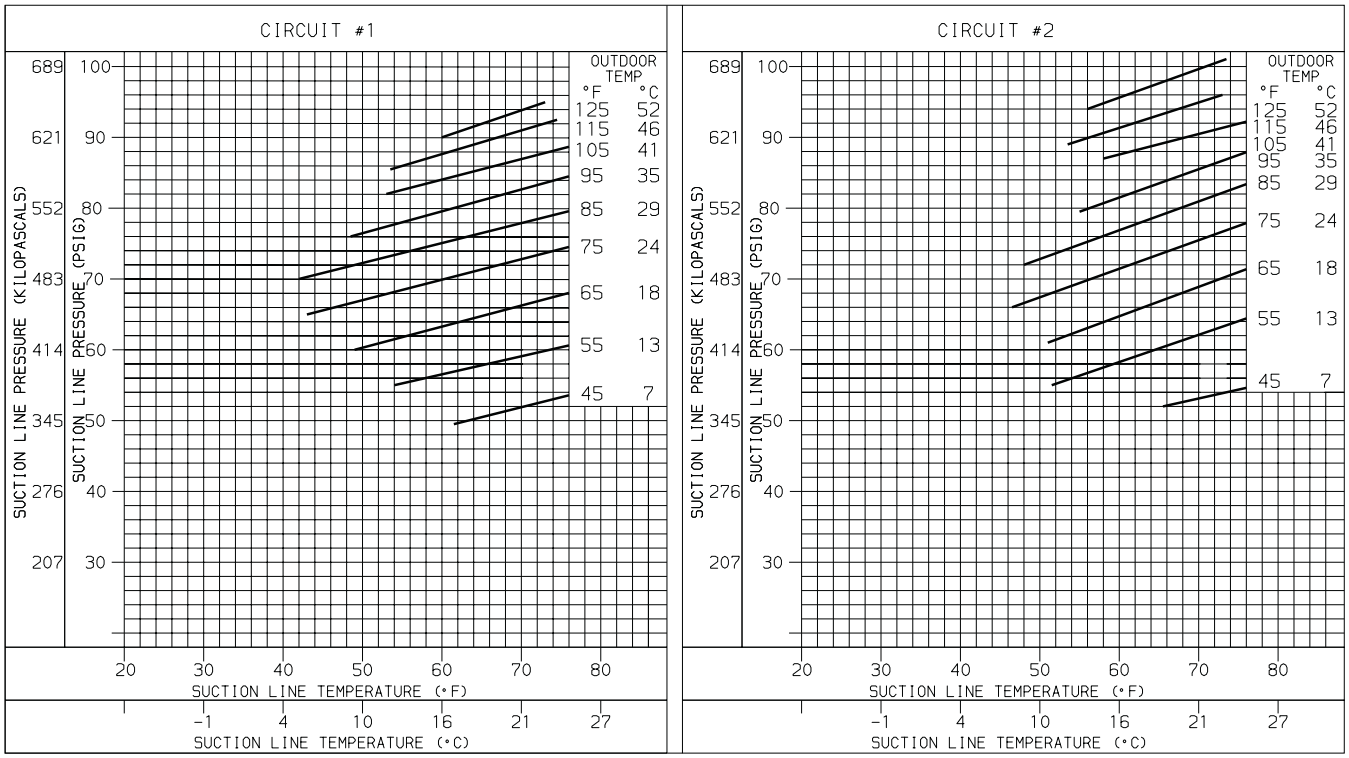


Fig. 53 — Cooling Charging Chart; 48TM012

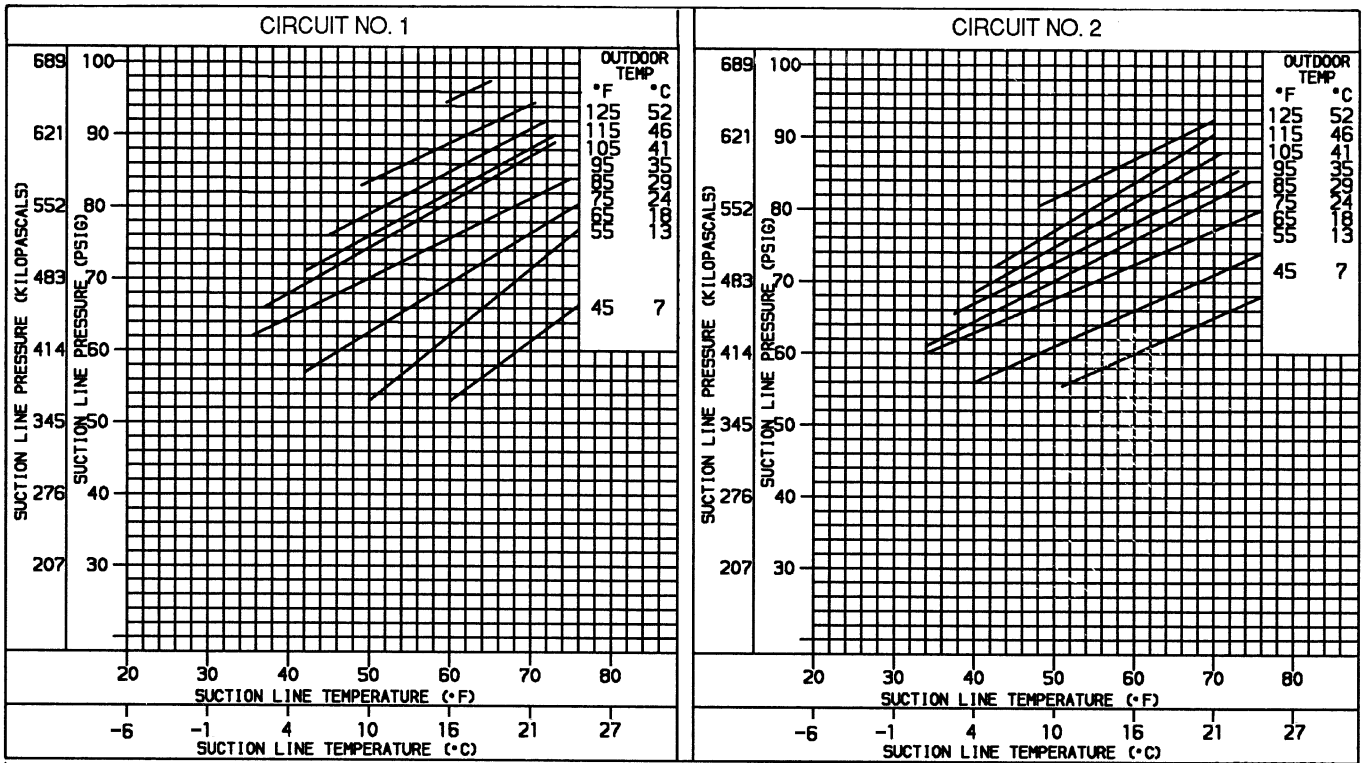


Fig. 54 — Cooling Charging Chart; 48TM014

Flue Gas Passageways — To inspect the flue collector box and upper areas of the heat exchanger:

1. Remove the combustion blower wheel and motor assembly according to directions in Combustion-Air Blower section below.
2. Remove the flue cover to inspect the heat exchanger.
3. Clean all surfaces as required using a wire brush.

Combustion-Air Blower — Clean seasonally to assure proper airflow and heating efficiency. Inspect blower wheel every fall and periodically during heating season. For the first heating season, inspect blower wheel bimonthly to determine proper cleaning frequency.

To inspect blower wheel, shine a flashlight into draft hood opening. If cleaning is required, remove motor and wheel as follows:

1. Slide burner access panel out.
2. Remove the 6 screws that attach induced-draft motor housing to vestibule plate (Fig. 55).
3. The blower wheel can be cleaned at this point. If additional cleaning is required, continue with Steps 4 and 5.
4. To remove blower from the motor shaft, remove 2 setscrews.
5. To remove motor, remove the 4 screws that hold blower housing to mounting plate. Remove the motor cooling fan by removing one setscrew. Then remove nuts that hold motor to mounting plate.
6. To reinstall, reverse the procedure outlined above.

Limit Switch — Remove blower access panel (Fig. 6A and 6B). Limit switch is located on the fan deck. Verify operation of limit by temporarily blocking return air until limit trips.

Burner Ignition — Unit is equipped with a direct spark ignition 100% lockout system. Integrated Gas Unit Controller (IGC) is located in the control box (Fig. 11). Module contains a self-diagnostic LED. A single LED on the IGC provides a visual display of operational or sequential problems when the power supply is interrupted. When a break in power occurs, the module will be reset (resulting in a loss of fault history) and the indoor (evaporator) fan ON/OFF times will be reset. For additional information, refer to the Start-Up, Heating section on page 44. The LED error code can be observed through the viewport. See Fig. 11. During servicing refer to the label on the control box cover or Table 22 for an explanation of LED error code descriptions.

If lockout occurs, unit may be adjusted by interrupting power supply to unit for at least 5 seconds.

Main Burners — At the beginning of each heating season, inspect for deterioration, blockage due to corrosion or other causes. Observe the main burner flames and replace burners if necessary.

⚠ CAUTION

When working on gas train, do not hit or plug orifice spuds.

REMOVAL AND REPLACEMENT OF GAS TRAIN (Fig. 55 and 56)

1. Shut off manual gas valve.
2. Shut off power to unit and install lockout tag.
3. Slide out burner section side panel (not shown).
4. Disconnect gas piping at unit gas valve using backup wrench on the flats of the valve body where the gas pipe enters the gas valve. See Fig. 56.
5. Remove wires connected to gas valve. Mark each wire.
6. Remove wires from ignitor and sensor wires at the Integrated Gas Unit Controller (IGC).
7. Remove the 2 screws that attach the burner rack to the vestibule plate.
8. Slide the burner tray out of the unit (Fig. 55).
9. To reinstall, reverse the procedure outlined above.

CLEANING AND ADJUSTMENT

1. Remove burner rack from unit as described in Removal and Replacement of Gas Train section, above.
2. Inspect burners; if dirty, remove burners from rack.
3. Using a soft brush, clean burners and cross-over port as required.
4. Adjust spark gap. See Fig. 57.
5. Reinstall burners on rack.
6. Reinstall burner rack as described in Removal and Replacement of Gas Train section, this page.

Replacement Parts — A complete list of replacement parts may be obtained from any Carrier distributor upon request.

Table 22 — LED Error Code Description*

LED INDICATION	ERROR CODE DESCRIPTION
ON	Normal Operation
OFF	Hardware Failure
1 Flash†	Evaporator Fan On/Off Delay Modified
2 Flashes	Limit Switch Fault
3 Flashes	Flame Sense Fault
4 Flashes	4 Consecutive Limit Switch Faults
5 Flashes	Ignition Lockout Fault
6 Flashes	Induced-Draft Motor Fault
7 Flashes	Rollout Switch Fault
8 Flashes	Internal Control Fault

LEGEND

LED — Light-Emitting Diode

*A 3-second pause exists between LED error code flashes. If more than one error code exists, all applicable codes will be displayed in numerical sequence.

†Indicates a code that is not an error. The unit will continue to operate when this code is displayed.

IMPORTANT: Refer to Troubleshooting Tables 22-24 for additional information.

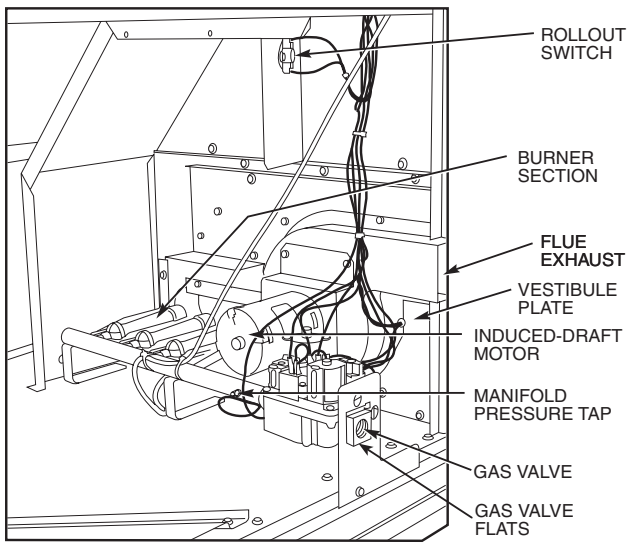


Fig. 55 — Burner Section Details

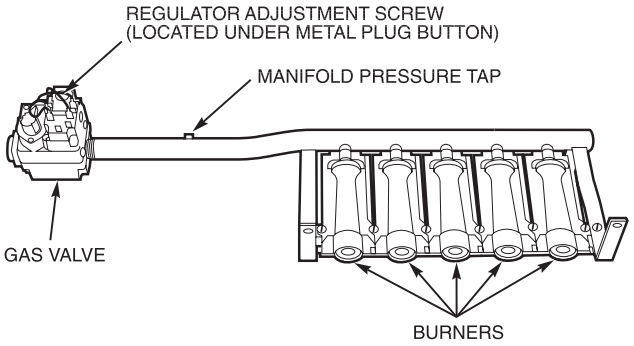
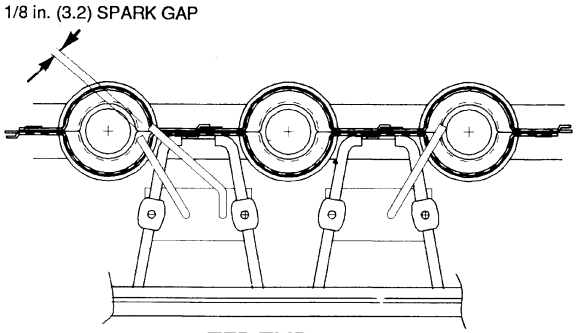
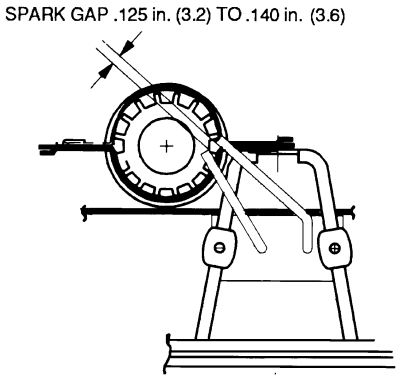


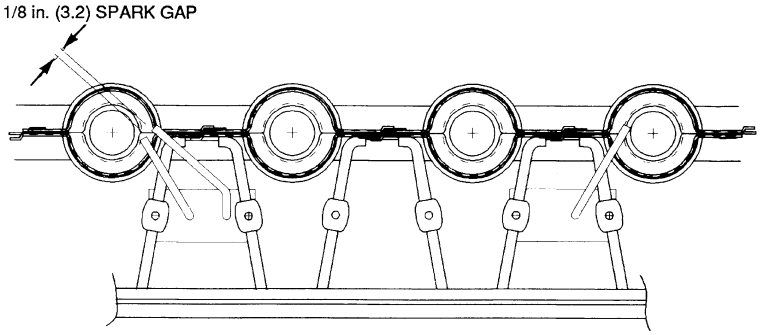
Fig. 56 — Burner Tray Details



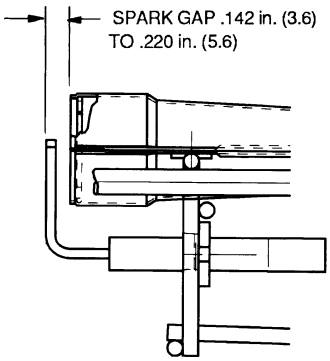
48TFD,TMD008,009
125,000 BTUH INPUT

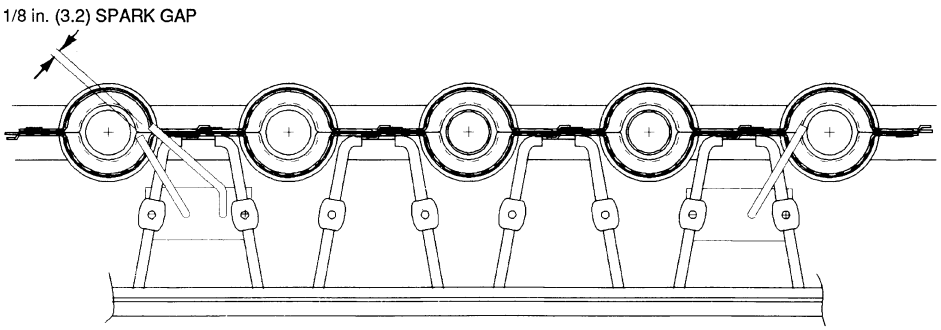


NOTE: Dimensions in ( ) are millimeters.



48TFD,TMD012, 48TFE,TME008,009
180,000 BTUH INPUT





48TFD,TMD014, 48TFE,TME012, 48TFF,TMF008,009, 48TFE,TME014, 48TFF,TMF012
220,000 BTUH INPUT
250,000 BTUH INPUT

Fig. 57 — Spark Gap Adjustment

TROUBLESHOOTING

Refer to Tables 23-25 for unit troubleshooting. See Fig. 58 for typical unit wiring.

EconoMi$er2 Troubleshooting

ECONOMI$ER2 CHECKOUT PROCEDURE — Prepare the EconoMi$er2 for troubleshooting by performing the following procedure:

1. Disconnect power at TR and TR1. Free Cool, unit control, indoor fan, and exhaust control contacts should be open.
2. Disconnect device at P and P1.
3. Disconnect device at Q and Q1.
4. Jumper P to P1.
5. Jumper Q to Q1.
6. Turn Minimum Position potentiometer fully CCW (counterclockwise).
7. Turn Maximum Position potentiometer fully CW (clockwise).
8. Place 6.8 kohm resistor across T and T1.
9. Jumper TR to 1.
10. If connected, remove enthalpy sensor from terminals S_O and +. Connect 1.2 kohm 4074EJM checkout resistor across terminals S_O and +.
11. Set both ISI and Exhaust potentiometers fully CCW.
12. Put 620-ohm resistor across terminals S_R and +.
13. Set enthalpy potentiometer to D.
14. Apply power (24 vac) to terminals TR and TR1.

DIFFERENTIAL ENTHALPY — To check differential enthalpy:

1. Make sure EconoMi$er2 checkout procedure has been performed.
2. Place 620-ohm resistor across S_O and +.
3. Place 1.2 Kohm resistor across S_R and +. The Free Cool LED should be lit.
4. Remove 620-ohm resistor across S_O and +. The Free Cool LED should turn off.
5. Return EconoMi$er2 settings and wiring to normal after completing troubleshooting.

SINGLE ENTHALPY — To check single enthalpy:

1. Make sure EconoMi$er2 checkout procedure has been performed.
2. Set the enthalpy potentiometer to A (fully CCW). The Free Cool LED should be lit.
3. Set the enthalpy potentiometer to D (fully CW). The Free Cool LED should turn off.
4. Return EconoMi$er2 settings and wiring to normal after completing troubleshooting.

ISI (INDOOR AIR QUALITY) AND POWER EXHAUST — To check ISI and Power Exhaust:

1. Make sure EconoMi$er2 checkout procedure has been performed.
2. Ensure terminals AQ and AQ1 are open. The LED for both ISI and Power Exhaust should be off. The actuator should be fully closed.
3. Connect a 9V battery to AQ (positive node) and AQ1 (negative node). The LED for both ISI and the Exhaust Fan should turn on. The actuator should drive to 90 to 95% open.
4. Turn the exhaust potentiometer CW until the Exhaust LED turns off. The LED should turn off when the potentiometer is approximately 90%. The actuator should remain in position.
5. Turn ISI potentiometer CW until ISI LED turns off. The ISI LED should turn off when the potentiometer is approximately 9V. The actuator should drive fully closed.
6. Turn ISI and Exhaust potentiometers CW until the Exhaust LED turns on. Forty-five seconds after the Exhaust LED turns on, the exhaust contacts will close.
7. Return EconoMi$er2 settings and wiring to normal after completing troubleshooting.

FREEZE PROTECTION — To check freeze protection:

1. Make sure EconoMi$er2 checkout procedure has been performed.
2. Connect a 9V battery to AQ (positive node) and AQ1 (negative node). The LED for both ISI and the Exhaust Fan should turn on. The actuator should drive to 90 to 95% open.
3. Remove the 6.8 Kohm resistor from T and T1. The Exhaust LED should turn off and the actuator should drive fully closed.
4. Replace the 6.8 Kohm resistor across T and T1. The Exhaust LED should turn on and the actuator should drive to 90 to 95% open.
5. Return EconoMi$er2 settings and wiring to normal after completing troubleshooting.

MINIMUM AND MAXIMUM POSITION — To check minimum and maximum position:

1. Make sure EconoMi$er2 checkout procedure has been performed.
2. Connect a 9V battery to AQ (positive node) and AQ1 (negative node). The LED for both ISI and the Exhaust Fan should turn on. The actuator should drive to 90 to 95% open.
3. Turn the Maximum Position potentiometer to midpoint. The actuator should drive to 20 to 80% open.
4. Turn the Maximum Position potentiometer to fully CCW. The actuator should drive fully closed.
5. Turn the Minimum Position potentiometer to midpoint. The actuator should drive 20 to 80% open.
6. Turn the Minimum Position Potentiometer fully CW. The actuator should drive fully open.
7. Return EconoMi$er2 settings and wiring to normal after completing troubleshooting.

AIR CHANGE, SHUTDOWN, AND PURGE — To check air change, shutdown, and purge:

1. Make sure EconoMi$er2 checkout procedure has been performed.
2. Ensure terminals AQ and AQ1 are open.
3. Turn the Minimum Position Potentiometer to midpoint. The actuator should drive 20 to 80% open.
4. Make sure that the Air Change, Purge, and Shutdown terminals are empty. The Exhaust LED should turn on. The Unit Control and Indoor Fan contacts will close.
5. Jumper the Air Change terminals. The exhaust LED turns on. The Unit Control contacts open. The Indoor Fan contacts remain closed. The actuator drives fully open.

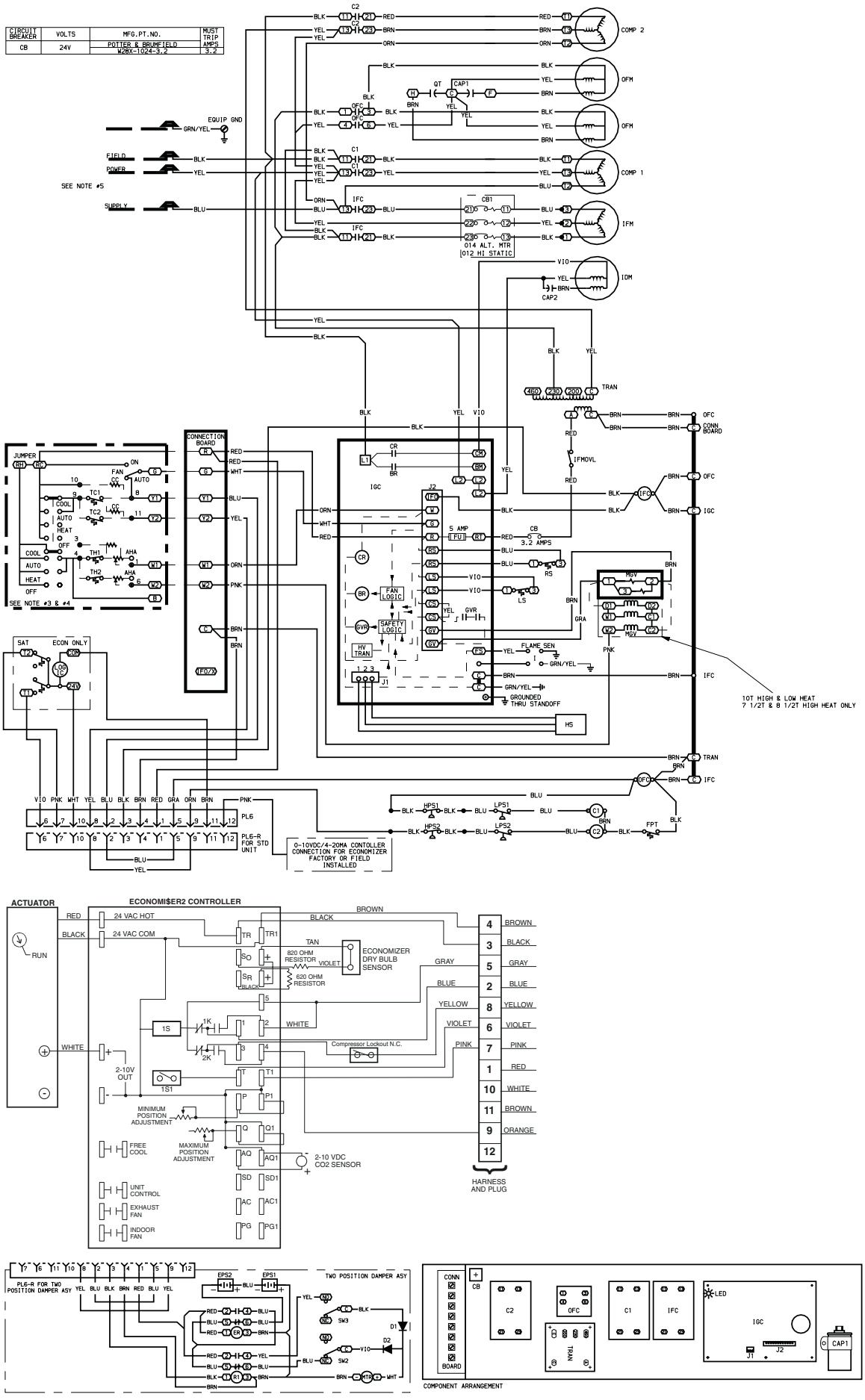


Fig. 58 — Typical Wiring Schematic and Component Arrangement (208/230-3-60 Unit Shown)

LEGEND FOR FIG. 58 — TYPICAL WIRING SCHEMATIC AND COMPONENT ARRANGEMENT

IMPORTANT: Refer to unit wiring label for actual unit wiring information.

AHA — Adjustable Heat Anticipator
C — Contactor, Compressor
CAP — Capacitor
CB — Circuit Breaker
CC — Cooling Compensator
COMP — Compressor Motor
D — Diode
ECON — Economizer
EPS — Emergency Power Supply (9 V Battery)
EQUIP — Equipment
FPT — Freeze Up Protection Thermostat
GND — Ground
HPS — High-Pressure Switch
HS — Hall-Effect Sensor
I — Ignitor
IDM — Induced-Draft Motor
IFC — Indoor Fan Contactor
IFM — Indoor (Evaporator) Fan Motor
IFMOVL — Indoor Fan Motor Overload
IGC — Integrated Gas Unit Controller
LED — Light-Emitting Diode
LPS — Low-Pressure/Loss-of-Charge Switch
LS — Limit Switch
MGV — Main Gas Valve
MTR — Motor
OFC — Outdoor Fan Contactor
OFM — Outdoor (Condenser) Fan Motor
PL — Plug Assembly
QT — Quadruple Terminal
R — Relay
RS — Rollout Switch

SAT — Supply-Air Thermostat
SEN — Sensor
SW1 — Switch Fully Open
SW2 — Switch Fully Closed
SW3 — Switch Min. Vent Position
SW4 — Switch Max. Vent Position
TC — Thermostat-Cooling
TH — Thermostat-Heating
TRAN — Transformer


Field Splice



Marked Wire



Terminal (Marked)


Terminal (Unmarked)



Terminal Block


Splice


Splice (Marked)

Factory Wiring

Field Control Wiring

Field Power Wiring

Accessory or Optional Wiring

To indicate common potential only; not to represent wiring.

NOTES:
1. If any of the original wire furnished must be replaced, it must be replaced with type 90 C wire or its equivalent.
2. Three-phase motors are protected under primary single-phasing conditions.
3. Thermostat: HH07AT170, 172, 174 and P272-2783 Subbase: HH93AZ176, 178 and P272-1882, 1883.
4. Set heat anticipator for first stage at .14 amp. Set second stage at .2 amp.
5. Use copper conductors only.
6. TRAN is wired for 230-v unit. If unit is to be run with 208-v power supply, disconnect BLK wire from 230-v tap (ORN) and connect to 208-v tap (RED). Insulate end of 230-v tap.

6. Remove the jumper from the Air Change terminals. Jumper the Shutdown terminals. The exhaust LED turns off. The Unit Control, Indoor Fan, and Exhaust Fan contacts open. The actuator drives fully closed.
7. Remove the jumper from the Shutdown terminals. Jumper the Purge terminals. The exhaust LED turns off. The Unit Control and Indoor Fan contacts open. The Exhaust Fan contacts close. The actuator drives fully closed.
8. Return EconoMi$er2 settings and wiring to normal after completing troubleshooting.

MIXED AIR INPUT — To check mixed air input:

1. Make sure EconoMi$er2 checkout procedure has been performed.
2. Set the Enthalpy potentiometer to A. The Free Cool LED turns on.
3. Replace the 6.8 Kohm resistor across T and T1 with a 5.6 Kohm resistor. The actuator should drive 20 to 80% open.
4. Remove the 5.6 Kohm resistor and jumper T to T1. The actuator should drive fully open.
5. Remove the jumper across T and T1. The actuator should drive fully closed.
6. Replace the 6.8 Kohm resistor across T and T1. The actuator should remain fully closed.
7. Return EconoMi$er2 settings and wiring to normal after completing troubleshooting.

MIXED-AIR SENSOR — The mixed-air sensor has a negative coefficient. As the ambient temperature increases, the resistance decreases. The sensor changes 70 ohms for every 1 degree change in temperature. See Fig. 59 for a resistance vs. temperature chart which can be used to check sensor calibration and operation.

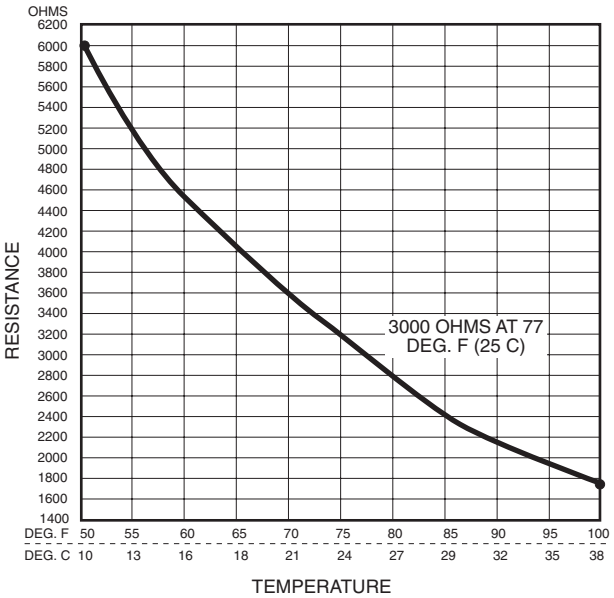


Fig. 59 — Mixed Air Sensor Resistance vs. Temperature Chart

Table 23 — LED Troubleshooting — Error Code

SYMPTOM	CAUSE	REMEDY
Hardware failure. (LED OFF)	Loss of power to control module (IGC).	Check 5 amp fuse on IGC, power to unit, 24-v circuit breaker, and transformer. Units without a 24-v circuit breaker have an internal overload in the 24-v transformer. If the overload trips, allow 10 minutes for automatic reset.
Fan ON/OFF delay modified. (LED/FLASH)	High limit switch opens during heat exchanger warm-up period before fan-on delay expires.	Ensure unit is fired on rate and temperature rise is correct.
	Limit switch opens within three minutes after blower-off delay timing in Heating mode.	Ensure unit's external static pressure is within application guidelines.
Limit switch fault. (LED 2 flashes)	High temperature limit switch is open.	Check the operation of the indoor (evaporator) fan motor. Ensure that the supply-air temperature rise is in accordance with the range on the unit nameplate.
Flame sense fault. (LED 3 flashes)	The IGC sensed flame that should not be present.	Reset unit. If problem persists, replace control board.
4 consecutive limit switch faults. (LED 4 flashes)	Inadequate airflow to unit.	Check operation of indoor (evaporator) fan motor and that supply-air temperature rise agrees with range on unit nameplate information.
Ignition lockout. (LED 5 flashes)	Unit unsuccessfully attempted ignition for 15 minutes.	Check ignitor and flame sensor electrode spacing, gaps, etc. Ensure that flame sense and ignition wires are properly terminated. Verify that unit is obtaining proper amount of gas.
Induced-draft motor fault. (LED 6 flashes)	IGC does not sense that induced-draft motor is operating.	Check for proper voltage. If motor is operating, check the speed sensor plug/IGC Terminal J2 connection. Proper connection: PIN 1— White, PIN 2 — Red, PIN 3 — Black.
Rollout switch fault. (LED 7 flashes)	Rollout switch has opened.	Rollout switch will automatically reset, but IGC will continue to lock out unit. Check gas valve operation. Ensure that induced-draft blower wheel is properly secured to motor shaft. Reset unit at unit disconnect.
Internal control fault. (LED 8 flashes)	Microprocessor has sensed an error in the software or hardware.	If error code is not cleared by resetting unit power, replace the IGC.

⚠ WARNING

If the IGC must be replaced, be sure to ground yourself to dissipate any electrical charge that may be present before handling new control board. The IGC is sensitive to static electricity and may be damaged if the necessary precautions are not taken.

IMPORTANT: Refer to Table 25, Heating Service troubleshooting chart for additional troubleshooting analysis.

LEGEND

IGC — Integrated Gas Unit Controller
LED — Light-Emitting Diode

Table 24 — Cooling Service Troubleshooting

PROBLEM	CAUSE	REMEDY
Compressor and condenser fan will not start.	Power failure.	Call power company.
	Fuse blown or circuit breaker tripped.	Replace fuse or reset circuit breaker.
	Defective thermostat, contactor, transformer, or control relay.	Replace component.
	Insufficient line voltage.	Determine cause and correct.
	Incorrect or faulty wiring.	Check wiring diagram and rewire correctly.
	Thermostat setting too high.	Lower thermostat setting below room temperature.
Compressor will not start but condenser fan runs.	Faulty wiring or loose connections in compressor circuit.	Check wiring and repair or replace.
	Compressor motor burned out, seized, or internal overload open.	Determine cause. Replace compressor.
	Defective run/start capacitor, overload, or start relay.	Determine cause and replace.
	One leg of 3-phase power dead.	Replace fuse or reset circuit breaker. Determine cause.
Compressor cycles (other than normally satisfying thermostat).	Refrigerant overcharge or undercharge.	Recover refrigerant, evacuate system, and recharge to nameplate.
	Defective compressor.	Replace and determine cause.
	Insufficient line voltage.	Determine cause and correct.
	Blocked condenser.	Determine cause and correct.
	Defective run/start capacitor, overload, or start relay.	Determine cause and replace.
	Defective thermostat.	Replace thermostat.
	Faulty condenser-fan motor or capacitor.	Replace.
	Restriction in refrigerant system.	Locate restriction and remove.
Compressor makes excessive noise (scroll compressors only).	Compressor rotating in wrong direction.	Reverse the 3-phase power leads as described on page 43.
Compressor operates continuously.	Dirty air filter.	Replace filter.
	Unit undersized for load.	Decrease load or increase unit size.
	Thermostat set too low.	Reset thermostat.
	Low refrigerant charge.	Locate leak, repair, and recharge.
	Leaking valves in compressor.	Replace compressor.
	Air in system.	Recover refrigerant, evacuate system, and recharge.
	Condenser coil dirty or restricted.	Clean coil or remove restriction.
Excessive head pressure.	Dirty air filter.	Replace filter.
	Dirty condenser coil.	Clean coil.
	Refrigerant overcharged.	Remove excess refrigerant.
	Air in system.	Recover refrigerant, evacuate system, and recharge.
	Condenser air restricted or air short-cycling.	Determine cause and correct.
Head pressure too low.	Low refrigerant charge.	Check for leaks, repair, and recharge.
	Compressor valves leaking.	Replace compressor.
	Restriction in liquid tube.	Remove restriction.
Excessive suction pressure.	High heat load.	Check for source and eliminate.
	Compressor valves leaking.	Replace compressor.
	Refrigerant overcharged.	Recover excess refrigerant.
Suction pressure too low.	Dirty air filter.	Replace filter.
	Low refrigerant charge.	Check for leaks, repair, and recharge.
	Metering device or low side restricted.	Remove source of restriction.
	Insufficient evaporator airflow.	Increase air quantity. Check filter and replace if necessary.
	Temperature too low in conditioned area.	Reset thermostat.
	Field-installed filter drier restricted.	Replace.
Compressor no. 2 will not run.	Unit in economizer mode.	Proper operation; no remedy necessary.

Table 25 — Heating Service Troubleshooting

PROBLEM	CAUSE	REMEDY
Burners will not ignite.	Misaligned spark electrodes.	Check flame ignition and sensor electrode positioning. Adjust as needed.
	No gas at main burners.	Check gas line for air purge as necessary. After purging gas line of air, allow gas to dissipate for at least 5 minutes before attempting to relight unit.
		Check gas valve.
	Water in gas line.	Drain water and install drip leg to trap water.
	No power to furnace.	Check power supply, fuses, wiring, and circuit breaker.
	No 24 v power supply to control circuit.	Check transformer. Transformers with internal overturerent protection require a cool down period before resetting.
	Miswired or loose connections.	Check all wiring and wirenut connections.
	Burned-out heat anticipator in thermostat.	Replace thermostat.
	Broken thermostat wires.	Run continuity check. Replace wires, if necessary.
Inadequate heating.	Dirty air filter.	Clean or replace filter as necessary.
	Gas input to unit too low.	Check gas pressure at manifold. Clock gas meter for input. If too low, increase manifold pressure, or replace with correct orifices.
	Unit undersized for application.	Replace with proper unit or add additional unit.
	Restricted airflow.	Clean filter, replace filter, or remove any restrictions.
	Blower speed too low.	Use high speed tap, increase fan speed, or install optional blower, as suitable for individual units.
	Limit switch cycles main burners.	Check rotation of blower, thermostat heat anticipator settings, and temperature rise of unit. Adjust as needed.
	Too much outdoor air.	Adjust minimum position.
		Check economizer operation.
Poor flame characteristics.	Incomplete combustion (lack of combustion air) results in:	Check all screws around flue outlets and burner compartment. Tighten as necessary.
	Aldehyde odors, CO, sooting flame, or floating flame.	Cracked heat exchanger.
		Overfired unit — reduce input, change orifices, or adjust gas line or manifold pressure.
		Check vent for restriction. Clean as necessary.
		Check orifice to burner alignment.
Burners will not turn off.	Unit is locked into Heating mode for a one minute minimum.	Wait until mandatory one minute time period has elapsed or power to unit.

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START-UP CHECKLIST
(Remove and Store in Job File)

I. PRELIMINARY INFORMATION

MODEL NO.:_____ SERIAL NO.: _____

DATE:_____ TECHNICIAN: _____

UNIT NO.: _____ JOB LOCATION: _____

JOB NAME: _____

II. PRE-START-UP (insert checkmark in box as each item is completed)

- ☐ VERIFY THAT ALL PACKING MATERIALS HAVE BEEN REMOVED FROM UNIT
- ☐ REMOVE SHIPPING TIE DOWN BANDS ON COMPRESSOR (SIZE 014 ONLY) PER INSTALLATION INSTRUCTIONS
- ☐ VERIFY THAT CONDENSATE CONNECTION IS INSTALLED PER INSTALLATION INSTRUCTIONS
- ☐ CHECK ALL ELECTRICAL CONNECTIONS AND TERMINALS FOR TIGHTNESS
- ☐ CHECK GAS PIPING FOR LEAKS
- ☐ CHECK THAT INDOOR-AIR FILTERS ARE CLEAN AND IN PLACE
- ☐ VERIFY THAT UNIT INSTALLATION IS LEVEL
- ☐ CHECK FAN WHEELS AND PROPELLERS FOR LOCATION IN HOUSING/ORIFICE AND SETSCREW TIGHTNESS
- ☐ ENSURE BELT TENSION IS CORRECT AND BLOWER PULLEYS ARE PROPERLY ALIGNED.
- ☐ VERIFY INSTALLATION OF ECONOMI$ER2 HOOD (IF EQUIPPED).

III. START-UP

ELECTRICAL

SUPPLY VOLTAGE	L1-L2	_____	L2-L3	_____	L3-L1	_____
COMPRESSOR AMPS	L1	_____	L2	_____	L3	_____
COMPRESSOR AMPS	L1	_____	L2	_____	L3	_____
INDOOR-FAN AMPS	L1	_____	L2	_____	L3	_____

TEMPERATURES AND PRESSURES

OUTDOOR-AIR TEMPERATURE	_____	DB		
RETURN-AIR TEMPERATURE	_____	DB	_____	WB
COOLING SUPPLY AIR	_____	DB	_____	WB
GAS HEAT SUPPLY AIR	_____	DB		

GAS INLET PRESSURE	_____	IN. WG		
GAS MANIFOLD PRESSURE	_____	IN. WG (HI FIRE)	_____	IN. WG (LO FIRE)
REFRIGERANT SUCTION PRESSURE	_____	PSIG — CIRCUIT NO. 1	_____	PSIG — CIRCUIT NO. 2
REFRIGERANT TEMP. (SUCTION) PRESSURE	_____	CIRCUIT NO. 1	_____	CIRCUIT NO. 2
REFRIGERANT DISCHARGE	_____	PSIG — CIRCUIT NO. 1	_____	PSIG — CIRCUIT NO. 2
DISCHARGE TEMPERATURE	_____	°F/C — CIRCUIT NO. 1	_____	°F/C — CIRCUIT NO. 2

- ☐ VERIFY REFRIGERANT CHARGE USING CHARGING TABLES
- ☐ VERIFY THAT 3-PHASE SCROLL COMPRESSOR ROTATING IN CORRECT DIRECTION (48TF014 AND 48TM009-014 ONLY)

GENERAL

- ☐ SET ECONOMI$ER2 MINIMUM VENT POSITION AND CHANGEOVER SETTINGS TO JOB REQUIREMENTS (IF EQUIPPED)

CUT ALONG DOTTED LINE

CUT ALONG DOTTED LINE

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